

Definitions:

- **Continuous spectrum:** spectrum consisting of **all wavelengths/frequencies** from EM spectrum
- **Line spectrum:** spectrum consisting of **discrete coloured lines**
- **Emission spectrum:** produced by atoms **emitting E as photons** when **e⁻ in excited states** return to **lower E levels/ground state**
- **Absorption spectrum:** produced when **radiation (white light)** is passed through sample, atom/ion/molecule becomes **excited & e⁻ move to higher E level**
- **Ground state:** **lowest E state** of atom
- **I.E.:** amount of E required to **remove 1 mole of e⁻ from 1 mole of isolated gaseous atoms** → form **1 mole of gaseous unipositive ion** under standard thermodynamic conditions (**n = ∞**)
- **1st I.E.:** **min.** E required to remove **1st e⁻ from gaseous atom (kJmol⁻¹)** ($X_{(g)} \rightarrow X^+_{(g)} + e^-$)
- **Convergence limit:** **f/λ** corresponding to **ionisation**
- **Isotope:** atoms of **same element** with **same no. of protons** but **different no. of neutrons**
- **Empirical formula:** **lowest whole number ratio** of atoms of each element present in compound
- **Molecular formula:** **actual no.** of atoms of elements present in compounds
- **Aufbau's principle:** e⁻ will always first fill **lowest E level orbital** available
- **Pauli's exclusion principle:** each orbital can hold **2 e⁻ of opposite spins**
- **Hund's rule:** each orbital must get **1 e⁻ of same spin** before any orbital gets **second e⁻**
- **Atomic orbitals:** region of space of **high probability** of finding an e⁻
- **Limiting reagent:** reactant that get **used up first** during reaction
- **Ideal gas:** theoretical gas made up of **randomly moving particles** with **no vol.** subjected to **interparticle interactions (IMF)**
- **Avogadro's law:** **equal vol.** of different gases at **same temp. & pressure** contain **equal no. of molecules**
- **Rate of reaction:** **change in conc.** of particular **reactant/product** per unit **time**

- **Activation energy: min. K.E.** colliding particles must possess for **successful** collision
- **Catalyst:** substance that **lowers E_A** without being **permanently changed/used up**
- **Intermediate: short-lived, stable** specie formed during chemical reaction which can be **isolated**
- **Activated complex:** unstable association of atoms of **highest E** formed during **transition state**
- **Homogeneous catalyst:** catalyst in **same phase** as reactants
- **Heterogeneous catalyst:** catalyst in **different phase** as reactants
- **Rate-determining step: slowest elementary reaction (highest E_A)** in **multi-step** reaction that determines **overall rate**
- **Electronegativity: relative** measure of tendency of atom in **covalent** bond to attract **shared pair of e^-** towards itself
- **e^- affinity: ΔH** when 1 mol of **gaseous** atoms each acquire 1 e^- to form 1 mol of **gaseous ions ($^-$)**
- **Oxidation state:** charge element would have if **e^- in bond** were assigned to atom of more **electronegative** element
- **Polarisation: deformation** of **e^- cloud of anion** by charge of **cation**
- **Formal charge:** charge assigned to atom in molecule assuming e^- are shared **equally**
- **Ligand:** atom/molecule containing **lone pairs of e^-** $\rightarrow$ form **coordinate covalent** bonds with **transition metals**
- **Coordinate covalent bond: both e^-** come from **1 atom**
- **Allotrope: ≥ 2 different physical form** of one element (e.g. graphite & diamond $\rightarrow$ C)
- **Av. bond enthalpy: ΔH** to break **1 mol** of a bond in **gaseous** molecule **averaged** over **similar** compounds
- **Standard enthalpy of combustion: ΔH** when **1 mol** of substance reacts **completely** with **O_2** under **STP** (100 kPa, 298.15 K) (reactants & products in **standard states** with **allotropes defined** (i.e. diamond differs from graphite))
- **Standard enthalpy of formation: ΔH** when **1 mol** of compound is **formed** from its **constituent elements** in **standard states**, under **STP**
- **Standard lattice enthalpy: ΔH** when **1 mol of solid ionic compound** forms into **ions** under **STP**

- **Standard enthalpy of atomisation:** formation of **1 mol of separate gaseous atoms** under STP
- **Transesterification:** organic reaction in which R group of **alcohol** $\leftrightarrow$ R' group of **ester**
- **Homologous series:** class of **organic** compounds with same **general formula, functional group & chemical properties** but **differ** in **physical properties** by a **CH₂** group
- **Biofuel:** produced by **biological fixation of C** over **short** period of time through **photosynthesis**
- **Fuel:** substances that react with **oxidising agent (O₂)** - combustion that **releases E** & forms new chemicals
- **Alkane:** **saturated hydrocarbons** containing only **single** bonds
- **Saturated:** contains **C-C (single bond only)**
- **Cracking:** process of producing **alkenes & smaller alkanes** by breaking of **larger alkanes**
- **Polymer:** long chain (**macro**)molecules formed from **monomers**
- **Addition polymerisation:** short chains of **alkenes** $\rightarrow$ long chain of **alkanes** by removing **C=C $\rightarrow$ C-C (unsaturated $\rightarrow$ saturated)**
- **Structural isomers:** structures with **same molecular** formula but **differ** in **arrangement**
- **Hess's law:** ΔH for reaction is **independent of pathway** between initial & final states
- **Oxidation:** **gain of oxygen; loss of hydrogen; loss of e⁻; increase in oxidation state**
- **Reduction:** **loss of oxygen; gain of hydrogen; gain of e⁻; decrease in oxidation state**
- **Oxidation state:** charge element would have if **e⁻ in bond** were assigned to atom of more **electronegative** element
- **Oxidation number:** written in **Roman numerals** (represent **oxidation state**) (i.e. copper (II) oxide)
- **Standard hydrogen electrode (SHE):** **standard electrode** used to **compare** all other half cells against (standard electrode potential = 0V at **298.15K, 1 mol dm⁻³ H⁺_(aq) & H_{2(g)} at 100 kPa**); inert **Pt** electrode = site for e⁻ transfer $\rightarrow 2H^{+}_{(aq)} + 2e^{-} \rightarrow H_{2(g)}$
- **Electrodes:** site of **e⁻ transfer**
- **Standard electrode potential:** **voltage** obtained when **half-cell** connected to **SHE** (standard hydrogen electrode) under STP (**298K, 1 mol dm⁻³**)

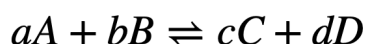
- **Standard reduction potential $E^\ominus$** : tendency for chemical species to be **reduced** (volts at **standard conditions**) (more (+) = better **oxidising** agent; more (-) = better **reducing** agent)
- **Lewis base**: lone pair/ e^- **donor**
- **Lewis acid**: lone pair/ e^- **acceptor**
- **Nucleophile**: ion/molecule with **lone pairs of e^-**
- **Fuel cell**: voltaic cell in which **fuel** becomes **oxidised** $\rightarrow$ **donating e^- at 1 electrode** & setting up **p.d. within the cell** at same time
- **Primary cell**: (**single-use**) electrochemical cell that converts E from **spontaneous redox reactions (chemical)** $\rightarrow$ **electrical E**
- **Secondary cell**: (**rechargeable**) cells involve **redox** reactions that can be **reversed** using **electrical E**
- **Resonance energy**: difference in E between **actual, delocalised** structure & most **stable theoretical Lewis structure**
- **LDF**: IMF between **instantaneous dipole-induced dipole**
- **Polar**: **uneven** distribution of e^- **density** in **covalent** bond
- **Electron domain**: **location** of e^- in **valence** shell (lone/single/double/triple bonded pairs)
- **Dipole**: **polar** molecule with **2 separate & opposite** electrical charges $\rightarrow \delta^+ \& \delta^-$
- **Molecular geometry**: shape of arrangement of bonded **atoms** surrounding central atom
- **Electron domain geometry**: shape of arrangement of e^- **domains** surrounding central atom
- **Dipole-dipole**: IMF between **polar** molecules (**permanent** dipole)
- **Solvent front**: line on chromatogram where **mobile** phase reaches before analysis
- **Mobile phase**: **moves** on **stationary phase** & moves **components of sample** at **different** rates
- **Stationary phase**: **adsorbs** sample & remains **unmoving**
- **σ bond**: **direct overlap** of **atomic** orbital along **internuclear axis**; e^- **density** is along this axis
- **π bond**: **sideway** overlap of p orbitals; e^- **density** **above & below internuclear axis**
- **Dissolution**: process of forming a **solution (solute + solvent)**
- **Dynamic eq.**: state in which **rate of forward reaction = backward**; **no net change** in **conc.** of reactants/products

- **Gibb's free E:** E of chemical reaction that can be used to do **non-mechanical work**
- **Entropy:** related to no. of **microstates** of a system (no. of ways the system can be arranged); measure of **disorder**
- **Spontaneous:** process which occurs **without** need for continual **input E**
- **Homogenous:** mixture in which substances involved are in **same phase**
- **pH:** measure of **acidity** using **logarithmic** scale
- **Amphoteric:** can **react** as **acid/base**
- **Amphiprotic:** can **donate/receive** p^+
- **Endpoint:** point at which **indicator changes colour** during acid-base titration
- **Equivalence point:** **stoichiometrically equal** amounts of **acid & base** & solution contains only **salt & water** ($mol_{acid} = mol_{base}$)
- **Half-equivalence point:** **pH = pKa**; vol. titrant = $\frac{1}{2}$ vol. of titrant at eq. point
- **Indicators:** **WAs** with **different colour** to **conjugate base**
- **Buffers:** substance that **resists change in pH** on addition of **small amounts** of acid/alkali; made from **WA/WB + conjugate A/B** (often added as **salt**)
- **Radical:** highly **reactive** molecular entity that has **one unpaired e^-** (e.g. $\bullet Cl$)
- **Homolytic fission:** **each** bonded atom in **covalent bond** ends up with one **unpaired e^-** & form **radicals**; **uncommon** & requires **specific amount of E (from UV)**
- **Heterolytic fission:** **one** of bonded atoms in **covalent bond** becomes an **anion** with lone **pair** of e^-
- **Base ion:** ion in mass spec with **highest peak**
- **Stereoisomer:** **same constitution** (atom identities, connectivities, bond multiplicities) with **different spatial arrangements** of atoms
- **Chiral carbon:** C with **4 different groups** attached

Equations:

- $c = f\lambda$ (speed of light (ms^{-1}) = frequency (Hz)*wavelength)
- $E = hf$ (energy (J) = planck's constant ($6.63 \times 10^{-34}(\text{Js})$)*frequency)
- $m = nM$ (mass(g) = no. of mol*Mr)
- $n = cv$ (no. of mol = conc.*vol.)
- **Relative atomic mass** = $\Sigma(\text{mass no.} \times \text{isotopic abundance}) / \Sigma(\text{isotopic abundance})$
- **%yield** = $100(\text{actual/theoretical})$
- **Atom economy** = $\text{Mr}_{\text{useful product}} / \text{Mr}_{\text{tot. product}}$
- **%error** = $100 \times |(\text{experimental-theoretical}) / \text{theoretical}|$
- $V = 22.7n$ (volume of gas at STP (dm^3) = no. of mol*22.7)
- $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$ (pressure₁*vol.₁/temp.₁ = pressure₂*vol.₂/temp.₂)
- $PV = nRT$ (pressure(kPa)*vol.(m^3) = no. of mol*8.31($\text{JK}^{-1}\text{mol}^{-1}$)*temp.(K))
- **RoR** = $|\Delta[\text{reactant or product}](\text{mol dm}^{-3})| / \Delta\text{time(s)}$
- $r = k[A]^x[B]^y$ (rate of reaction($\text{mol dm}^{-3}\text{s}^{-1}$) = rate constant*[A_{conc.}]^{orderA}*[B_{conc.}]^{orderB} (mol dm^{-3}))
- $\ln\left(\frac{k_1}{k_2}\right) = \frac{E_A}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)$ (determine E_A from 2 temps.)
- $\Delta\chi = |\chi_A - \chi_B|$ (electronegativity difference = |electronegativity₁ - electronegativity₂|)
- $\Sigma\chi = \frac{\chi_A + \chi_B}{2}$ (av. electronegativity = (electronegativity₁ + electronegativity₂)/2)
- $\Delta H = \Sigma(H_{\text{bonds broken}}) - \Sigma(H_{\text{bonds made}})$
 $\rightarrow \Delta H = \Sigma(\text{bond enthalpies of reactants}) - \Sigma(\text{bond enthalpies of products})$
- $\Delta H = - \frac{Q}{n_{\text{limiting}}}$
- $\Delta H_r^0 = \Sigma(\Delta H_f^0(\text{products})) - \Sigma(\Delta H_f^0(\text{reactants}))$
- $\Delta H_r^0 = \Sigma(\Delta H_c^0(\text{reactants})) - \Sigma(\Delta H_c^0(\text{products}))$
- $E_{\text{cell}}^{\ominus} = E_{\text{cathode}}^{\ominus} - E_{\text{anode}}^{\ominus}$ (overall cell potential (V) = cell potential_{cathode} (V) - cell potential_{anode} (V))

- $R_f = \frac{d \text{ travelled by substance}}{d \text{ travelled by solvent front}}$ (retardation factor)
- $\Delta G^0 = - nFE_{cell}^0$ (Gibb's free E (J) = mol. of e^- (mol)*Faraday's cst. (9.65×10^4) (mol⁻¹)*overall cell potential (V))
- $\Delta G^0 = \Delta H^0 - T\Delta S^0$ (Gibb's free E (J) = enthalpy change (kJmol⁻¹) - temp. (K)*entropy (JK⁻¹mol⁻¹)) *Must convert entropy from J $\rightarrow$ kJ*
- $\Delta S = \Sigma(\Delta S_{products}) - \Sigma(\Delta S_{reactants})$



$$K_c = \frac{[C]_{eq}^c \times [D]_{eq}^d}{[A]_{eq}^a \times [B]_{eq}^b}$$

- $\Delta G = \Delta G^0 + RT \ln Q$
- $\Delta G^0 = - RT \ln K$
- $K_c = e^{-\frac{G^0}{RT}}$
- $pH = -\log_{10}[H_3O^+] = -\log_{10}[H^+]$
- $pOH = -\log_{10}[OH^-]$
- $pH + pOH = 14$ (at **298.15K**)
- $K_w = [H^+][OH^-]; = 1 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ at **298.15K**; neutral $\therefore [H^+] = [OH^-]$
- $K_a = \frac{[H^+_{(aq)}][A^-_{(aq)}]}{[HA_{(aq)}]}$
- $pK_a = -\log_{10} K_a$
- $K_b = \frac{[BH^+_{(aq)}][OH^-_{(aq)}]}{[B_{(aq)}]}$
- $pK_b = -\log_{10} K_b$
- $K_a \times K_b = K_w$ (for **conjugate pair**)
- $pK_a + pK_b = pK_w = 14$ (for **conjugate pair** at **298.15K**)

- $\text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$

- $\text{pH} = \text{pK}_a + \log(\text{base/acid})$

- $\text{pOH} = \text{pK}_b + \log\left(\frac{[\text{M}^+]}{[\text{MOH}]}\right)$

- $\text{pOH} = \text{pK}_b + \log(\text{acid/base})$

- $\text{Wave number} = \frac{1}{\lambda} \text{ (cm}^{-1}\text{)}$

General experimental knowledge:

Percentage yield:

- **100(actual/theoretical)**
- Reasons for **non-100% yield**:
 - **Reversible** reactions
 - **Unwanted** reaction (e.g. metal oxidation)
 - **Incomplete** reactions
 - Loss of product during **transfer**
 - **Escape** of products (gases)

Atom economy:

- % of **useful** product
- $\frac{\text{Mr}_{\text{useful product}}}{\text{Mr}_{\text{tot. product}}}$

Standard solution:

1. Measure mass of weighing boat
2. Measure mass of solute + weighing boat
3. Transfer solute to **beaker**:
 - Easier to **stir & transfer**
 - If hard to dissolve → **heat/volumetric flask** are calibrated for **RT**

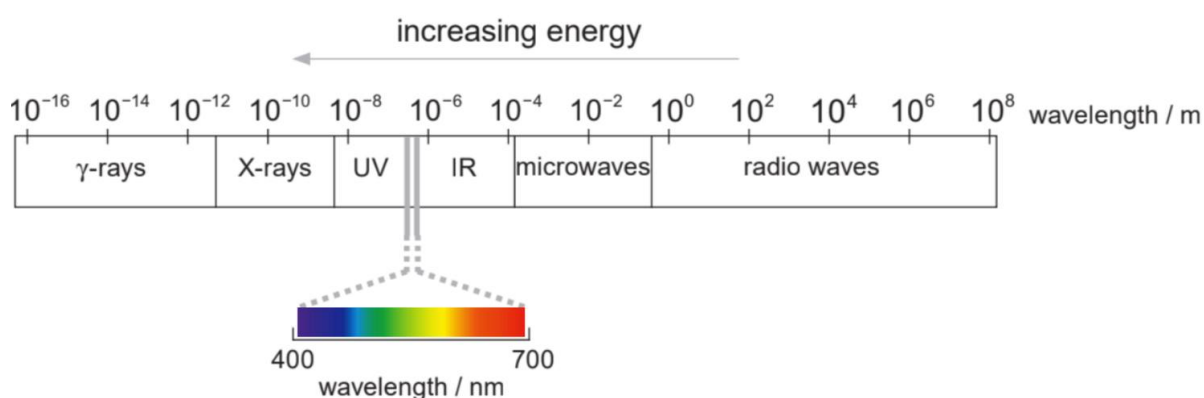
4. Measure mass of weighing boat again → get **true mass** of solute (accounts for solids **left in weighing boat**)
5. Add small amount of water & stir to fully dissolve → ensure vol. **doesn't exceed final vol.**
6. Transfer to **volumetric flask** using **funnel**
7. **Rinse** glass rod & beaker with **distilled water** → ensure **none remains**
8. Add distilled water until **bottom of meniscus**
9. **Invert/shake** flask

Titration:

- Add acid slowly to **equivalence point** (just enough to **neutralise** solution)

Atomic structure:

EM spectrum:



- $c = f\lambda$ (speed of light (ms^{-1}) = frequency (Hz)*wavelength)
- $E = hf$ (energy (J) = planck's constant ($6.63 \times 10^{-34}(\text{Js})$)*frequency)

Spectra:

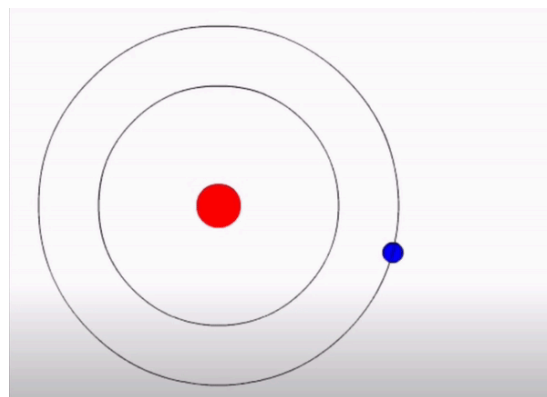
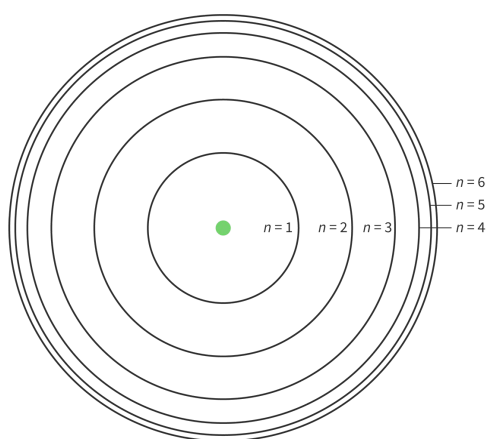
Continuous (white light): 

- **Emission spectrum** consisting of **continuous wavelengths** (consists of all frequencies in EM spectrum - **R** → **V**)

Line (emission): 

- Produced by atoms **gaining & emitting E as photons** when **electrons in excited states** return to **lower E levels/ground state**
- Produced when sample passes through **hot gas (gains E)**
- Lines **converge** at **higher frequencies**:

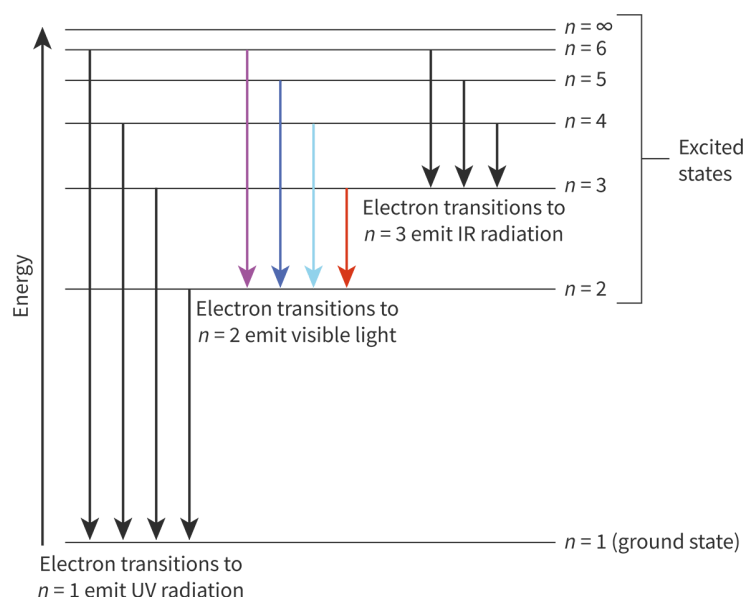
Bohr's model:



n = principal quantum number; $n = \infty \rightarrow$ ionised (completely removed)

- **Ground state = lowest E level ($n = 1$)**
- Each E level can hold up to **$2(n)^2$ e^-** (e.g. $n = 3 \rightarrow 18 e^-$)
- Principle E levels **converge** at **higher E (shielding)**, as shown on **emission spectrum**
- Electrons **transition between E levels in discrete/quantitative** steps, by **emitting/absorbing E as photons**
 - If **absorbed** $\rightarrow$ **move up E level & in excited state (unstable relative to ground state)**
 - **Unstable e^-** emits same amount of E as it absorbed & return to **lower & more stable levels**
 - Each transition corresponds to **specific wavelength on EM spectrum**

Hydrogen:



- Higher $\rightarrow n = 1$ emits **UV**
- Higher $\rightarrow n = 2$ emits **visible light** (bigger jump = higher frequency (e.g. $n = 6$ - violet; $n = 3$ - red))
- Higher $\rightarrow n = 3$ emits **IR**

Line (absorption):

- When **radiation (white light)** is passed through sample, atom/ion/molecule becomes **excited & electrons** move to **higher E level**
- Produced when sample passes through **cold** gas
- Only **specific frequencies** absorbed

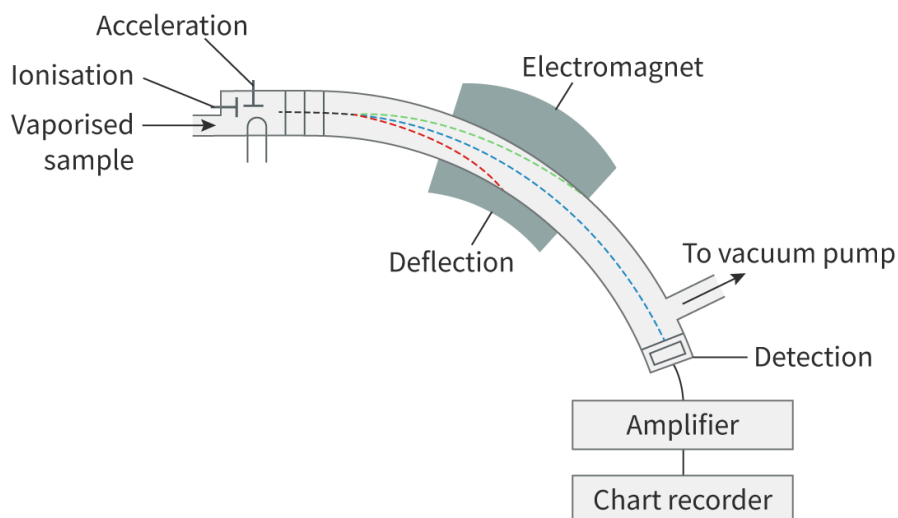
Mass spectra:

- Determine **Mr** from **isotopic compositions**
- **Relative atomic mass** = $\frac{\Sigma(\text{mass no.} \times \text{isotopic abundance})}{\Sigma(\text{isotopic abundance})}$

Isotopes:

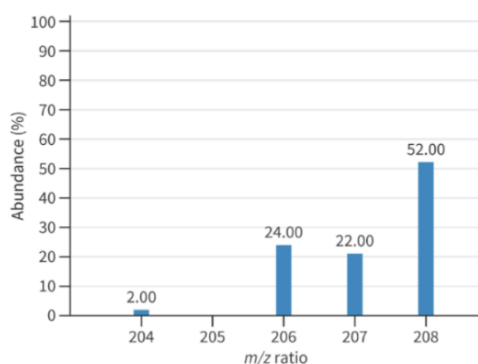
- Atoms of **same element** with **same no. of protons** but **different no. of neutrons**
- **Same chemical properties** $\rightarrow$ same **electronic configuration** (~~NOT same no. of valence e^-~~)
- **Different physical properties** $\rightarrow$ different **masses**

Mass spectrometer:



1. **Vaporisation:** $X_{(l)} \rightarrow X_{(g)}$
2. **Ionisation:** high E e^- fired $\rightarrow$ ionises atoms & forms cations/fragments
3. **Acceleration** by electric field: **voltage** across parallel plates
4. **Deflection** by magnetic field: ions deflected by **electromagnet** - degree of deflection depends on **m:z (mass:charge)**
 - **Lower mass** $\rightarrow$ deflected **more**
5. **Detection:** **current** generated with **size α abundance** of each ion

Interpreting mass spectrum:



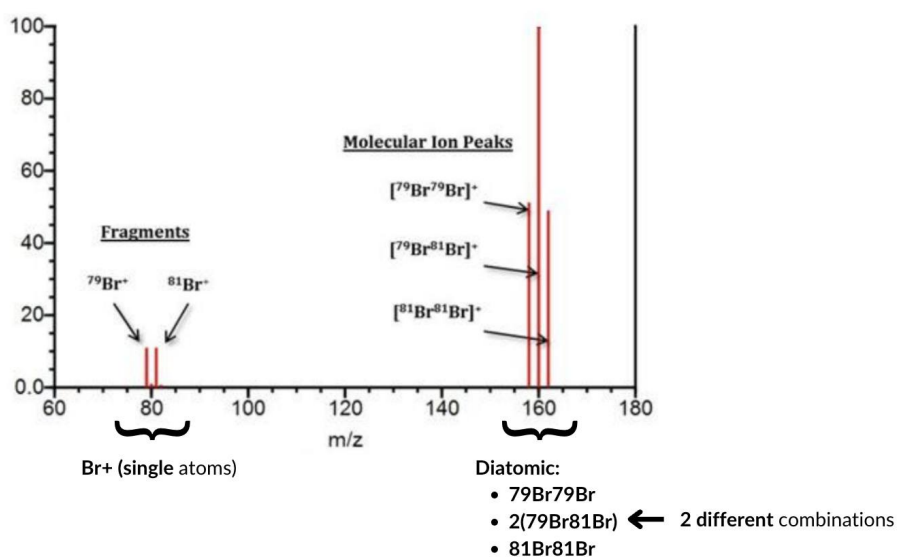
x-axis $\rightarrow$ divide m by z

Isotope:	Abundance (%):
^{204}Pb	2
^{206}Pb	24
^{207}Pb	22
^{208}Pb	52



- Small amount of atoms will get $2e^-$ knocked off:
 - As $1e^-$ gets knocked off $\rightarrow$ electrostatic forces of attraction will hold on to other e^- tighter
 - $\therefore$ Peaks are smaller

Diatomic elements:



Empirical formula:

- Lowest whole number ratio of atoms of each element present in compound
- E.g. $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow \text{CH}_2\text{O}$

Deriving from data:

1. Convert % $\rightarrow$ mass
2. Convert mass $\rightarrow$ mol. ($\div$ Ar)
3. Divide by **smallest mol. no.**

E.g. Calculate **molecular formula** of ibuprofen:

- % composition: 75.69% C, 8.80% H, 15.51% O
- Mr = 206.29 g/mol

Steps:	C:	H:	O:
% $\rightarrow$ mass	75.69g	8.80g	15.51g
Mass $\rightarrow$ mol. ($\div$ Ar)	75.69/12.011 = 6.30	8.80/1.008 = 8.73	15.51/15.999= 0.97 (smallest)
$\div$ by smallest (0.97)	6.30/0.97 = 6.5	8.73/0.97 = 9	1
x2 (eliminate decimal)	6.5*2 = 13	9*2 = 18	1*2 = 2

$\therefore$ Molecular formula = **C₁₃H₁₈O₂**

Electron configuration - quantum model:

Rules:

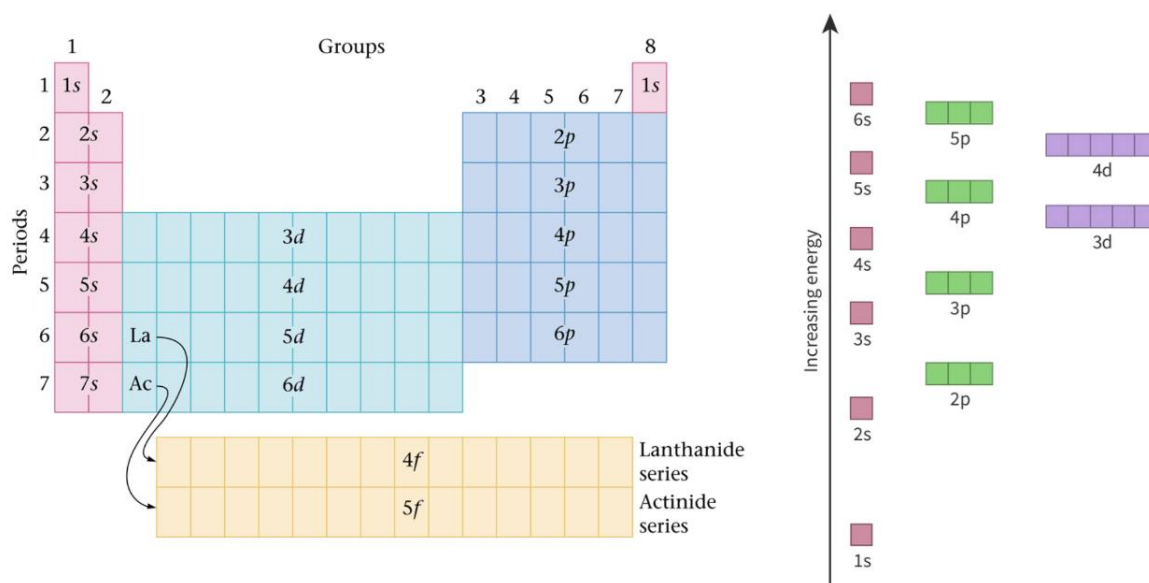
1. **Aufbau's principle:** e⁻ will always first fill **lowest E level orbital** available
 - *Once **filled with e⁻** $\rightarrow$ **3d<4s** (e⁻ from **4s** gets removed first)*
 - E.g. Ni $\rightarrow$ [Ar]4s² 3d⁸; Ni²⁺ $\rightarrow$ [Ar]3d⁸ - (e⁻ in **4s** gets taken out first)
2. **Pauli's exclusion principle:** each orbital can hold **2 e⁻** of **opposite spins**
3. **Hund's rule:** each orbital must get **1 e⁻** of **same spin** before any orbital gets **second e⁻**

Energy level (n):

- Principle E level
- Each E level can hold up to $2(n)^2$ electrons (e.g. $n = 3 \rightarrow 18 e^-$)
- Each main E level are divided into **sublevels**

Sub-level (l):

Sublevels	s (sharp)	p (principal)	d (diffuse)	f (fundamental)
Diagram:				
# of orbitals:	1	3	5	7
capacity of # of e- $[2 * (\# \text{ of orbitals})]$	2	6	10	14



- Lower sublevels gets **filled first** (i.e. 4s gets filled before 3d)

Exceptions:

1. 4s gets **filled before 3d**; e^- are **removed** from 4s first
 - E.g. $\text{Fe}^+ = [\text{Ar}]4s^1 3d^6$
- 2.

~~[Ar]4s² 3d⁴~~~~[Ar]4s² 3d⁹~~

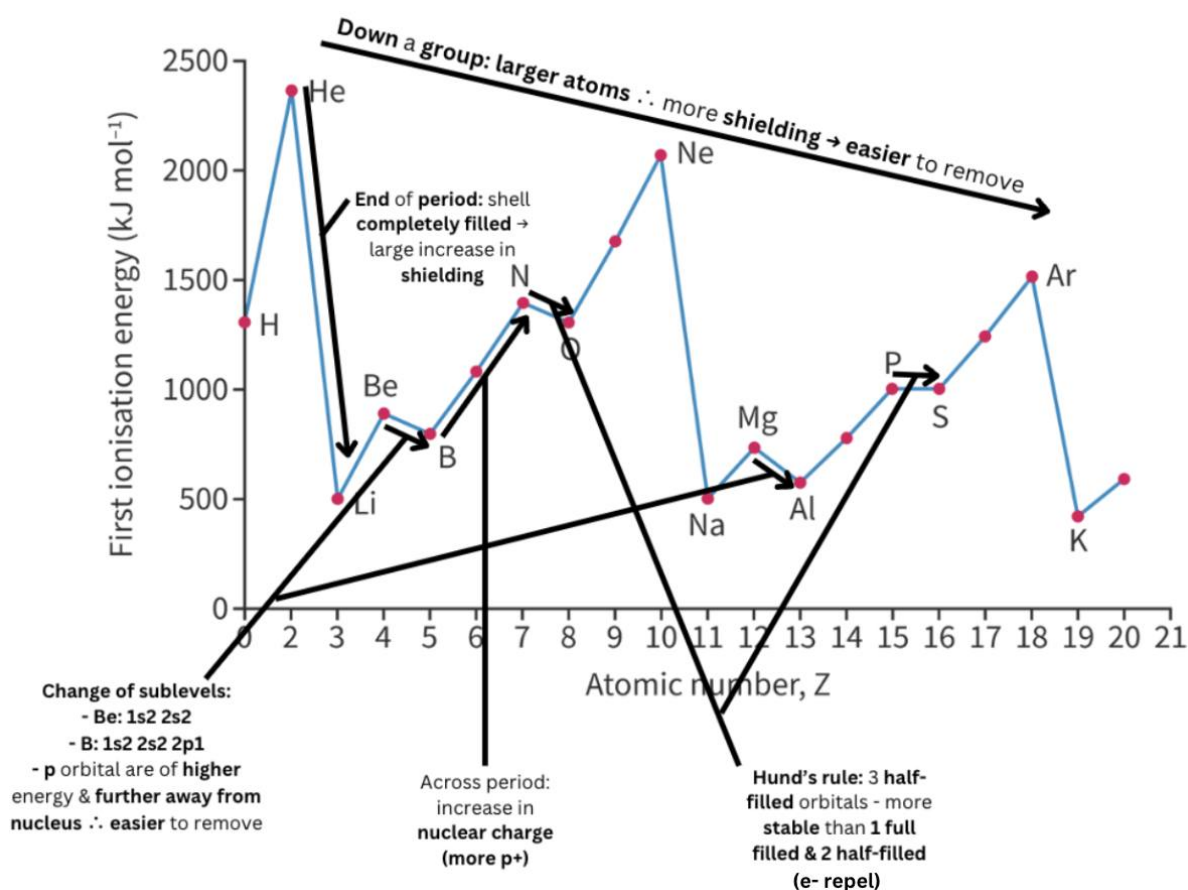
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- Amount of E required to **remove 1 mole of e^- from 1 mole of isolated gaseous atoms**
 → form **1 mole of gaseous unipositive ion** under standard thermodynamic conditions
($n = \infty$)

- $\text{kJ mol}^{-1} = \text{kJ} \cdot \text{avogadro's}$

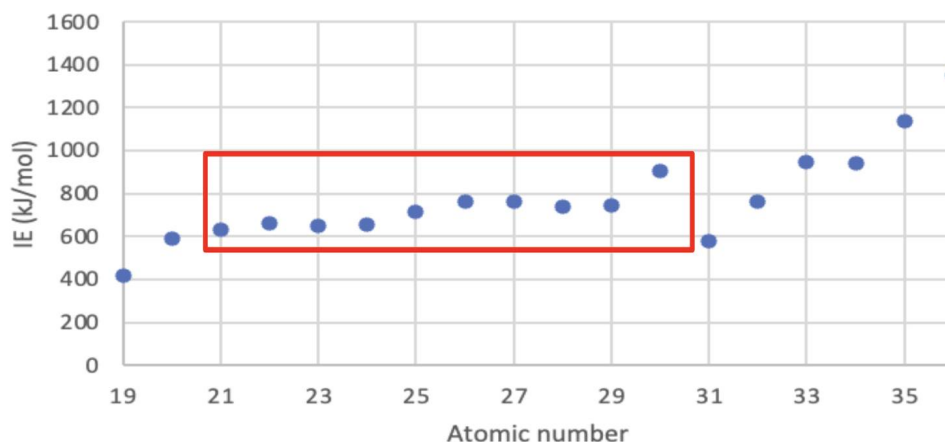
Factors:	What is it?	Effect on I.E.
Nuclear charge	Attractive force between p^+ & e^-	- Atoms with more protons $\rightarrow$ strong nuclear charge $\therefore$ harder to remove
Shielding	Repulsion between e^- in inner shell & valence e^- in outer shell	- Larger atoms have more shielding $\rightarrow$ easier to remove

Trend in periodic table:



- Shows evidence for existence of **main & sub E levels**
- **Main:**
 - **Highest** values for **noble gases** (nuclear charge - more p^+)
 - General **increase across period**
- **Sub:**

- **Drop:** Be $\rightarrow$ B ($2s \rightarrow 2p$ - easier to remove from **higher E** orbitals)
- **Drop:** N $\rightarrow$ O (3 half-filled (**more stable**) $\rightarrow$ 1 fully filled & 2 half-filled)



- **Small change** in d-block elements $\rightarrow$ increase in **nuclear charge** only has small effect on 4s sublevel $\rightarrow$ **shielding** by e^- in 3d

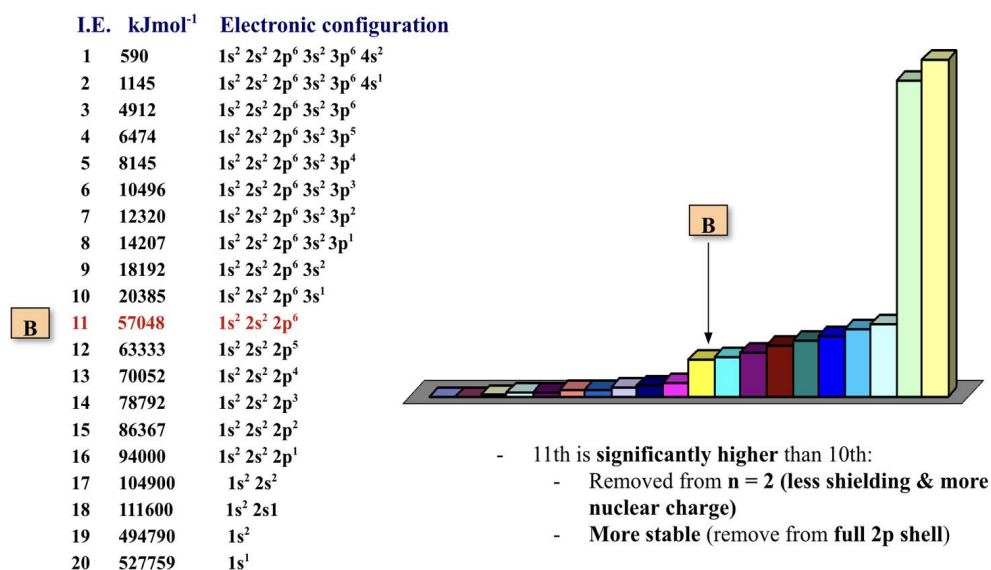
Successive I.E.:

- **1st I.E.:** min. E required to remove **1st e^-** from **gaseous atom** (kJmol^{-1}) $\rightarrow X_{(g)} \rightarrow X^+_{(g)} + e^-$
- **2nd I.E.:**
 - **Min. E** required to remove **2nd e^-** from **gaseous atom** (kJmol^{-1}) $X^+_{(g)} \rightarrow X^{2+}_{(g)} + e^-$
 - $> 1^{\text{st}}$ $\rightarrow$ **electrostatic force of attraction**

Trends:

1. **Higher** than previous $\rightarrow$ **electrostatic force of attraction** (same nuclear charge, fewer remaining e^-)
 - General **increase** $\rightarrow$ same **subshell** ($\sim$ same amount of **shielding**)
2. **Large increases** with change of **E level** (e.g. $n = 2 \rightarrow n = 1$) $\rightarrow$ large decrease in **shielding**

E.g. successive I.E. of **Ca**:



Gas laws:

Ideal gases:

- Theoretical gas made of **random moving particles** with **all vol.** subjected to **interparticle interactions (intermolecular forces)**

<i>Ideal:</i>	<i>Real:</i>
Constant PV/T	Non-constant PV/T
Negligible vol. (compared to container occupied)	Non-zero vol. (high pressure - vol. of empty space decreases)
No IMF → no attraction/repulsion between particles	Has IMF (low temp. → IMF)
Elastic collisions (KE conserved)	Inelastic collisions

K.E. $\propto$ temp.	Real gas with highest molar mass (IMF attraction) deviates most from ideal gas model
--	---

- **Limitations:** very few behave like one - especially at **low temp./high pressure**

Avogadro's law:

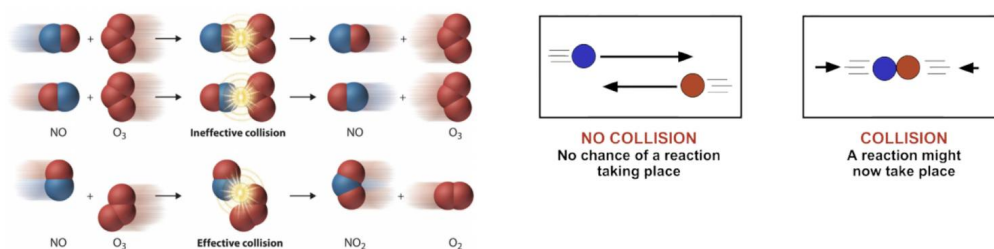
- **1 mole of any gas** at particular temp. & pressure $\rightarrow$ **same molar vol. & no. of particles (6.02×10^{23}) (constant V/n)**
- **$V = 22.7n$** (volume of gas at STP (dm^3) = no. of moles*22.7)

Combined gas laws:

- **$P_1V_1/T_1 = P_2V_2/T_2$**
- **$PV = nRT$** (pressure(kPa)*vol.(m^3) = no. of mol* $8.31(\text{JK}^{-1}\text{mol}^{-1})$ *temp.(K))

Kinetics:

Collision theory:



- Reaction occur when particles **collide**
- Doesn't proceed if reactants in **solid** form $\rightarrow$ limited to **vibrational motion** & don't have enough **K.E.** to **collide & overcome E_A**
- More **frequent & successful** collisions $\rightarrow$ **faster RoR**
- **Successful** collisions:
 - Sufficient **E** (min = E_A)
 - Correct **orientation (steric factor)**:
 - Reactive atoms/functional groups are **aligned**

Factors of RoR:

Increase/decrease frequency of collisions

1. **SA:**
 - **Smaller** particle (**larger SA**) $\rightarrow$ **faster**
 - Only particles on **outside** are available to react (collisions) $\rightarrow$ more **sites for collision**
2. **Conc:**
 - More **collision sites/area**
 - **Increased conc.** < **increased temp.** $\rightarrow$ no increase in **fraction** of particles with **sufficient E**
3. **Pressure:**
 - Increase in **temp.**/decrease in **vol.**

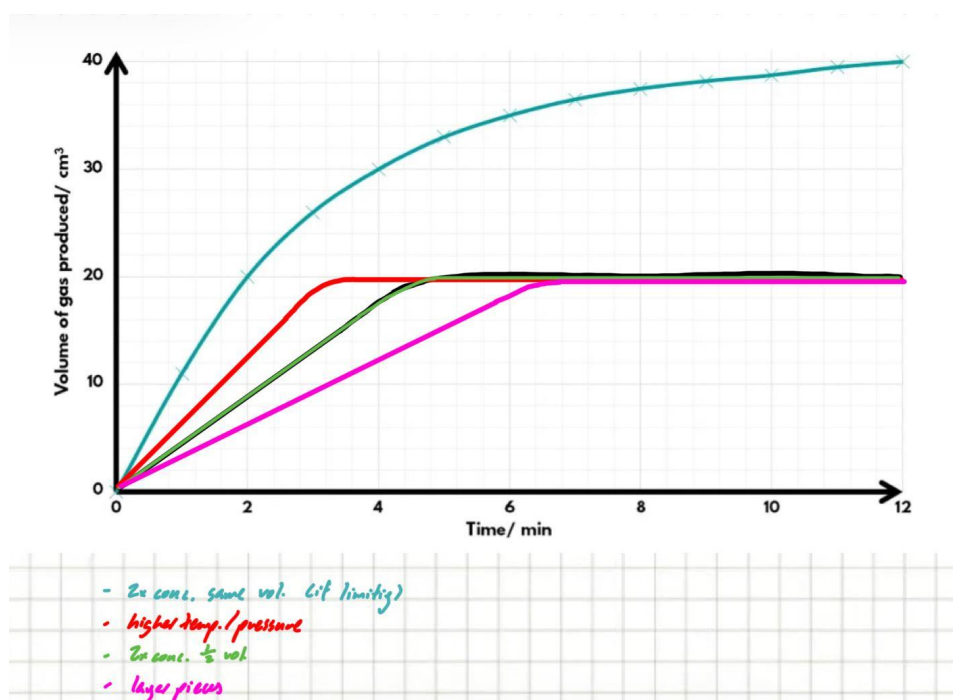
- More **collision sites/area** & more particles with E_A

4. Temp:

- Higher **K.E.** → **higher proportion** of particles with $K.E. > E_A$ & **frequency** of collisions

5. Catalyst:

- **Lowers E_A** without being **permanently changed/used up**



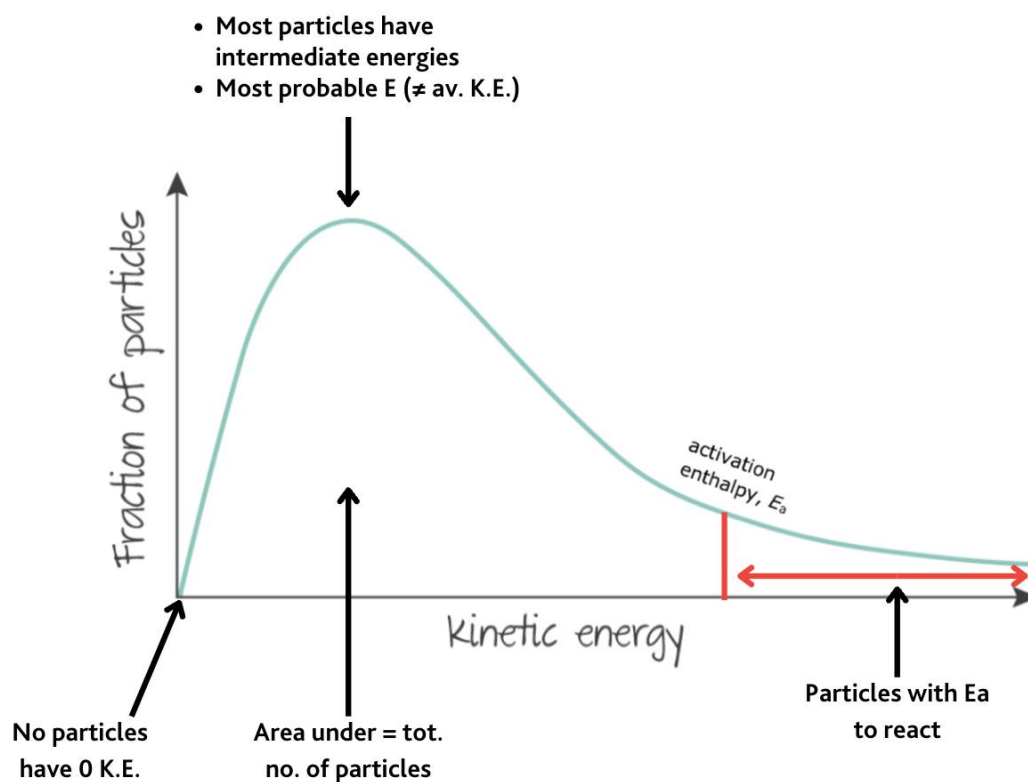
Exo v.s. Endo:

Exo:	Endo:
Potential Energy (stored in bonds of substance) Reactants (H_1) Products (H_2) Activation energy $\Delta H = -ve$ $\Delta H = E(p) - E(r)$ Reaction Progress	Potential Energy (stored in bonds of substance) Reactants (H_1) Products (H_2) Activation energy $\Delta H = +ve$ $\Delta H = E(p) - E(r)$ Reaction Progress
Gives out E ($\Delta H = (-)$)	Takes in E ($\Delta H = (+)$)

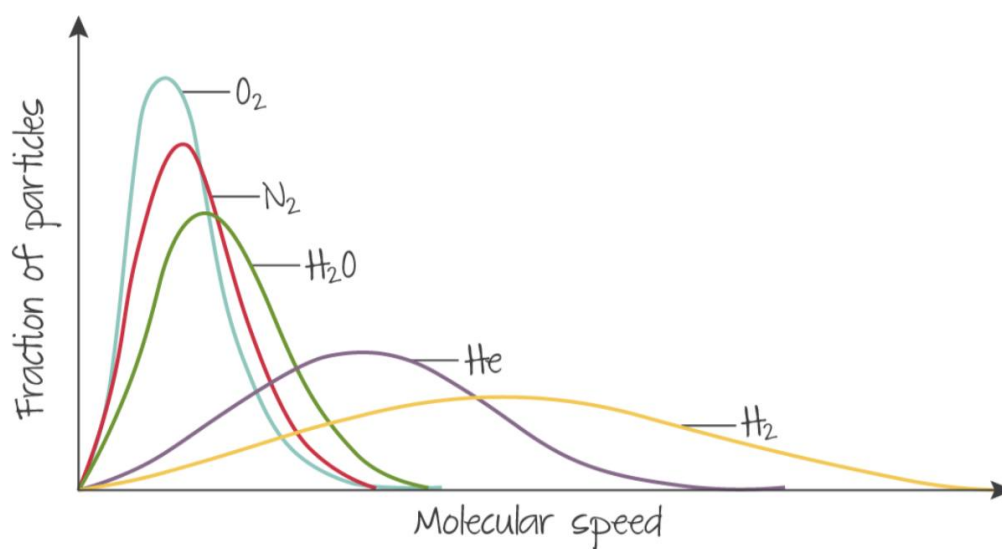
Formation of bonds	Breaking of bonds
--------------------	-------------------

Maxwell-Boltzmann E distribution curve:

Properties:

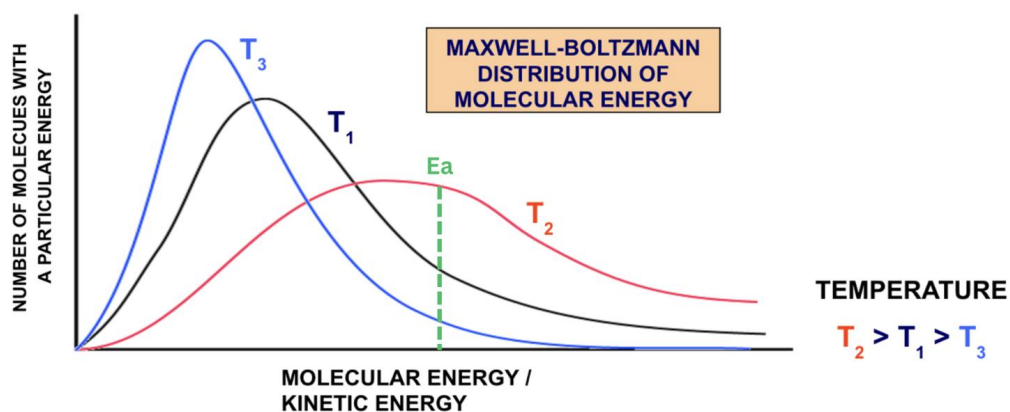


Gases:



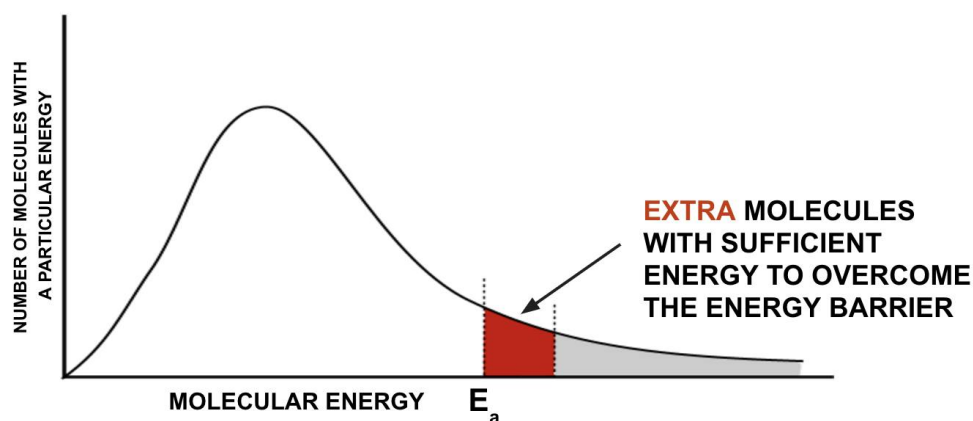
- Same temp. → same av. K.E.
- Dependant on Mr - lighter → faster

Temp.:



- Higher temp. → higher proportion of particles with K.E. $> E_a$ & vice versa

Catalyst:

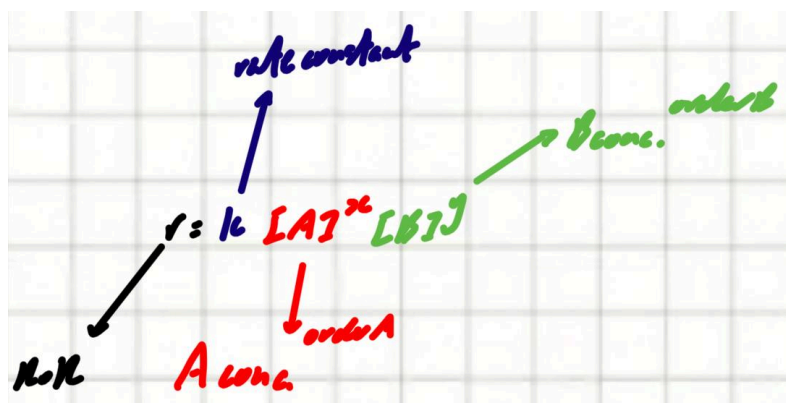


Reaction mechanisms:

Rules:

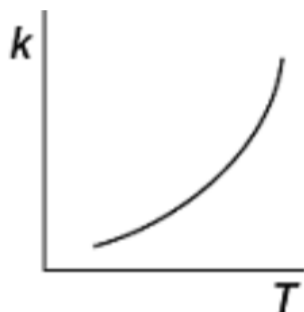
1. Σ elementary steps = overall balanced equation
2. All steps leading up & including RDS must agree with rate law

Rate equations:



*Substances **not** in stoichiometric equation can be in rate equation (**catalysts**)*

*Substances **in** stoichiometric equation sometimes aren't in rate equation $[A]^0$ *



E.g.

[A]	[B]	Rate
0.1	0.1	5
0.4	0.1	20
0.8	0.2	80

Find:

- Rate equation
- Value & units of k

1. Work out **order A** with **same [B]**

[A]	[B]	Rate
0.1	0.1	5

0.4	0.1	20
0.8	0.2	80

- Same [B]; [A] $\rightarrow \times 4$
- Rate (in terms of increase of [A]) $\rightarrow \times 4^1$
- $\therefore$ order A = 1

2. Find order B

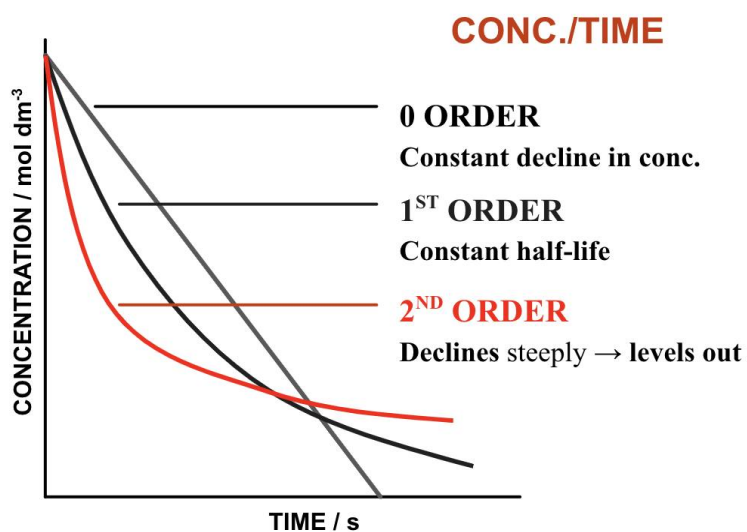
[A]	[B]	Rate
0.1	0.1	5
0.4	0.1	20
0.8	0.2	80

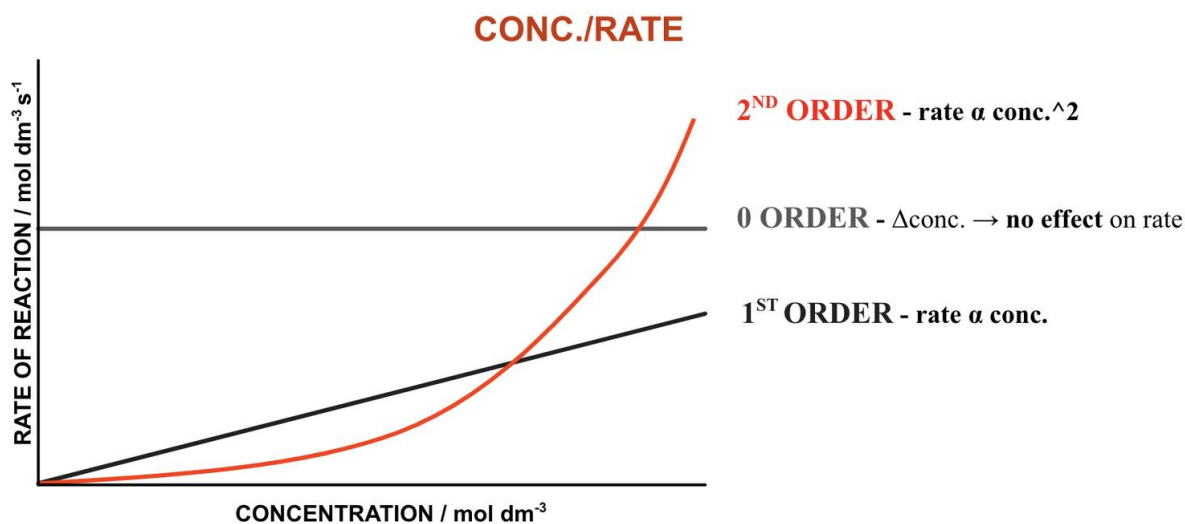
- [A] $\rightarrow \times 2$, rate $\times 2^2$
- $\text{Increase}_{\text{rate}} / \text{Increase}_{[A]} = \times 2$
- [B] $\rightarrow \times 2$, rate (after cancellation) $\rightarrow \times 2^1$
- $\therefore$ order B = 1

3. Find k by substitution:

- $5 = k \cdot 0.1^1 \cdot 0.1^1$
- $k = 500$
- $\text{mol}^{-1} \text{dm}^3 \text{s}^{-1}$

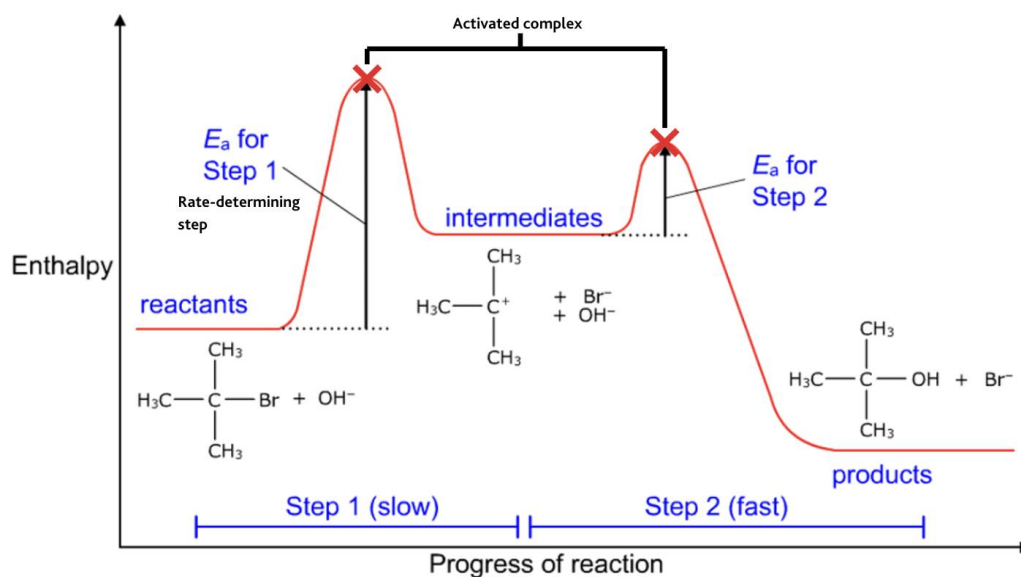
Graphical determination:





RDS:

- Slowest elementary reaction in **multi-step** reaction that determines **overall rate**
- Highest E_a
- Last **reversible** step in reaction

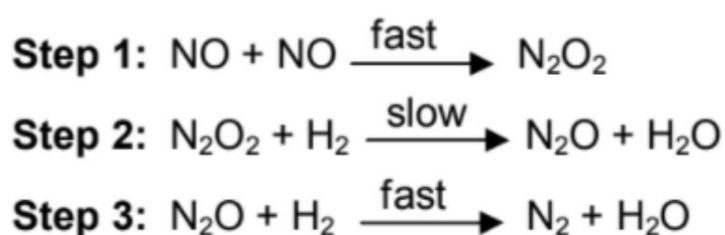


Molecularity	Elementary Reaction	Rate Law
Unimolecular	$A \longrightarrow \text{products}$	$\text{Rate} = k[A]$
Bimolecular	$A + A \longrightarrow \text{products}$	$\text{Rate} = k[A]^2$
Bimolecular	$A + B \longrightarrow \text{products}$	$\text{Rate} = k[A][B]$
Termolecular	$A + A + A \longrightarrow \text{products}$	$\text{Rate} = k[A]^3$
Termolecular	$A + A + B \longrightarrow \text{products}$	$\text{Rate} = k[A]^2[B]$
Termolecular	$A + B + C \longrightarrow \text{products}$	$\text{Rate} = k[A][B][C]$

Termolecular is rare $\rightarrow$ low probability of 3 particles colliding with enough E, correct orientation & at same time

- E.g.

Identify rate-determining step, rate equation & overall chemical equation:



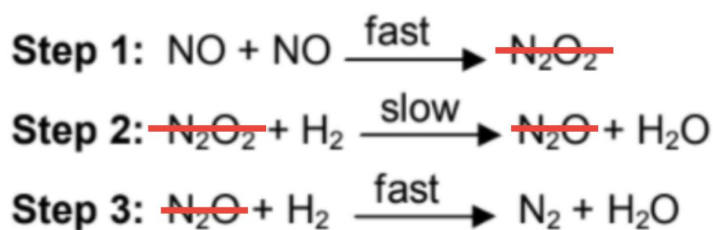
1. Rate-determining step:

a) Identify **slow** step $\rightarrow$ **rate-determining step**

$\therefore$ Rate-determining step = $\text{N}_2\text{O}_2 + \text{H}_2 \rightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$

2. Rate equation:

a) Cancel out **intermediates**:



b) **Reactants** of **rate-determining step** $\rightarrow$ rate equation:

- (if **reactant** = **intermediate**, use **reactants** of **intermediate**):

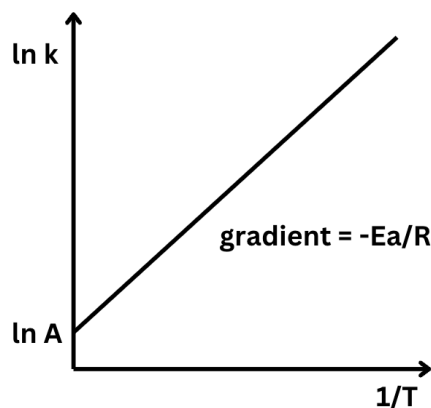
$$\begin{aligned}
 r &= k[\text{NO}][\text{NO}][\text{H}_2] \\
 &= k[\text{NO}]^2[\text{H}_2]
 \end{aligned}$$

3. Overall chemical equation:

a) $2\text{NO} + 2\text{H}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$

Temp.

- $k = Ae^{\frac{-E_A}{RT}}$ (rate constant = frequency factor * $e^{((-EA)(J)/(8.31(JK-1mol-1)*temp.(K))})$)
- $\ln k = \frac{-E_A}{RT} + \ln A$ (graph form)



- If **high** $E_A \rightarrow$ increase in **temp.** = **profound effect on RoR**
- **Unit A = unit k**

E_A from 2 temp.:

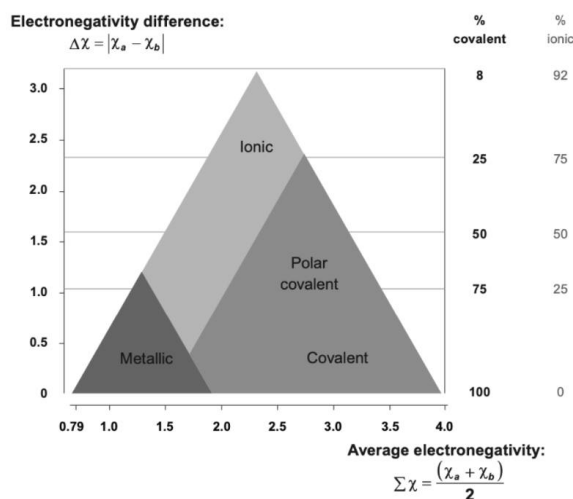
T1:	T2:
$\ln k_1 = \frac{-E_A}{RT_1} + \ln A$	$\ln k_2 = \frac{-E_A}{RT_2} + \ln A$
$\ln\left(\frac{k_1}{k_2}\right) = \frac{E_A}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)$	

Periodicity:

Electronegativity:

- **Relative** measure of tendency of atom in **covalent** bond to attract **shared pair of e⁻** towards itself
- Atoms in **covalent** bonds don't always **share e⁻ equally** → difference in **electronegativity**
- **No units - relative scale**

Bonding triangle:



- E.g. CuCl_2 :
 - $\Sigma\chi = \frac{\chi_{\text{Cu}} + \chi_{\text{Cl}}}{2} = \frac{1.9 + 3.2}{2} = 2.55$
 - $\Delta\chi = |\chi_{\text{Cu}} - \chi_{\text{Cl}}| = |1.9 - 3.2| = 1.3$

*When an element is in A_n form, just use $\chi_{\text{A}} \rightarrow n\chi_{\text{A}}$ *

IMF:

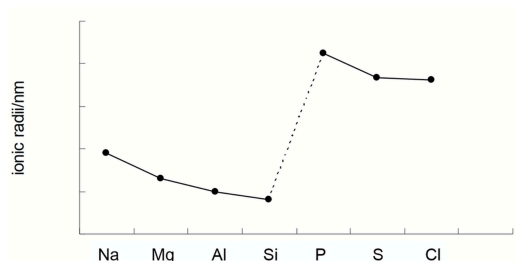
- Dependent on:
 - **Valence e⁻**
 - **SA**

Periodicity:

Atomic radius:

Horizontal:	Vertical:
- ↓ - Same amount of E levels ∴ shielding but increase in nuclear charge due to more p⁺ - Similar for d-block → Filling d sublevel (can take 10 e⁻) → Same E level ∴ shielding → Small effect from nuclear charge due to large atom size	- Increases - Increase in E levels ∴ shielding - E levels > nuclear charge

Ionic radius:

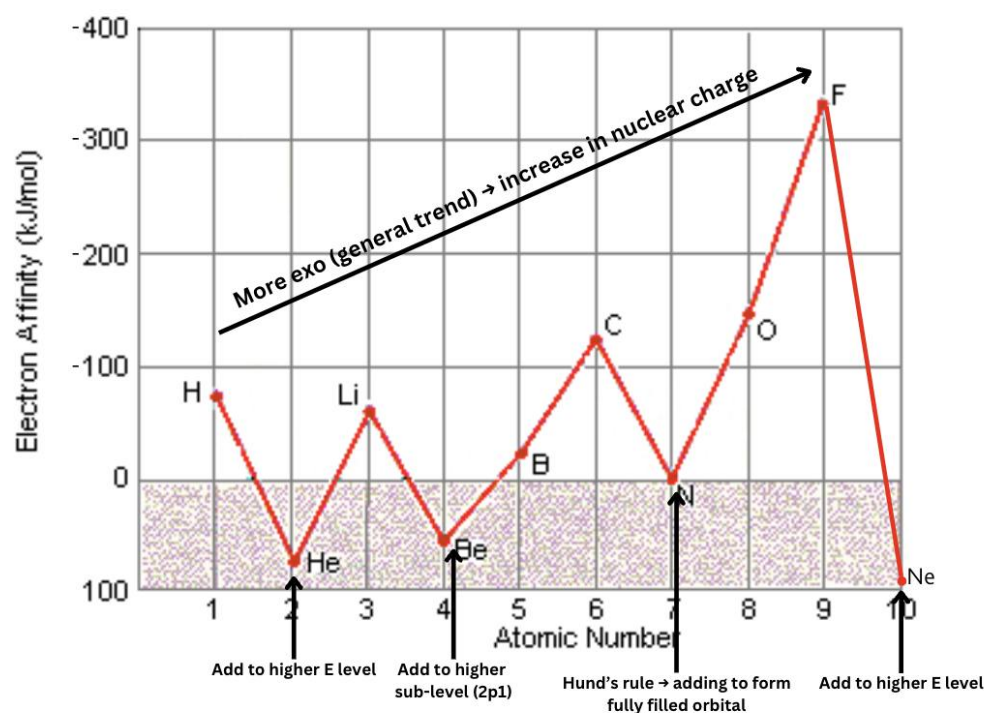


Horizontal:	Vertical:
- Cation → ↓ (isoelectronic & ↑ in nuclear charge) - Significant ↑ for 1st anion then ↓ (isoelectronic & increase in nuclear charge)	- ↑ - ↑ in E levels ∴ shielding - E levels > nuclear charge

1st I.E.:

Horizontal:	Vertical:
- $\uparrow \rightarrow \uparrow$ in nuclear charge	- $\downarrow \rightarrow \uparrow$ in shielding & further from nucleus (less attraction)

e⁻ affinity:



- ΔH when 1 mol of **gaseous** atoms each acquire **an e⁻** to form 1 mol of **gaseous ions** (⁻¹)
- $X_{(g)} + e^- \rightarrow X_{(g)}^-$
- **1st = exo** (bond forming)
- **2nd = endo** (overcome electrostatic force of attraction - forced)

Horizontal (general):	Vertical (general):
- $\uparrow$ exo (increase in nuclear charge $\therefore$ attractive force)	- $\downarrow$ endo ($\uparrow$ in shielding & larger atomic radius $\therefore$ $\downarrow$ in attractive force ($F = k \frac{q_1 q_2}{r^2}$))

Electronegativity:

Horizontal:	Vertical:
- $\uparrow \rightarrow \uparrow$ in nuclear charge yet shielding & distance from nucleus remains relatively same - Noble gas excluded	- $\downarrow \rightarrow \uparrow$ in E levels $\therefore$ shielding & further from nucleus ($F = k \frac{q_1 q_2}{r^2}$)

Exceptions:

- **G8 (period n) > G1 (period n+1)**

m.p.:

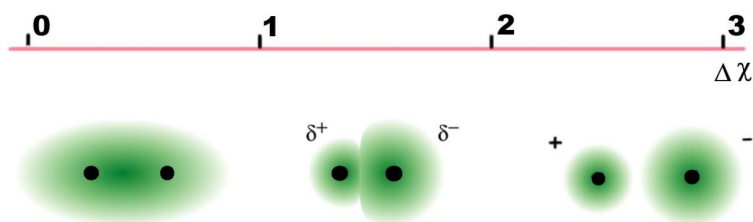
Horizontal:	Vertical:
- Metal $\rightarrow \uparrow$ (stronger electrostatic force of attraction between cation & sea of delocalised e⁻ as common ion charge increases) - Significant $\downarrow$ for 1st non-metal - Dependent on structure & bonding (IMF) (e.g. $S_8 > P_4 \rightarrow$ more SA)	- Metals: $\downarrow \rightarrow \uparrow$ in shielding $\therefore$ decrease in attractive forces - Halogens: $\uparrow \rightarrow \uparrow$ IMF as radius $\uparrow$

Metal v.s. non-metals:

Metallic character:

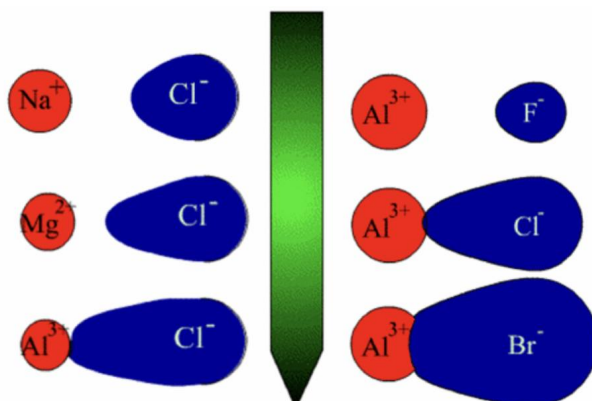
- Related to element's ability to be **oxidised**

Polarisation:



- **Deformation of e^- cloud of anion by charge of cation**
- **Ionic:covalent > 1:1** (i.e. more ionic character) $\rightarrow$ **stronger bonds** $\therefore \uparrow mp$
- **Ionic covalent < 1:1** (i.e. more covalent character) $\rightarrow$ **weaker bonds** $\therefore \downarrow mp$
- **High $\rightarrow$ lower solubility in polar solvents**
 - **Dispersion of charge $\rightarrow$ harder to attract to polar water molecules**

Factors:



- **Charge of cation - higher $\rightarrow$ increase**
- **Size of cation - smaller $\rightarrow$ increase (higher $^+$ charge density)**
- **Size of anion - larger $\rightarrow$ increase (larger e^- cloud)**

Bonding:

Ionic:

- **Electrostatic force of attraction** between **cation & anion**
- **Giant ionic lattice** → “**molecule**” - (only **formula units** (e.g. NaCl))

Properties:

1. **Hard & brittle:**
 - **Lattice: regular** arrangement → can be broken into pieces easily
2. **High mp/bp:**
 - Strong **intermolecular force** → requires **high E** to break them
 - Need to break **all bonds - throughout whole lattice structure** → between **each ion** (intertwining oppositely charged ions)
 - Factors:
 1. **Polarisability**
 2. **Charge & radius** (affected by geometry of unit cell) between →
$$F = k \frac{q_1 q_2}{r^2}$$
3. **Low volatility:**
 - Strong **electrostatic attraction** → requires lots of **E**
4. Tend to be **soluble in water:**
 - Water: **polar** → ions become **attracted to charges**
5. **Conductive** when **molten/aqueous:**
 - **Charged** ions gain **mobility** & can **move around freely**

Covalent:

- **Shared pair of e⁻**
- **Electrostatic force of attraction** between **shared e⁻ & nuclei**
- All **molecular** structure (except giant covalent)

Drawing Lewis diagrams:

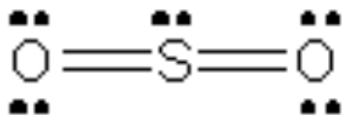
1. Add all **valence e⁻**
2. Account for **ionic charges** (i.e. if (1+) → subtract 1 e⁻)
3. Decide **central atom(s)** (generally = element with fewest atoms)
4. Draw **single bond** to connect central to surrounding
5. **Subtract bonding e⁻** to find **remaining lone pair e⁻** & draw
6. Turn all **lone pairs into bonds** → all atoms must have **octet**
 - Exceptions:
 - **Be & B** → **e⁻ deficient** (can have <octet)
 - **n ≥ 3** (≥ E level 3) → **hypervalent** (can have >octet)

Coordinate bond:

- **Both e⁻ come from 1 atom**
- Draw arrow → indicate **where e⁻** in bond comes from
- Common coordinate covalent bond structures:
 - NH₄⁺
 - H₃O⁺
 - CO
 - NO⁺
 - SO₂
 - H₂SO₄ (central S forms dative bonds with O)
 - HNO₃

Stability:

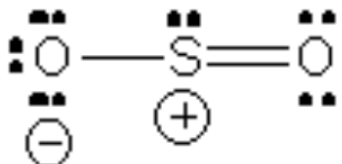
1. **Formal charge:**
 - Deduce **most stable structure**
 - Charge assigned to atom in molecule assuming e⁻ are shared **equally**
 - Structure with FC **closest to 0 = most stable**
 - **FC = no. of valence e⁻ - bonding e⁻ - lone pair**
 - E.g.
 - 1.



$$FC_O = 6 - 2 (\text{bonding}) - 4 (\text{lone pair}) = 0$$

$$FC_S = 6 - 4 (\text{bonding}) - 2 (\text{lone pair}) = 0$$

2.



$$FC_O = 6 - 1 (\text{bonding}) - 6 (\text{lone pair}) = -1$$

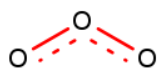
$$FC_S = 6 - 3 (\text{bonding}) - 2 (\text{lone pair}) = +1$$

$\therefore$ 1 = most stable

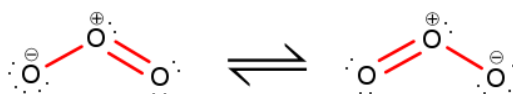
2. Electronegativity:

- If 2 structures have **same FC** $\rightarrow$ most **stable** = **negative FC** on more **electronegative** atom

Resonance structures:



Ozone Hybrid



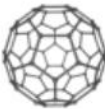
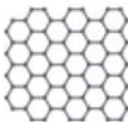
Ozone Resonance Structures

- Doubles + single bonds $\rightarrow$ **double** can go **anywhere**
- By **delocalising 1 pair of e^-**
- **Same stability**
- **Same position** of atoms ($\therefore$ **alkenes $\neq$ resonance** $\rightarrow$ H_2 positions change)
- Common resonance structures:
 - **O_3**

- NO_2^-
- NO_3^-
- CO_3^{2-}
- C_6H_6
- CH_3COO^-

Giant covalent networks:

- **Covalent** bonding usually results in **molecules**//some **carbon allotropes & SiO_2** (quartz) form **covalent network solids**
- Creates **lattice structure**
- Properties:
 - **High mp**
 - **Low electrical conductivity** (exceptions: **graphite - 1 delocalised valence e^-**)
 - **Insoluble**
 - **Hard** (exception: **graphite - $\rightarrow$ covalent BONDS** between C of same layer; **Van der Waal IMF** between layers)

	Graphite	Diamond	Fullerene	Graphene
Diagram				
Structure	Trigonal planar	Tetrahedral	Spherical	Single planar
Electrical conductivity	yes	no	no	yes
Thermal conductivity	low	high	high	high
Type of hybridization	sp^2	sp^3	sp^2 and sp^3	sp^2
Appearance	Dark grey with metallic luster	Colorless and if impure is colored	Black solid	black
Special properties	insoluble in water and is soft and slippery	Insoluble in water and brittle. Very hard substance	Not dissolve in water but can in some non-polar solvents (benzene) low mp	Very flexible and transparent
Uses	Pencil lead, lubricant	Jewelry, drilling	Gene and drug carriers (inclusion complexes)	LCD

Metal:

- Electrostatic force of attraction between **cations** & sea of **delocalised e^-** (can't tell which e^- belongs to which)
- **Giant metallic lattice**

Properties:

- **High mp/bp** → strong IMF due to **electrostatic force of attraction**
- **Malleable** → able to **roll** over into new positions **without breaking bonds** (due to free e^-)
- Good **conductors** of electricity & heat → free e^- move around
- **Strong:**

Dependent on:

- **Ionic radius** → **smaller** = **stronger** attractive forces
- **Charge** → **higher** = **stronger** (higher **charge density**)

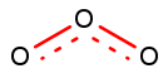
Ligands:

- Atom/molecule containing **lone pairs of e^-** → form **coordinate covalent** bonds with **transition metals**

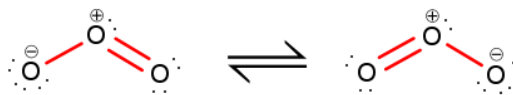
Alloys:

- **Mixture** of metals (**NOT compound**)
- **Stronger** than pure metal → **different** sized atoms **disrupts regular lattice** structure
∴ **harder to slide over**
- **Lower mp** → disrupts **regular lattice** structure

Resonance:



Ozone Hybrid

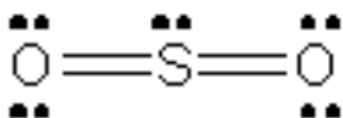


Ozone Resonance Structures

- Has to have **1 double O bond & 1 single O bond (O=X-O)**
- Doubles + single bonds → **double** can go **anywhere**
- By **delocalising 1 pair of e⁻**
- **Same stability**
- **Same position** of atoms (∴ **alkenes ≠ resonance** → H₂ positions change)
- Common resonance structures:
 - **O₃**
 - **NO₂⁻**
 - **NO₃⁻**
 - **CO₃²⁻**
 - **C₆H₆**
 - **CH₃COO⁻**

F.C.:

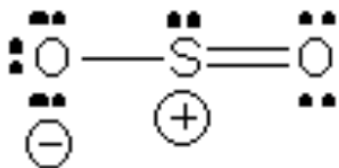
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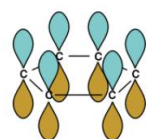
$$FC_S = 6 - 3 (\text{bonding}) - 2 (\text{lone pair}) = +1$$

$\therefore$ 1 = most stable

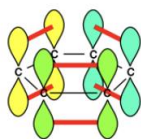
Benzene:

Delocalisation:

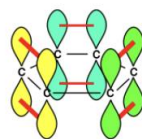
- Instead of **3 localised** (in 1 position) **double bonds** $\rightarrow$ **6 π e⁻** are **delocalised** (not in one particular position) around ring by **overlapping p orbitals**:



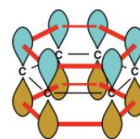
6 single bonds



one way to overlap adjacent p orbitals



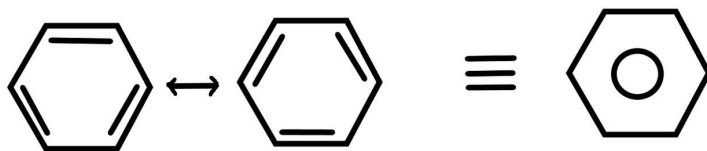
another possibility



delocalised pi orbital system

Kekulé structures:

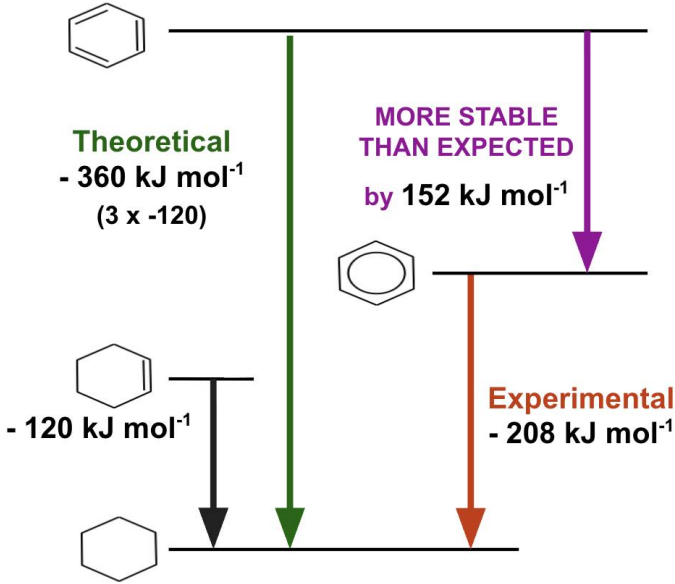
Kekule structures



Two equivalent resonating structures of benzene.

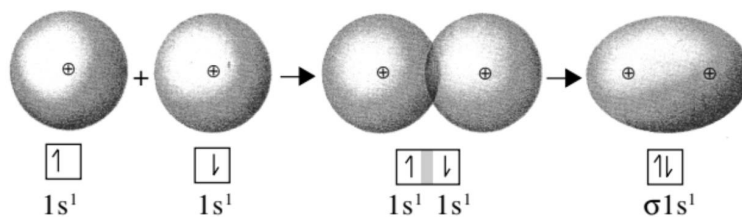
Resonance hybrid

- Initially thought benzene to be **hexagonal** with **alternating single & double bonds**
- However, physical & chemical properties supported **resonance hybrid** instead of Kekulé structure:

Physical:	Chemical:
Regular/symmetrical structure → similar 6 carbon bond lengths// $C=C < C-C$	More thermodynamically stable (unreactivity):  - Enthalpy of hydrogenation (unsaturated reduced to saturated) - Cyclohexene (1 C=C bond) → cyclohexane = 120kJ - Theoretically if benzene has 3 C=C bonds → enthalpy = 360kJ; yet actual release = 208kJ - Resonance E: 152kJmol⁻¹ more stable than expected

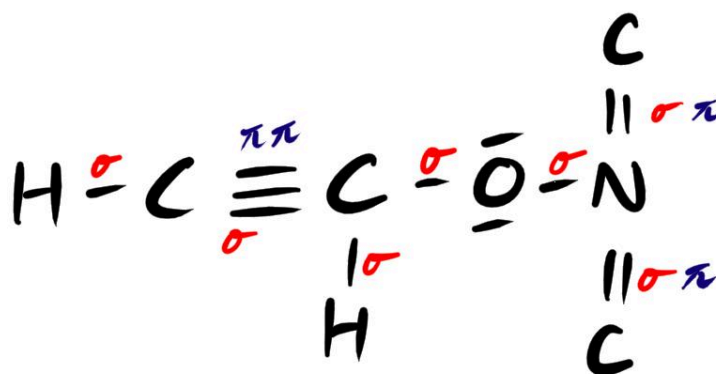
Molecular orbital theory:

- **Atomic orbitals merge** → **molecular orbitals** belonging to whole **molecule**
 - When 2 **atomic orbitals** merge → **attractive nuclei** ↔ e⁻ & **repulsion nucleus** ↔ nucleus/e⁻ ↔ e⁻
 - Creates 2 types of molecular orbitals:
 - **Bonding** → **lower E** than original atomic orbitals (**σ & π**)
 - **Antibonding** → **higher E** than original



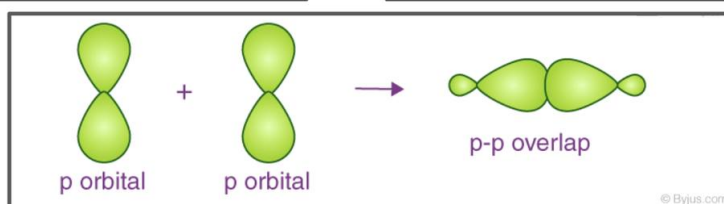
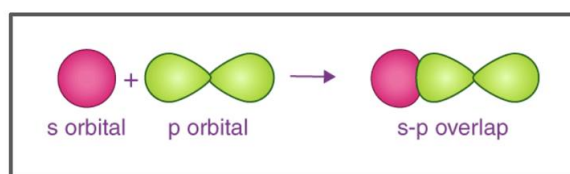
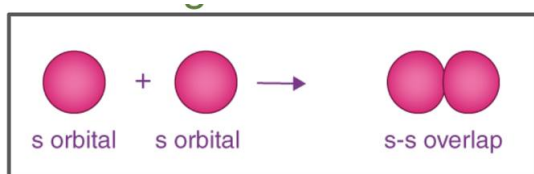
σ & π bonds:

Single:	1 σ
Double:	1 σ , 1 π
Triple:	1 σ , 2 π



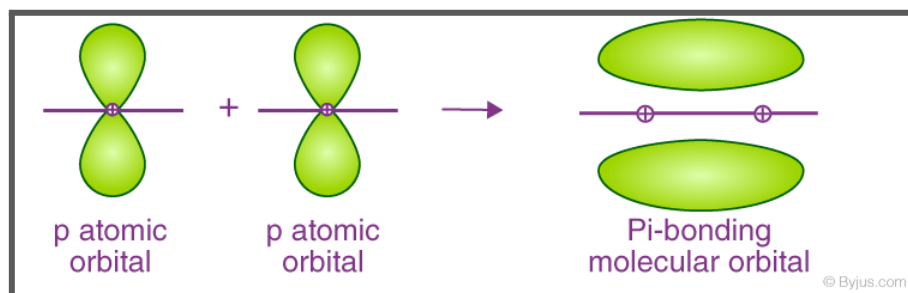
σ :

- Direct overlap of atomic orbital **along internuclear axis**; e^- density is **along** this axis



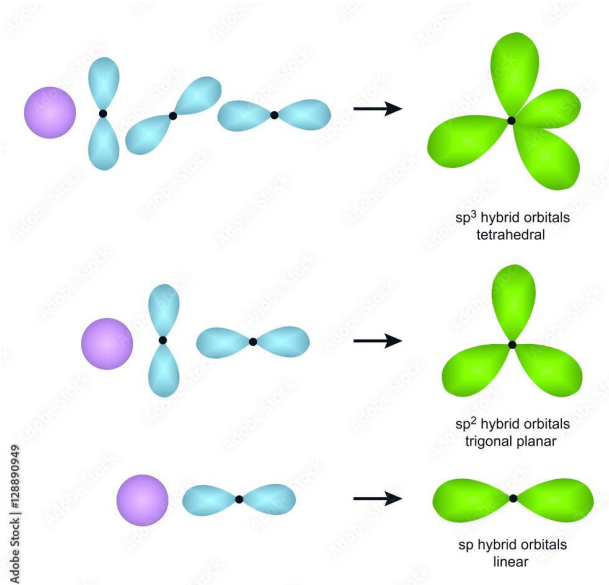
π :

- Sideway overlap of p orbitals; e^- density **above & below internuclear axis**

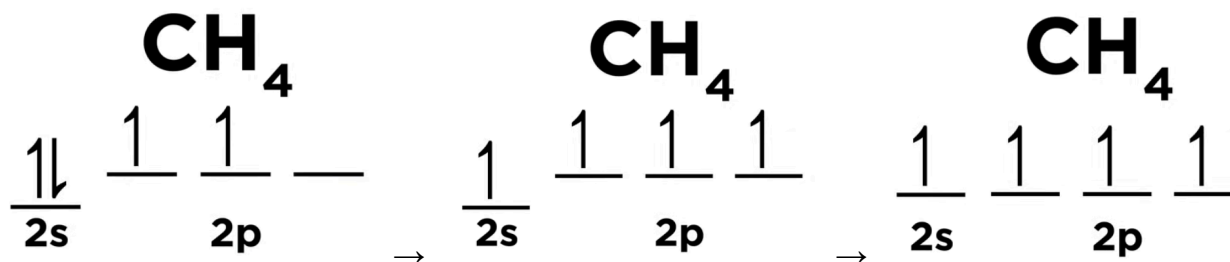


Hybridisation:

- Mixing of **atomic** orbitals $\rightarrow$ **hybrid** orbitals
- Can only **promote e^- /hybridise** for σ bonds; π leave p as is

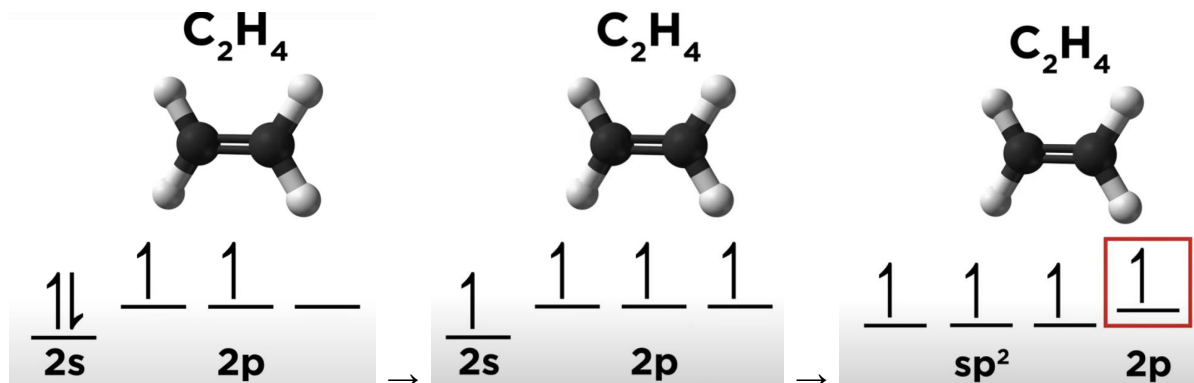


- E.g. C in CH_4 – sp^3 (1s & 3p):
 - C has 4 bonds $\therefore$ must have 4 unpaired e^- domains:



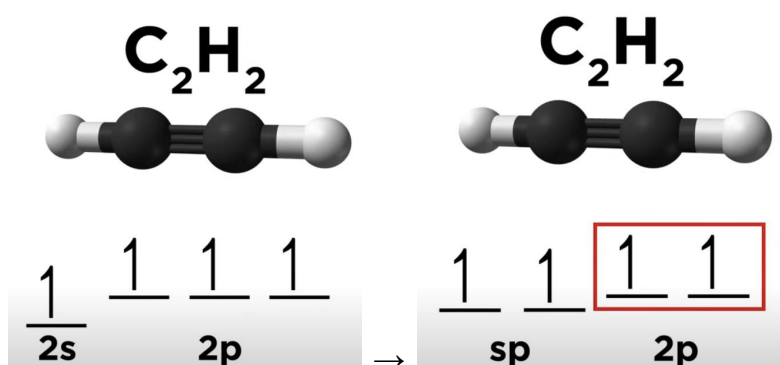
Original central C orbital arrangement	Promote one 2s $e^- \rightarrow 2p$ to create 4 unpaired	Hybridised to form orbitals identical in E (degenerate)
--	--	---

- C in $C_2H_4 - sp^2$:



Original	Promote e^- from $s \rightarrow p$	Hybridise & leave 1 p for 1 π
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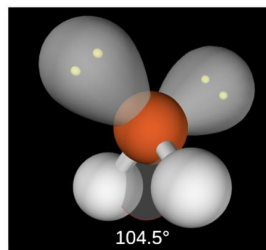
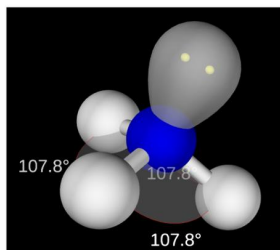
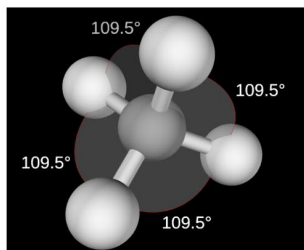
- $C_2H_2 - sp$:
 - $2 e^-$ in $p \rightarrow 2 \pi$ bonds



VESPR:

- e^- domains want to get as **far** away from each other as possible in 3D
- Shape of molecule depends on:
 - **Atoms/elements**
 - **Bond distance (i.e. single, double)**
 - **Bond angles**
- Bonding angles:
 - **Bp: nuclei on both sides $\therefore$ more confined & less space**

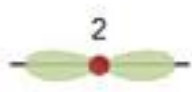
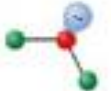
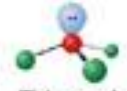
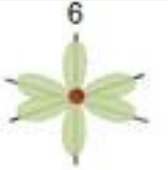
- **Lp:** nucleus only on **central** atom $\therefore$ **less confined** & more space; “squeezes” bp ($\sim 2.5^\circ$)



- **Hybridisation** tells us about VESPR shapes:

sp^3	sp^2	sp
Tetrahedral	Trigonal planar	Linear

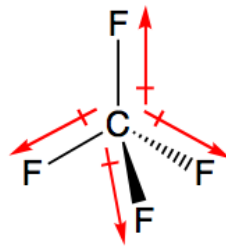
VSEPR Geometries					
Steric No.	Basic Geometry 0 lone pair	1 lone pair	2 lone pairs	3 lone pairs	4 lone pairs
2	 Linear				
3	 Trigonal Planar	 Bent or Angular			
4	 Tetrahedral	 Trigonal Pyramid	 Bent or Angular		
5	 Trigonal Bipyramid	 Sawhorse or Seesaw	 T-shape	 Linear	
6	 Octahedral	 Square Pyramid	 Square Planar	 T-shape	 Linear

Number of Electron Dense Areas	Electron-Pair Geometry	Molecular Geometry				
		No Lone Pairs	1 lone Pair	2 lone Pairs	3 lone Pairs	4 lone Pairs
2 	Linear	 Linear				
3 	Trigonal planar	 Trigonal planar	 Bent			
4 	Tetrahedral	 Tetrahedral	 Trigonal pyramidal	 Bent		
5 	Trigonal bipyramidal	 Trigonal bipyramidal	 Sawhorse	 T-shaped	 Linear	
6 	Octahedral	 Octahedral	 Square pyramidal	 Square planar	 T-shaped	 Linear

e⁻ domain/pair geometry:	Molecular geometry:
Take into account bp & lp	Only take into account bp

Polarity:

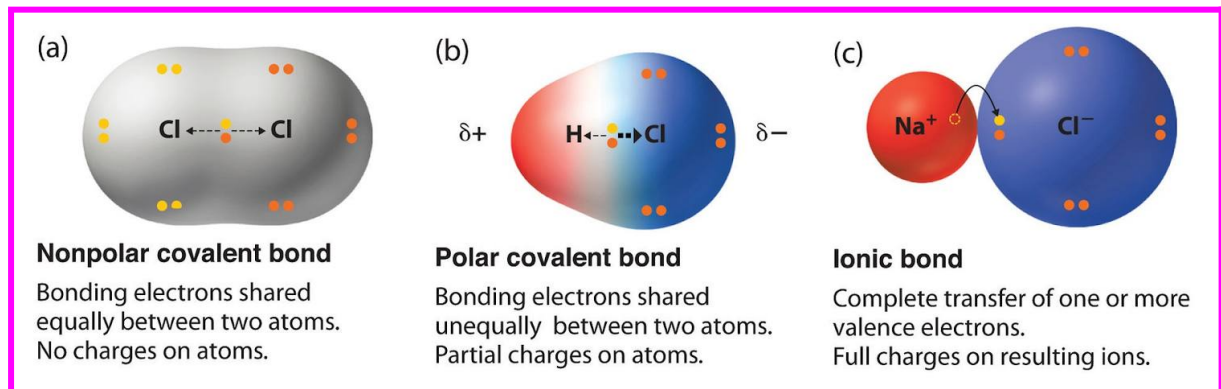
Bond:	Molecular:
Electronegativity between 2 atoms	e ⁻ distribution/net dipole moment of entire molecule
i.e. CF ₄ → each C-F = polar//molecule = non-polar:	



The individual dipole moments of the C-F bonds cancel out. CF_4 has a net dipole moment of zero. CF_4 is non-polar

Bond:

- To do with **electronegativity**:

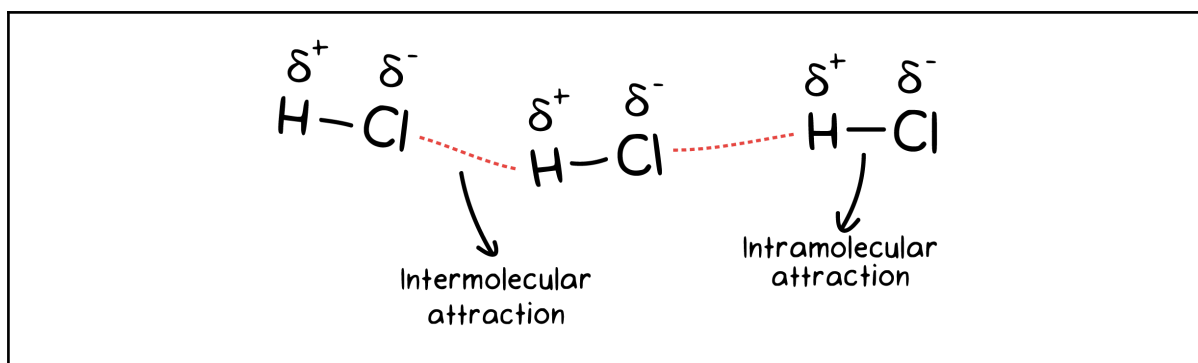


Molecular:

- For a **molecule** to be **polar**:
 1. **Polar bonds**
 2. Arranged in a way that **polarity isn't cancelled out** → **non-symmetrical** charge distribution hence has **net dipole moments**

IMF/Van der Waal:

Inter:	Intra:
Between molecules	Within/within molecules
Covalent bonds → weak intermolecular; strong intramolecular	

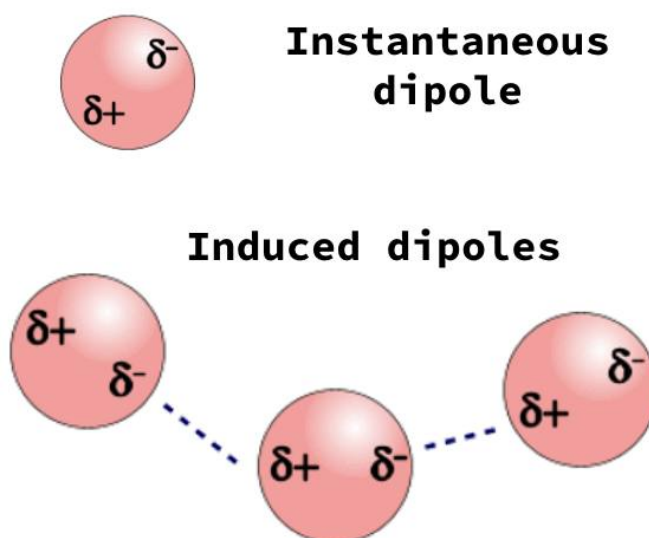


- Strength:

Hydrogen > permanent dipole-dipole > LDF

LDF (London dispersion forces)/induced dipoles:

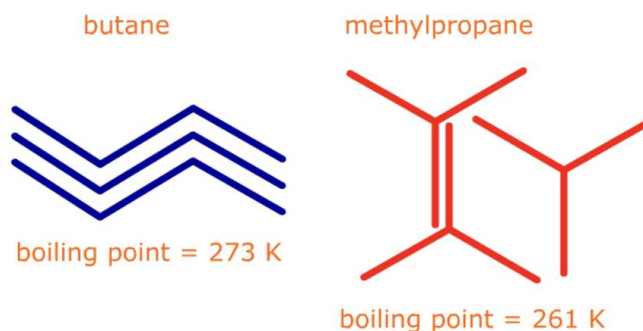
- e^- are **constantly moving** & can be **anywhere** in atom
- Some instances where **more e^-** are on **1 side** of atom than the other → **instantaneous dipole**
- Dipole on one atom can **induce** dipole on others
- Atoms now **attracted** by **weak LDF**:



Factors:

- **No. of e^- :**
 - Higher no. of e^- → **stronger attraction** ∴ more E needed to break
- **Molecular size/chain length:**

- **Larger size** → **stronger LDF**
- Increase in **atomic radius** = **valence e⁻ further from nucleus** ∴ less attracted
→ **easier to induce temporary dipoles**
- **Shape:**
 - **Straight** chains pack **closer**



(Permanent) dipole-dipole:

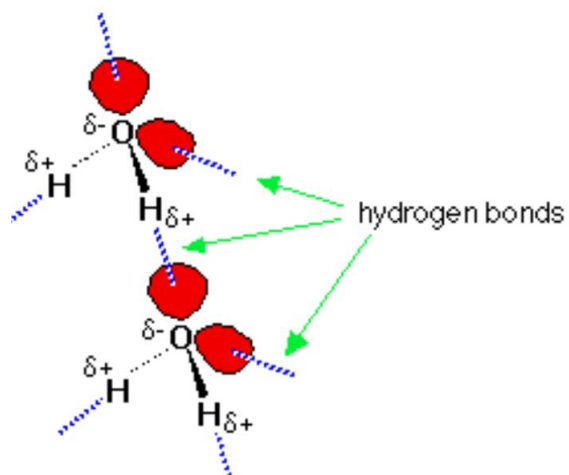
- Between **polar** molecules (**permanent dipoles**)
- **Addition** to LDF
- Extra attraction between dipoles → **higher mp/bp** than **non-polar molecule** with **same molecular mass**:

	Mr	°C	
CH ₄	16	-161	Non-polar
SiH ₄	32	-117	
GeH ₄	77	-90	
SnH ₄	123	-50	
NH ₃	17	-33	Polar
PH ₃	34	-90	
AsH ₃	78	-55	
SbH ₃	125	-17	

Hydrogen bonding:

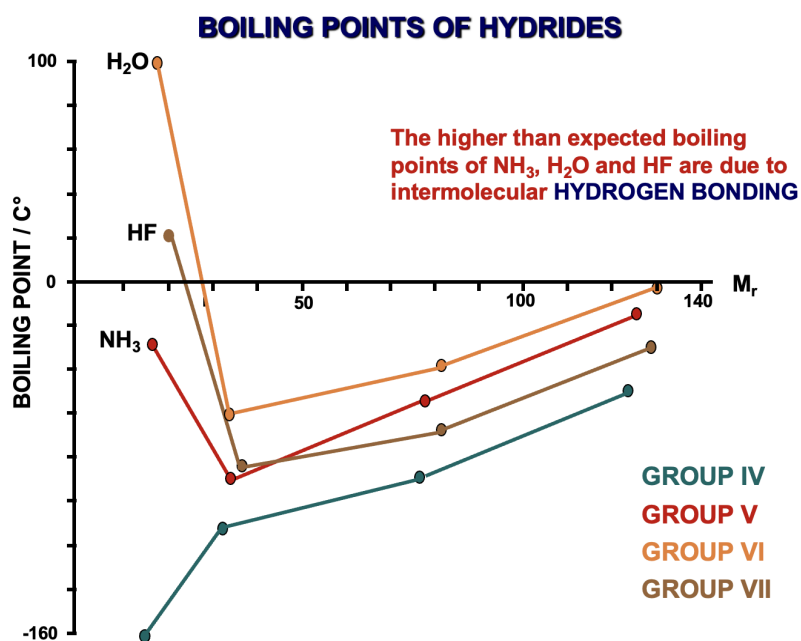
- **H** bonds with **N/O/F** → **larger dipoles** compared to other polar bonds:
 - Highly **electronegative** (**high nuclear charge & small size**), e⁻ are **withdrawn** from **H** → slightly (+)
 - H is very **small** ∴ (+) charge more **concentrated** → stronger attraction
- (+) **dipole** on **H** atom; **neighboring** molecule must have **N/O/F** with **lone pair**

HYDROGEN BONDING



Notice that the hydrogen bond is a dashed line from the hydrogen to the lone pair on the oxygen.

It's important to also show the δ^+ and δ^- charges on the molecule

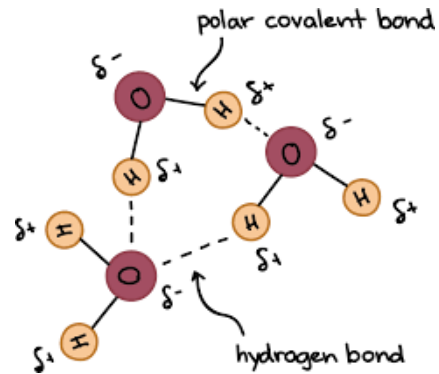


- Larger = higher molecular mass $\therefore$ LDF
- Group V, VI, VII hydrides have higher than expected value (NOF) (hydrogen bonding)

Covalent bonding:

Molecular/simple:

- Strong intra → insulators; weak inter → low mp/bp

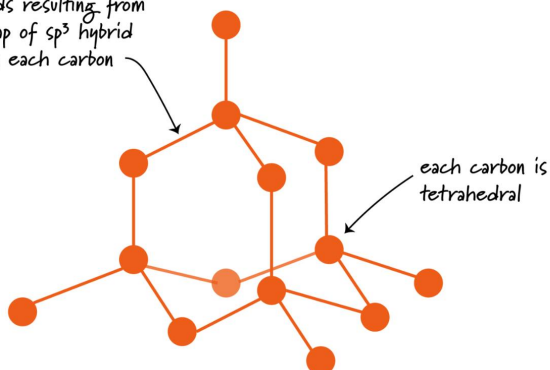


Network/macro:

- **Covalent** bonding usually results in **molecules**//some **carbon allotropes & SiO_2** (quartz) form **covalent network solids**
- Creates **lattice structures**

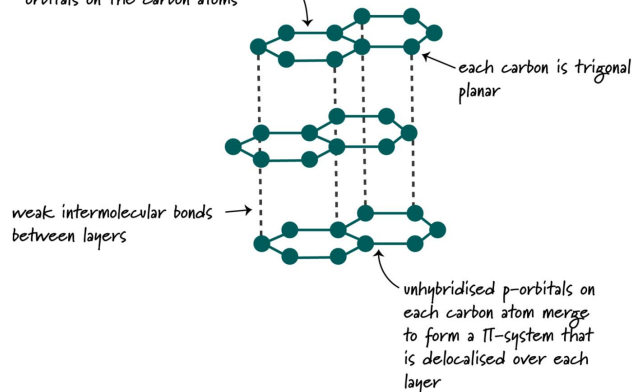
Diamond:

all C-C bonds are single, sigma bonds resulting from the overlap of sp^3 hybrid orbitals on each carbon



Graphite:

strong C-C single, sigma bonds within each layer, resulting from the overlap of sp^2 hybrid orbitals on the carbon atoms

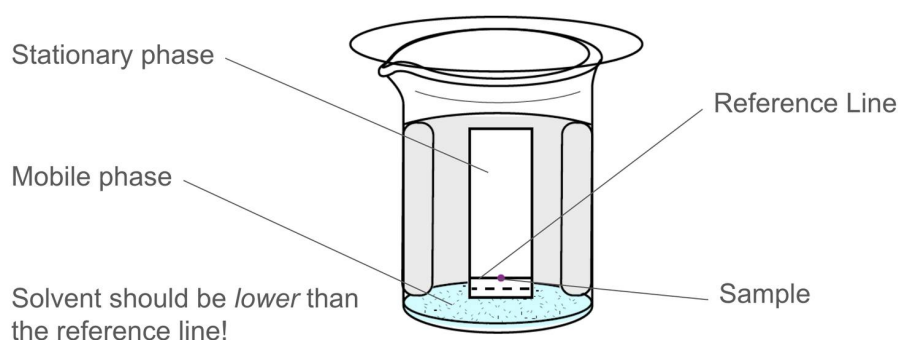


Chromatography:

- **Dissolution**: process of forming a **solution (solute + solvent)**
- General rule: **like dissolves like (same IMF)** → polar dissolves polar; non-polar dissolves non-polar:

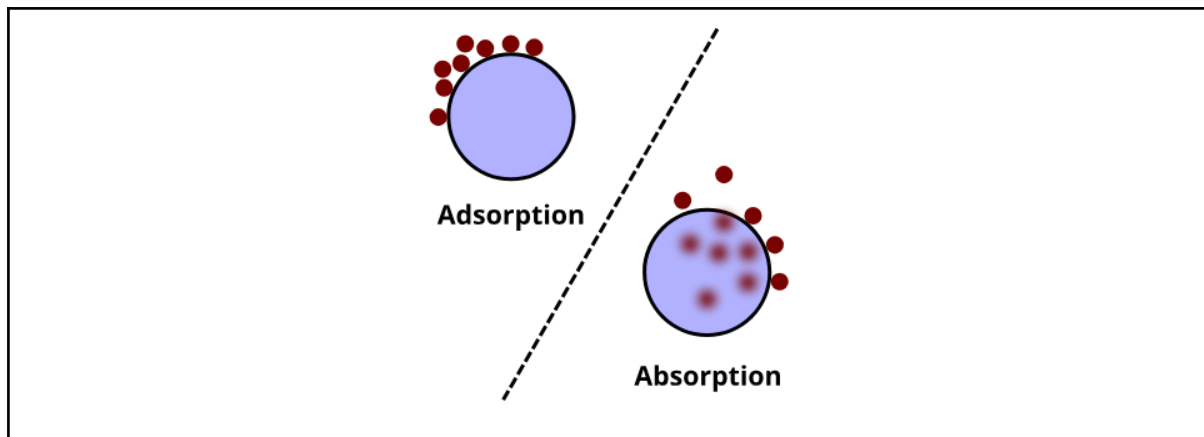
- 3 stages of **dissolution**: **breakdown of solute-solute attraction**, breakdown of **solvent-solvent, solute-solvent interactions**
- Similar IMF allows for **strong solvent-solute** interactions → dissolution more **energetically favourable**
- Some vocab...
 - **Solvent front**: line on chromatogram where **mobile** phase reaches before analysis
 - **Mobile phase**: moves on **stationary phase** & moves **components of sample** at **different** rates
 - **Stationary phase**: **adsorbs** sample & remains **unmoving**

Paper chromatogram:



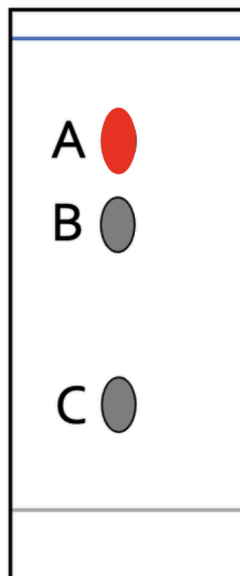
- **Stationary phase** = **paper**
- **Mobile phase** examples:
 - **Polar**: **water, methanol**
 - **Non-polar**: **hexane**

Absorption:	Adsorption:
Enters	Sticks



Process:

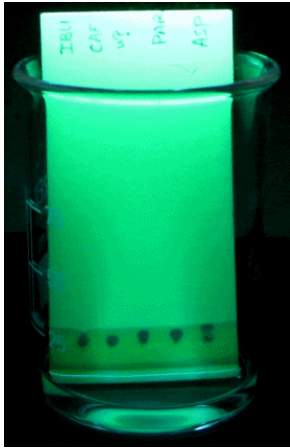
- Sample molecules **adsorbed** into surface **stationary phase**
- As **solvent front** moves up → sample components **desorb** from **stationary phase**
- Separation depends on **affinity** of component molecules for **mobile & stationary** phases:
 - **Faster** molecules = **stronger affinity/attraction** to **mobile** phase:



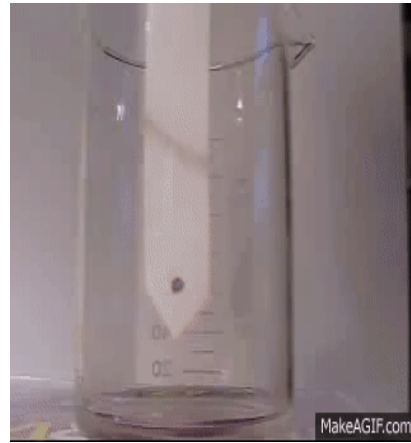
- $R_f = \frac{d \text{ travelled by substance}}{d \text{ travelled by solvent front}}$ (retardation factor)

TLC (thin layer chromatography):

- Contains **fluorescent** additive:

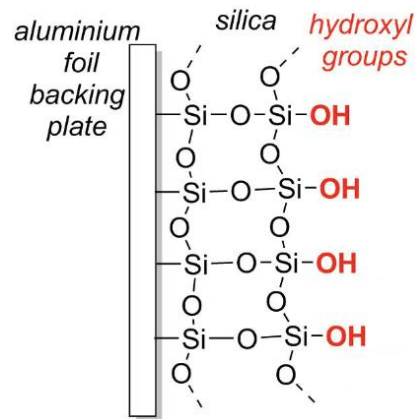


TLC



Paper

- Coated with **adsorbent** material → **stationary phase**
 - Usually **silica** gel (polar):



- Developed in **chamber** → **non-polar mobile** phase = highly volatile

Organic:

Homologous series:

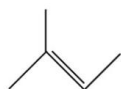
Features:

- Same general formula
- Same functional group
- Similar chemical properties
- Differ in physical properties by a CH_2 group

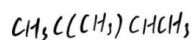
Representation:

2-methyl but-2-ene:

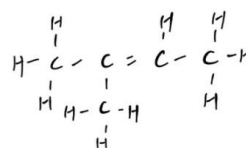
Skeletal



Condensed structural



Full structural

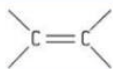
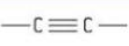
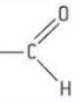
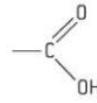


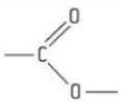
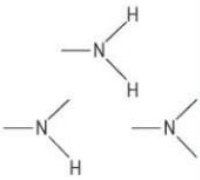
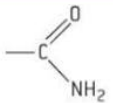
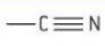
Nomenclature:

- No. of C in **longest chain** → **prefix**
- Identify **homologous** series
- Identify **position, number & name of substituents**
 - If substituent > 1 → **alphabetical** order (number follows)
- Functional groups **priority** list (highest → lowest):
 - **Carboxylic acids** ($-\text{COOH}$)
 - **Esters** ($-\text{COOR}$)
 - **Amides** ($-\text{CONH}_2$)
 - **Nitriles** ($-\text{C}\equiv\text{N}$)
 - **Aldehydes** ($-\text{CHO}$)
 - **Ketones** ($-\text{CO}-$)
 - **Alcohols** ($-\text{OH}$)

- **Amines** ($-\text{NH}_2$)
- **Alkenes** ($\text{C}=\text{C}$)
- **Alkynes** ($\text{C}\equiv\text{C}$)
- **Alkanes** ($\text{C}-\text{C}$)
- **Ether** ($\text{R}-\text{O}-\text{R}'$)

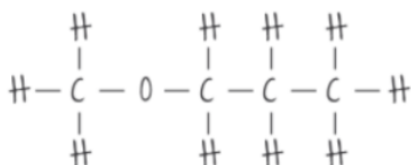
Functional groups:

Class	Functional group	Suffix	General formula	Example
alkanes	—	-ane	$\text{C}_n\text{H}_{2n+2}$	Propane C_3H_8
alkenes	alkenyl 	-ene	C_nH_{2n}	but-2-ene $\text{CH}_3\text{CH}=\text{CHCH}_3$
alkynes	 alkynyl	-yne	$\text{C}_n\text{H}_{2n-2}$	but-2-yne $\text{CH}_3\text{C}\equiv\text{CCCH}_3$
arenes	phenyl 	—	$\text{C}_n\text{H}_{2n-6}$	benzene C_6H_6
halogenoalkanes	 [X = F, Cl, Br, I]	—	$\text{C}_n\text{H}_{2n+1}\text{X}$	2-chlorobutane $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{CH}_3$
alcohols	 hydroxyl	-ol	ROH	butan-2-ol $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$
aldehydes	aldehyde 	-al	RCHO	ethanal CH_3CHO
ketones	 carbonyl	-one	$\text{RC(O)R}'$	propanone $\text{CH}_3\text{C(O)CH}_3$
carboxylic acids	 carboxyl	-oic acid	RCOOH	ethanoic acid CH_3COOH

esters	 ester	-oate	RCOOR'	methyl ethanoate CH ₃ COOCH ₃
ethers	 ether	—	ROR'	ethoxyethane CH ₃ CH ₂ OCH ₂ CH ₃
amines	 amino	-amine	RNH ₂ RNHR' RN(R')R''	propan-1-amine CH ₃ CH ₂ CH ₂ NH ₂ <i>N</i> -methylethanamine CH ₃ CH ₂ NHCH ₃
amides	 amido	-amide	RCONH ₂	ethanamide CH ₃ CONH ₂
nitriles	 cyano	-nitrile	RCN	propanenitrile CH ₃ CH ₂ CN

Ethers:

- FG = **alkoxy**
- Prefix = **-oxy-**
- GF = **C_nH_{2n+1}OC_mH_{2m+1}**
- Naming:



Methoxypropane

Alkanes:

- FG = **alkanyl**
- GF = **C_nH_{2n+2}**
- **Saturated hydrocarbons** containing only **single** bonds
- Properties:
 - **Unreactive** → **insoluble & doesn't react with acids/alkalis**

Small:

Big:

Lower m.p.	Higher m.p. (stronger IMF)
Lower viscosity	Higher viscosity (stronger IMF)
Easier to ignite	Harder to ignite (stronger IMF)
Cleaner flame	Dirtier flame

Alkynes:

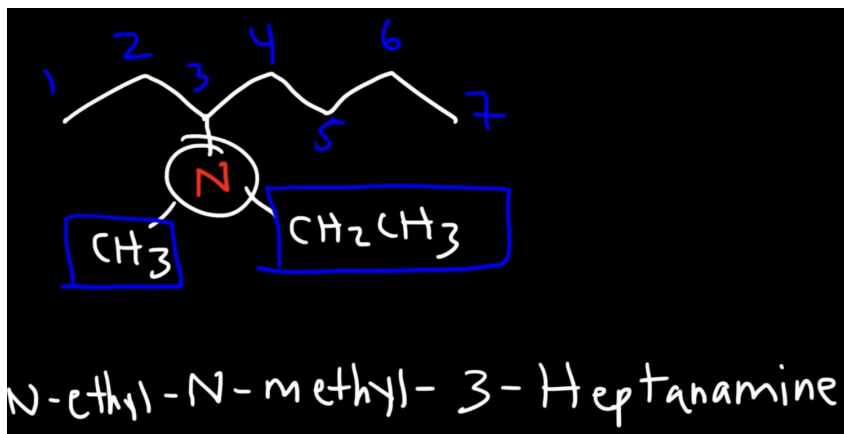
- FG = **alkynyl**
- GF = C_nH_{2n-2}

Alkenes:

- FG = **alkenyl**
- GF = C_nH_{2n}
- **Unsaturated** → double C=C bond
- Test: **bromine water** - **orange** → colourless
- Methods of production:
 - **Cracking**: process of producing **alkenes & smaller alkanes** by breaking of **larger alkanes**
 - **Dehydration of alcohol**

Amine:

- FG = **amino group** (NH_2)
- GF = $C_nH_{2n+3}N$
- Naming:



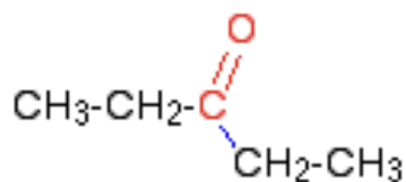
- If followed by FG with **higher priority** → **__amino__**

Alcohol:

- FG = **hydroxyl**
- GF = $\text{C}_n\text{H}_{2n+1}\text{OH}$
- Substituent name = **-oxyl-**

Ketone:

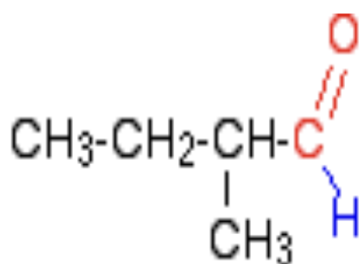
- FG = **carbonyl (R-CO-R')** ← C bonded to **2 other C chains**
- Suffix = **-one**
- Naming:



Pentan-3-one

Aldehydes:

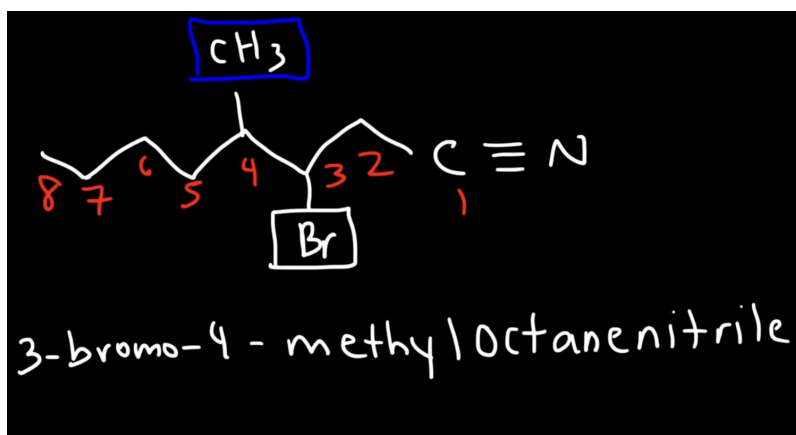
- FG = **carbonyl (-CHO)** ← always write at end
- Suffix = **-al**
- GF = $\text{C}_n\text{H}_{2n+1}\text{CHO}$
- Naming:



2-methyl-but-1-al

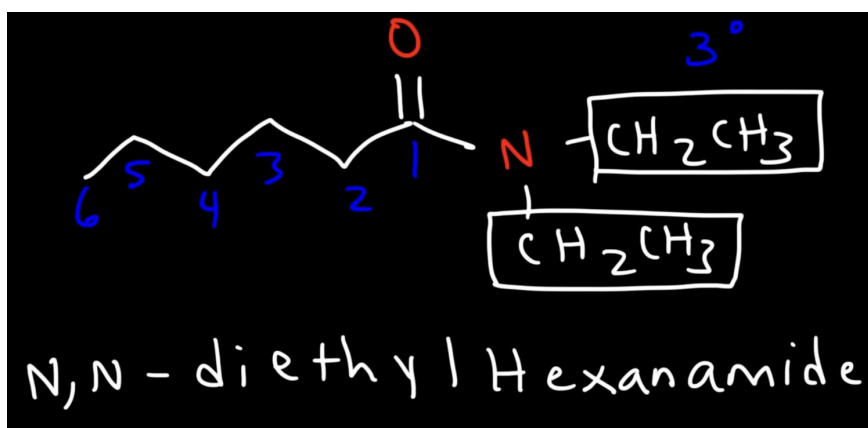
Nitriles:

- FG = **cyano group**
- GF = $\text{C}_n\text{H}_{2n-1}\text{N}$
- Naming:



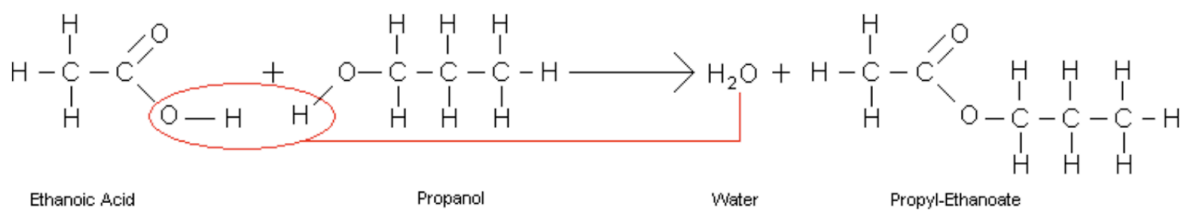
Amides:

- FG = **amide group** (-CONH₂)
- GF = **C_nH_{2n+1}NO**
- Naming:



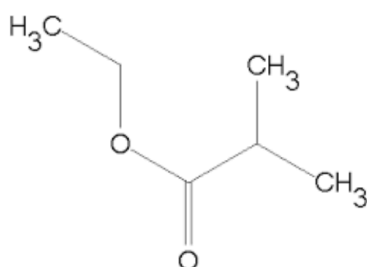
Esters:

- Production method:



- FG = **ester** (-COO-)
- GF = **C_nH_{2n+1}COOC_mH_{2m+1}**
- Suffix = **-yl -oate**
- Naming:

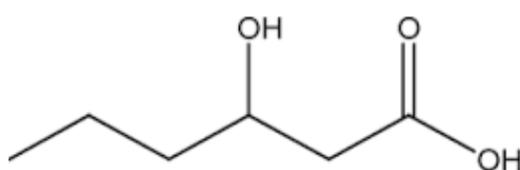
- Part from alcohol first (single bond O); part from carboxylic acid (double bond O)
- E.g.



2-methyl ethyl propanoate

Carboxylic acid:

- FG = **carboxyl (-COOH)**
- Suffix = **-oic acid**
- GF = $C_nH_{2n+1}COOH$
- Naming:



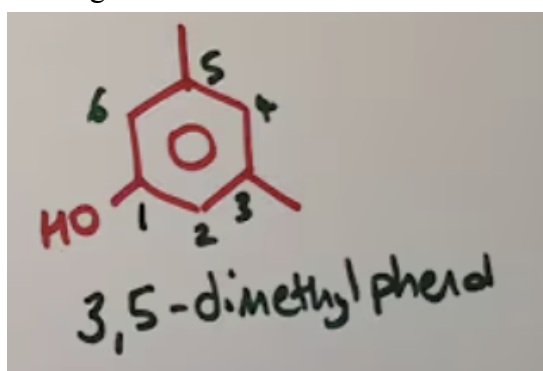
3-hydroxyl-hexanoic acid

Halogenoalkanes:

- Prefix = **fluoro-, chloro-...**
- GF = $C_nH_{2n+1}X$ (X = halogen)

Phenols:

- **Benzene** ring with **-OH** (substituent) attached
- Naming:



Phenyls:

- Benzene ring with **any** C_mH_n attached
- Naming:
 ___benzene

m.p.:

Carboxylic acid > alcohol > aldehyde > ketone > ester > ether > alkane > alkene > alkyne

Structural isomers:

- Same molecular formula

Isomers:

Structural:

- Structures with **same molecular** formula but **differ** in **arrangement**

Chain	Different arrangements of C → straight/branched chains
Positional	Different position of same FG: - Primary: C_{FG} attached to 1C - Secondary: C_{FG} attached to 2Cs - Tertiary: C_{FG} attached to 3Cs
Functional	- Aldehydes ↔ ketones - Acids ↔ esters - Cycloalkanes ↔ alkenes - Alcohol ↔ ether

Stereo:

Geometric:

- Different arrangement of groups about **C=C** group
- To form:

1. Must have **C=C/cycloalkane**
2. **Each C** must have 2 **different** groups attached

Cis	Trans
- Branches on longest chain on same side - Polar - Higher b.p. → polarity - dipole ↔ dipole interactions (stronger IMF)	- Branches on longest chain on different sides - Nonpolar - Higher m.p. (easier to pack as a solid)
Alkenes: 2H on same side of C=C bond	
Cycloalkanes: both groups facing same direction (same plane)	

Polymerisation:

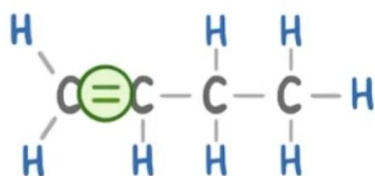
- Long chain (**macro**)**molecules** formed from **monomers**
- Requires **double C=C bonds**

Addition:

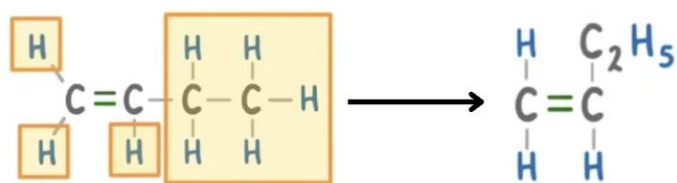
- Short chains of **alkenes** → long chain of **alkanes** by removing **C=C** → **C-C**
(**unsaturated** → **saturated**)

Diagrams:

1. Identify **double bond**:



2. Draw **condensed formula** around **double bond** (only **top & bottom**):



3. Get rid of **double bond** → **single**:



Curly arrows:



Fig 1. Double and Single-Headed Curly Arrows. A double-headed arrow represents the movement of a pair of electrons, whereas a single-headed arrow represents the movement of just one electron.

Radicals:

- Highly **reactive** molecular entity that has **one unpaired e⁻** (e.g. •Cl)
- Produced during **homolytic fission**:

Fission:

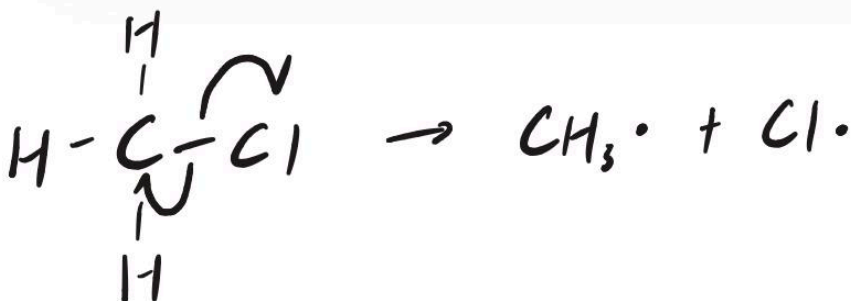
Homo vs. hetero:

Homo:	Hetero:
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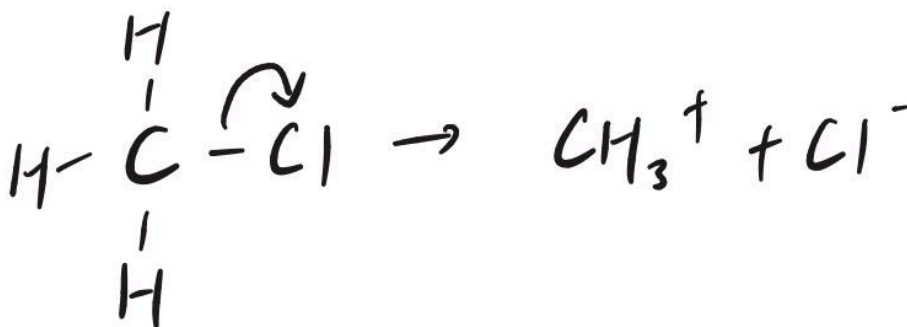
Each bonded atom in covalent bond ends up with one unpaired e⁻	One of bonded atoms in covalent bond gets all e⁻
Forms radicals	Cation + anion
Requires specific amount of E (from UV) ∴ least likely	

E.g. C-Cl bond:

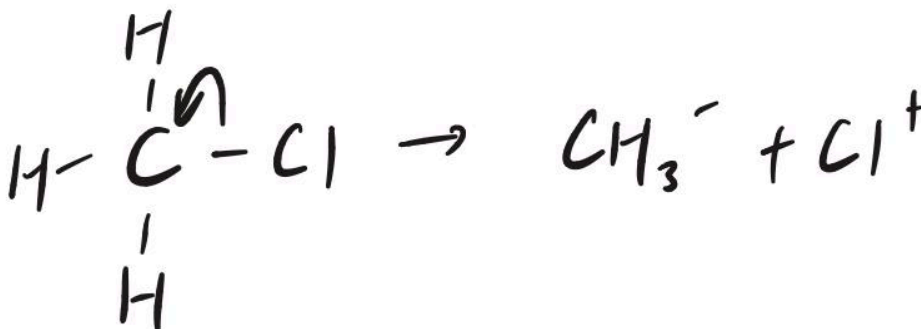
1. Homo:



2. Hetero (1):



3. Hetero (2):



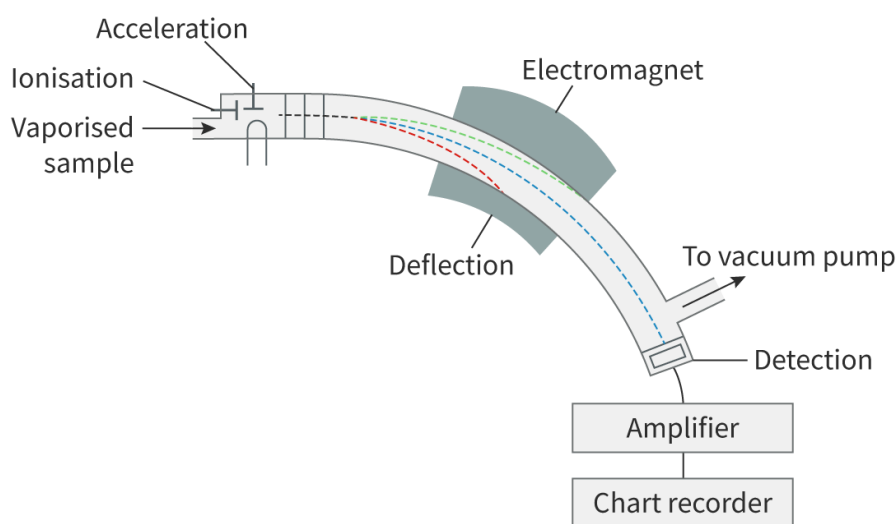
Nucleophiles:

Strength:

- **More available lps**
- **(-) charge > dipole**
 - E.g. $\text{OH}^- > \text{H}_2\text{O}$:
- **If all dipoles** → lp on the atom with **lowest electronegativity**

Mass spectrometry:

Mass spectrometer:



1. **Vaporisation:** $\text{X}_{(\text{l})} \rightarrow \text{X}_{(\text{g})}$
2. **Ionisation:** high E e^- fired → **ionises atoms & forms:**
 - **Cations:** $\text{X} + \text{e}^- \rightarrow \text{X}^+ + 2\text{e}^-$
 - **Fragments:** molecules **breaking into parts**; **charged fragments** appear as **peaks**//**free radicals** are **undetected** (e.g. $\text{CH}_3\text{CHO} \rightarrow \text{CH}_3^+$ (detected) + $\text{CHO}^\bullet$ (undetected free radical))
3. **Acceleration** by **electric** field: **voltage** across parallel plates
4. **Deflection** by **magnetic** field: ions deflected by **electromagnet** - degree of deflection depends on **m:z (mass:charge)**
 - **Lower mass** → deflected **more**
5. **Detection:** **current** generated with **size α abundance** of each ion

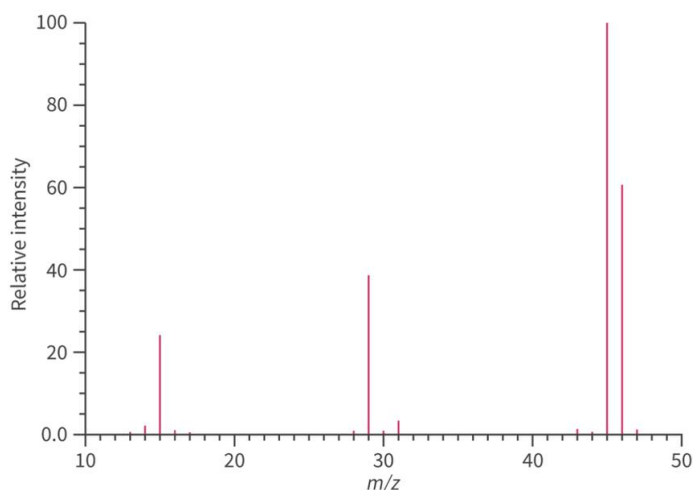
Mass spectrum:

- **Base ion peak:** most intense peak in a spectrum (**highest abundance** $\Rightarrow$ **most stable** fragment)
- Analysing - e.g.:

The mass spectrum is for a compound with the empirical formula C_2H_6O .

Use the mass spectrum and table x to determine the following:

1. Relative molecular mass, M_r , of the compound
2. m/z of the base ion peak
3. Formula of the molecular ion
4. Mass lost and possible neutral fragment formed for the peaks at $m/z = 15$ and $m/z = 31$



1. The molecular ion peak has a m/z of 46 so this is the relative molecular mass.
2. The base ion peak has an m/z of 45.
3. $C_2H_6O^+$
4. m/z 15: mass lost: $46 - 15 = 31$
Possible neutral fragment lost is $\bullet OCH_3$
 m/z 31: mass lost $46 - 31 = 15$
Possible neutral fragment lost is $\bullet CH_3$

Free radical rxns.:

Halogenation of alkanes:

1. Initiation:

- **Free radical** created through **homolytic** fission via **UV**
- **WEAKEST** bond is broken
- Creates **reactive** species

2. Propagation:

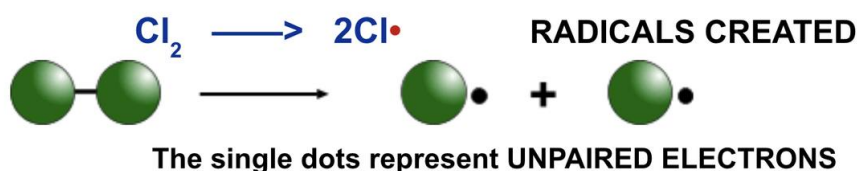
- Radicals used + regenerated (both as reactants & products)
- Producing **more reactive species** so chain reactions **don't stop**
- Will always likely form some **halogen hydroxide** (i.e. HCl, HBr)

3. Termination:

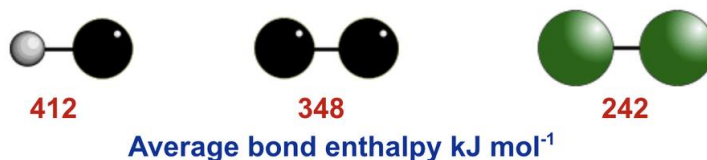
- Remove free radicals left to continue reacting
- May result in **increase in length of C chain**
- Removing reactive species & only forming **molecules**

E.g. Chlorination of ethane:

Initiation

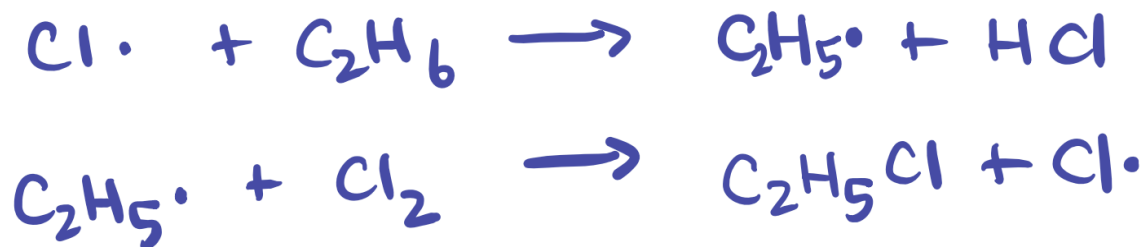


During initiation, the **WEAKEST BOND IS BROKEN** as it requires less energy. There are three possible bonds in a mixture of alkanes and chlorine.

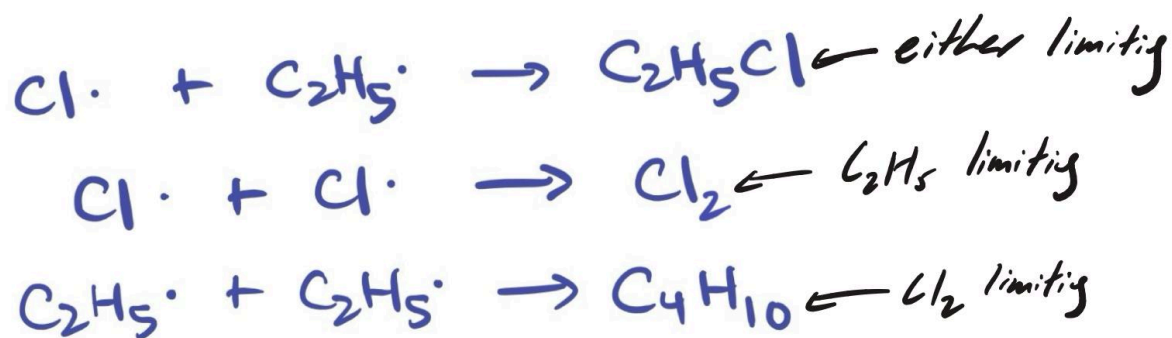


The Cl-Cl bond is broken in preference to the others as it is the weakest and requires less energy to separate the atoms.

Propagation:

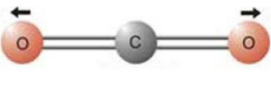
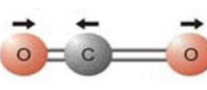


Termination (3 options, depending on **limiting** reagent):



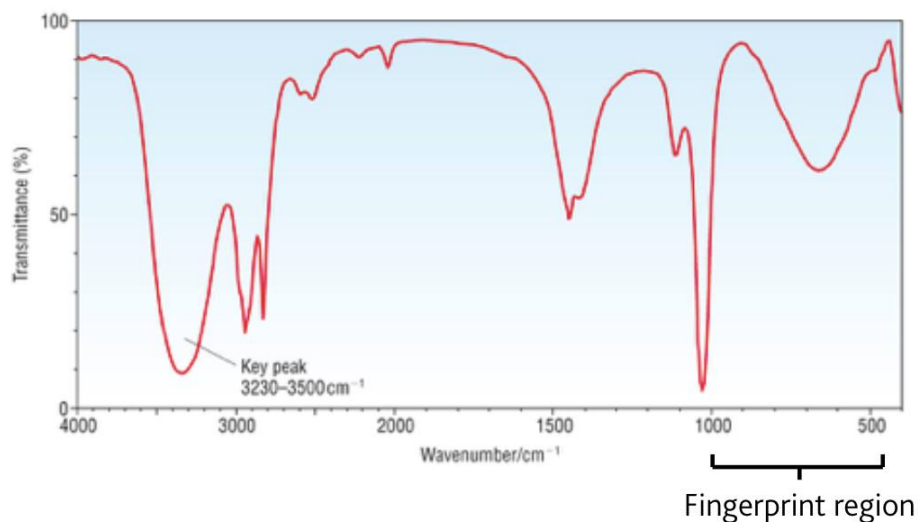
IR spectroscopy:

- Wave number = $\frac{1}{\lambda}$ (cm^{-1})
- Molecules can absorb IR due to **bending/stretching vibrations**
- Only vibrational motions causing **changes in dipole moment** causes IR absorption

 symmetric stretching	 asymmetric stretching	 bending
Infrared inactive	Infrared active	Infrared active

- IR frequency absorbed depends on:
 - Molecule **mass**
 - Bond **strength** $\Rightarrow$ stronger requires **higher E** (higher f)

Spectrograms:

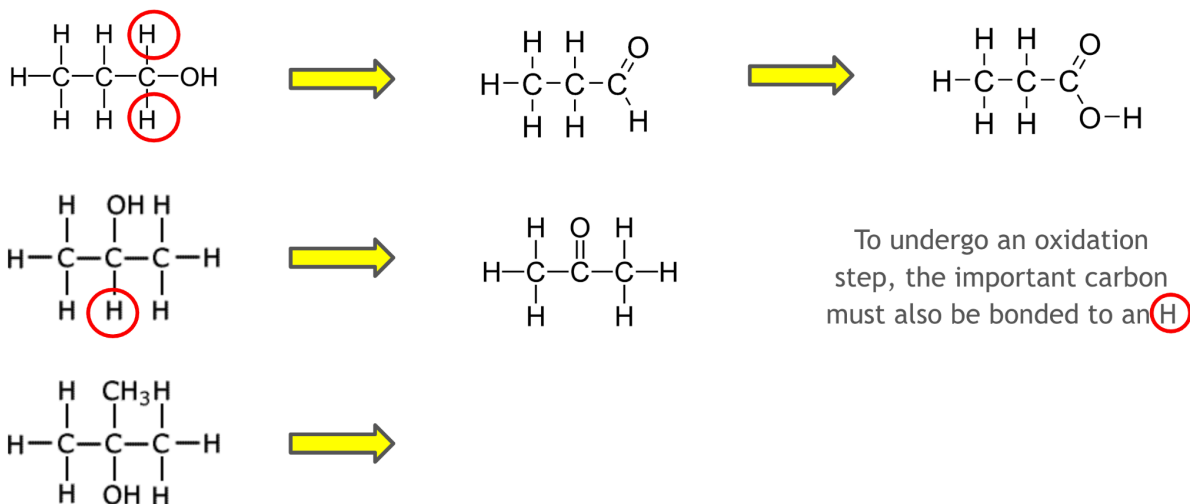


- **Strong, broad** = O-H bond
- **Fingerprint region:**
 - **500 - 1000cm⁻¹**
 - Complex due to **large number of possible vibrations**
 - Compare to **database**
- Compare to:

Bond	Organic molecules	Wavenumber (cm ⁻¹)	Intensity
C-I	iodoalkanes	490-620	strong
C-Br	bromoalkanes	500-600	strong
C-Cl	chloroalkanes	600-800	strong
C-F	fluoroalkanes	1000-1400	strong
C-O	alcohols, esters, ethers	1050-1410	strong
C=C	alkenes	1620-1680	medium-weak; multiple bands
C=O	aldehydes, ketones, carboxylic acids and esters	1700-1750	strong
C≡C	alkynes	2100-2260	variable
O-H	hydrogen bonding in carboxylic acids	2500-3000	strong, very broad
C-H	alkanes, alkenes, arenes	2850-3090	strong
O-H	hydrogen bonding in alcohols and phenols	3200-3600	strong, broad
N-H	primary amines	3300-3500	medium, two bands

Oxidation of alcohols:

- **Condition: acidified oxidising agent** (i.e. potassium dichromate (VI) + dilute H₂SO₄)



- Primary $\rightarrow$ aldehyde $\rightarrow$ carboxylic acid
- Secondary $\rightarrow$ ketone
- Tertiary $\rightarrow$ no reaction as no H attached to alpha C

Collecting products:

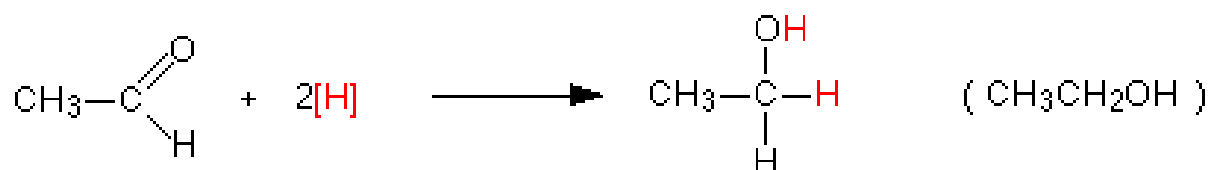
Aldehyde:	Carboxylic acid:
- Limiting [O] (oxidising agent)	- Excess [O] $\rightarrow$ ensure complete oxidation
DISTILLATION SETUP warmer H_2O out cold H_2O in aldehyde primary alcohol $\text{H}^+ / +[\text{O}]$ limiting heat	REFLUX SETUP warmer H_2O out cold H_2O in primary alcohol $\text{H}^+ / +[\text{O}]$ in excess heat

Undo reverse - reduction of carbonyls:

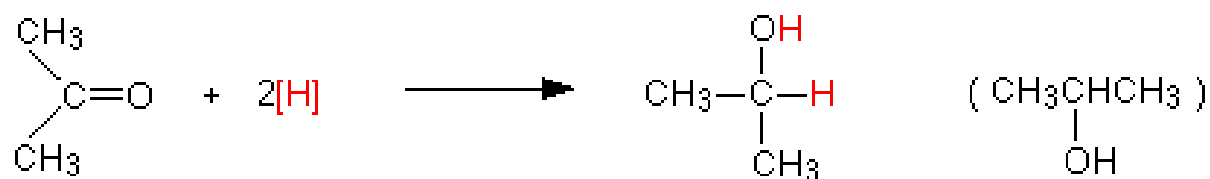
Reducing agents:

Name:	Lithium aluminum hydride (LiAlH₄):	Sodium borohydride (NaBH₄):
Condition:	Anhydrous	Aqueous
Reactivity:	- Violent reaction - Only used for reduction of carboxylic acid	- Safer reagent - Not reactive enough to reduce carboxylic acids

Aldehyde → 1° alcohol:

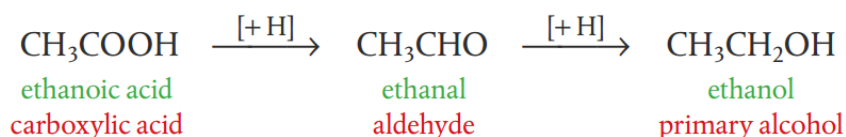


Ketone → 2° alcohol:



Carboxylic acid → 1° alcohol:

- Aldehyde → alcohol



Conditions: heat with LiAlH₄ in dry ether. The reaction cannot be stopped at the aldehyde as it reacts too readily with LiAlH₄.

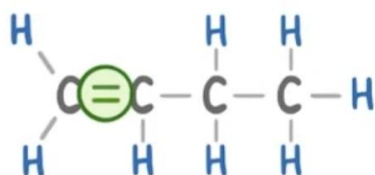
Polymerisation:

Addition:

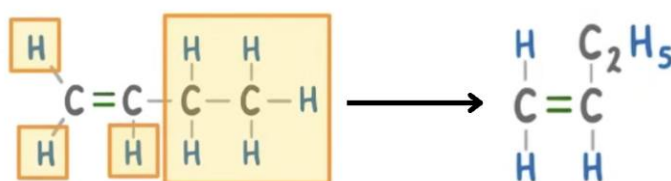
- Alkene monomers → one polyalkane
- No byproducts

Diagrams:

1. Identify **double bond**:



2. Draw **condensed formula** around **double bond** (only **top & bottom**):



3. Get rid of **double bond** → **single**:

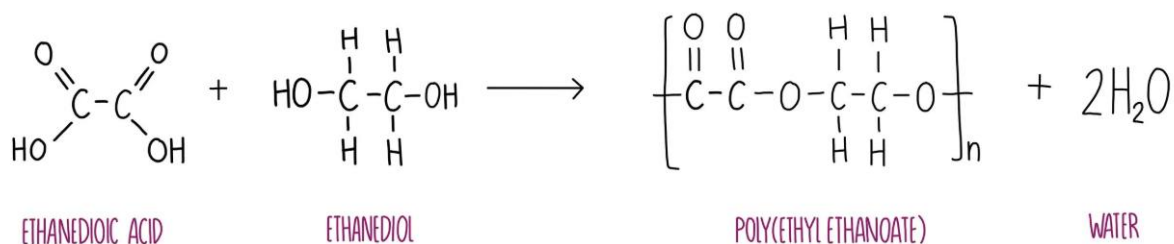


Condensation:

- 2 different monomers join to form either polyester + water/polyamide + water

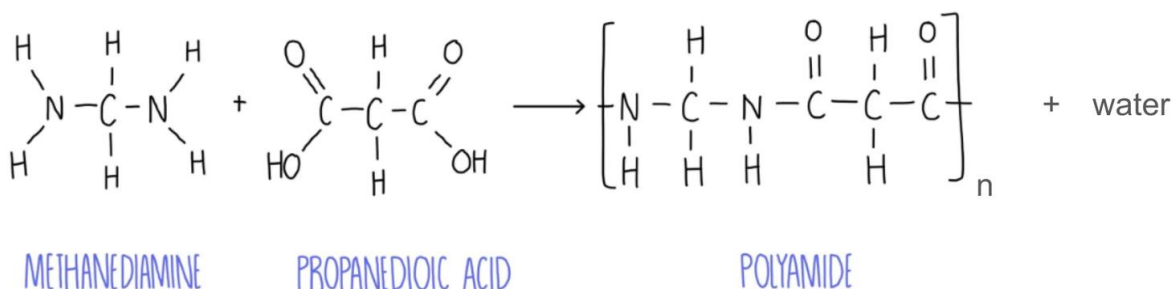
Polyester:

- Dicarboxylic acid (2 carboxylic acid groups) + diol (2 alcohol groups) → ester bond ($-\text{COO}-$) + water (removed)



Polyamide:

- Diamine + dicarboxylic acid $\rightarrow$ amide (-CONH) linkage + water (removed)

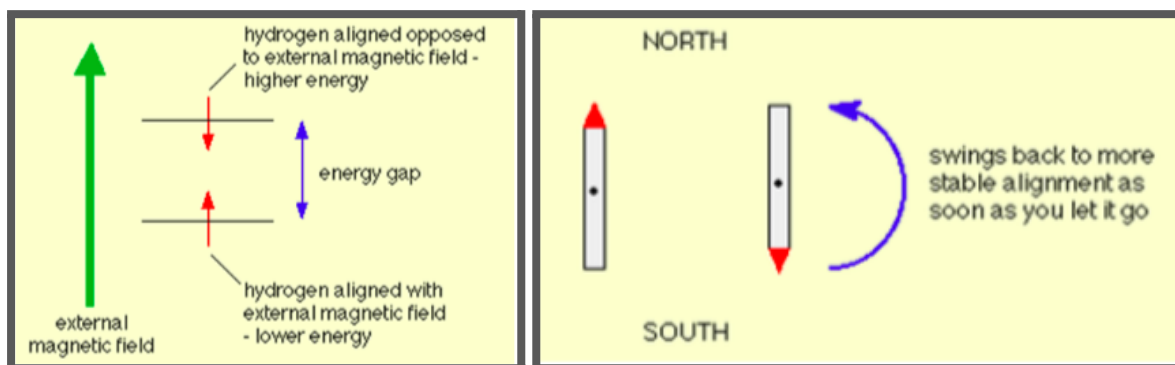


^1H -NMR:

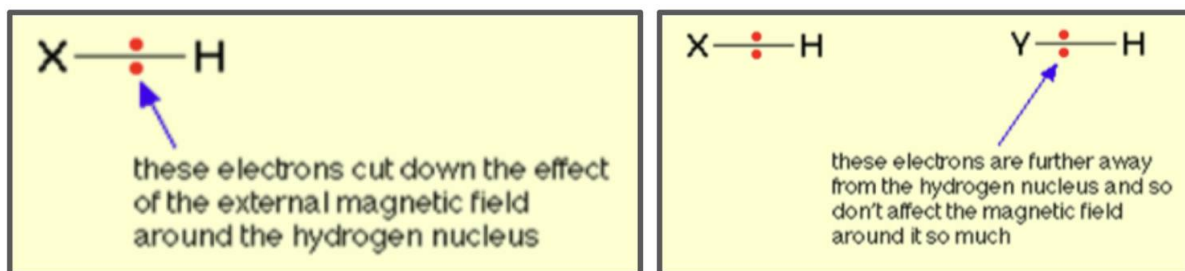
- Gives info on **different hydrogen environments** in a molecule

Functioning:

- Like e^- , nuclei can have a “**spin**”
- **Ground state:** spin of nucleus **align** with a magnetic field
- When exposed to **radio waves** $\rightarrow$ nucleus can **absorb E** & “**flip**”:



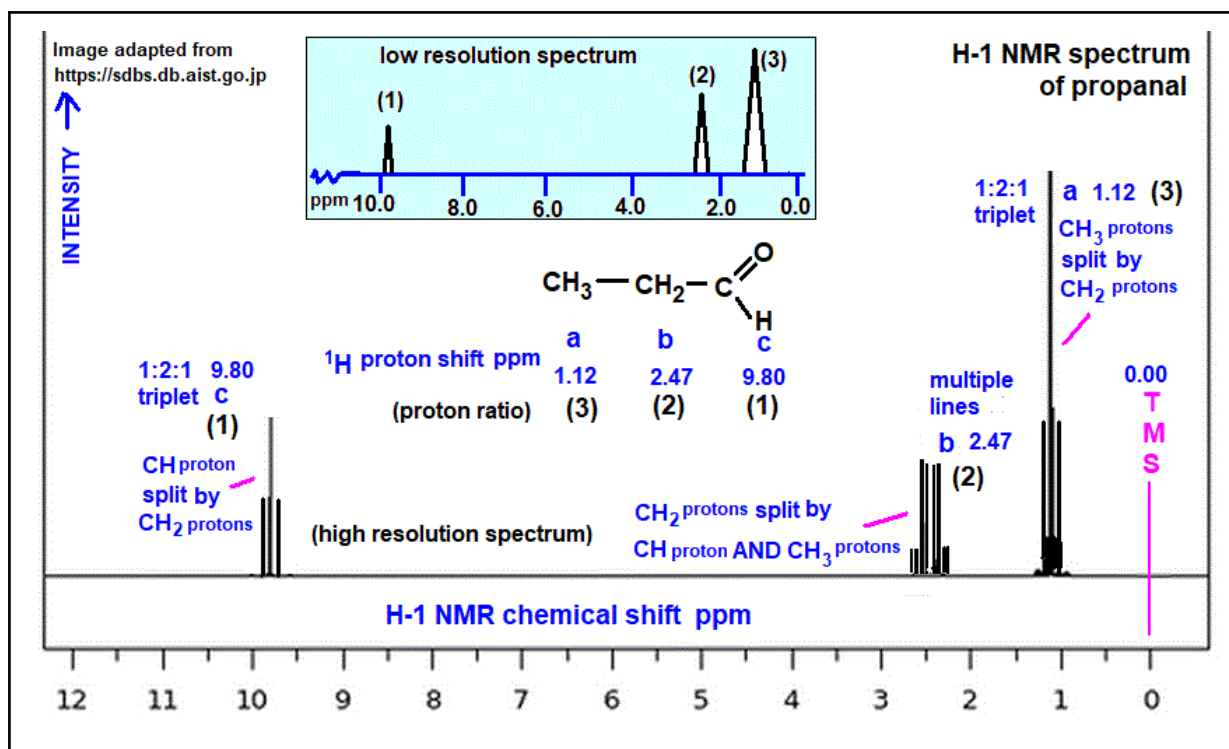
- e^- affects **B** $\Rightarrow$ electronegativity of neighbouring atoms affect how much E is required to flip the spin of a nucleus (different types of location absorb different f):



Interpreting spectra:

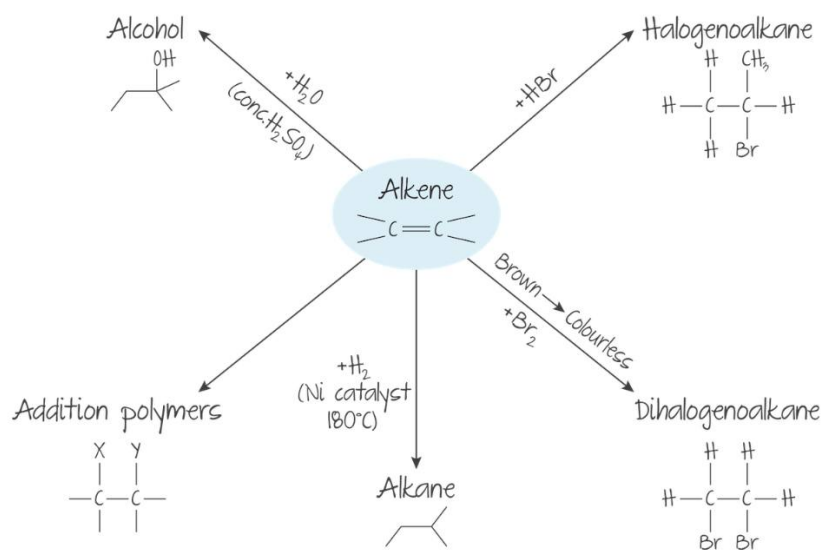
Low + high res:	No. of H environments :	- Non-identical places & neighbouring groups - Equals no. of peaks on spec - E.g.: $\begin{array}{c} \text{H} & \text{H} \\ & \\ \text{H}-\text{C}- & \text{C}-\text{O}-\text{H} \\ & \\ \text{H} & \text{H} \end{array}$ 3 $\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{Si}-\text{CH}_3 \\ \quad \diagup \quad \diagdown \\ \text{CH}_3 \quad \text{CH}_3 \quad \text{CH}_3 \end{array}$ 1 $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ 4 $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C}- & \text{C}- & \text{C}-\text{H} \\ & & \\ \text{H} & \text{H} & \text{H} \end{array}$ 2 $\begin{array}{c} \text{O} \\ \\ \text{H}_2\text{C}=\text{C}-\text{C}-\text{CH}_3 \\ \\ \text{H} \end{array}$ 3
	Chemical shift:	- Type of H environment - Downfield (neighbouring more electronegative elements) ↔ upfield (less electronegative)
	Integration:	- No. of Hs in one environment (area under peak)
	TMS:	- Tetramethylsilane $\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{Si}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$ - Ref peak to set chemical shift scale - 1 chemical environment & 12Hs → clear singlet signal that can be set to 0 ppm

High res:	Splitting:	<u>General:</u> - No. of Hs on neighbouring atoms - no. of splitting = $n + 1$ - n: no. of neighbouring Hs - I.e.: Splitting expected ($n+1$ rule) - I: triplet - II: sextet - III: triplet (O shields) - IV: singlet (O shields) <u>Splitting pattern:</u> - Pascal's triangle (combs) - E.g.: - Triplet $\rightarrow 1:2:1$ - Quartet $\rightarrow 1:3:3:1$
E.g.		



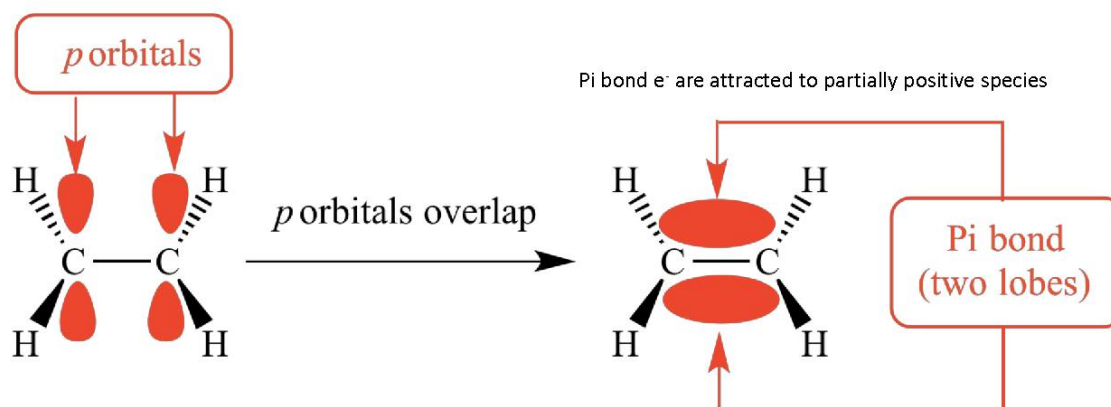
Alkene reactions:

- Alkanes = unreactive due to:
 - Strong bonds (high bond enthalpies)
 - Non-polar ∴ not susceptible to attacks



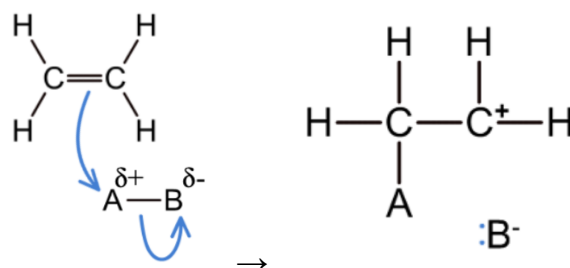
Electrophilic addition:

- **Alkenes** $\Rightarrow$ susceptible to **electrophilic attack** $\rightarrow$ **high e^- density** of **double bond** (pi bond) $\rightarrow$ lead to **electrophilic addition**
 - **π bond** = **weaker & high e^- density** $\rightarrow$ likes to react with **electrophiles** ((+) species)



Mechanism:

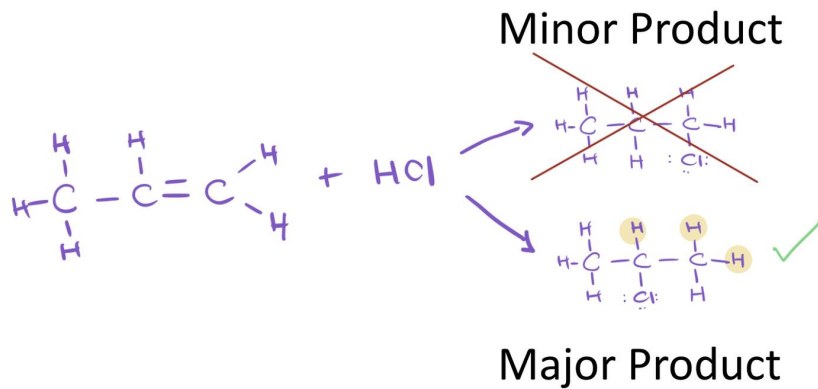
1. π bond attacks the **slightly (+) H (δ^+)**
2. H-X undergo **heterolytic fission** $\rightarrow H^+ + X^-$
3. 1 C gets H^+ ; other loses $e^- \therefore$ becomes **carbocation**



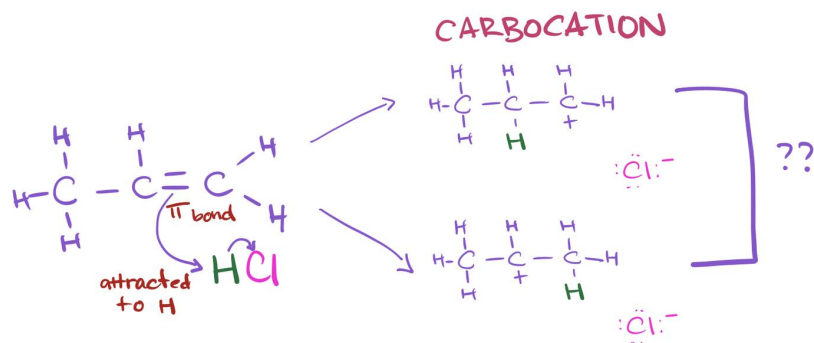
4. X^- acts as **nucleophile** & attacks **carbocation** $\rightarrow$ **coordinate covalent bond**

Asymmetric alkenes - stability (Markovnikov's):

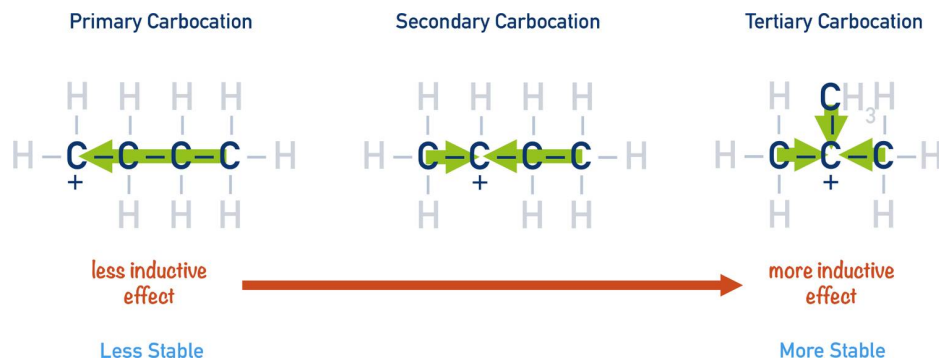
- Markovnikov's rule: when **asymmetric alkene** reacts with a **HX (hydrogen halide)**, the **major product** will have **H** added to C with **greater no. of H substituents**:



- **Polar** reagent (e.g. HX) will have **partial charges** on each atom:
 - $H = \delta^+$; $X = \delta^-$ (more **electronegative**)
- 2 **different** carbocations could form depending on where H^+ **attaches** $\Rightarrow$ reaction prefers the **most stable** one
 - Influenced by **alkyl** groups

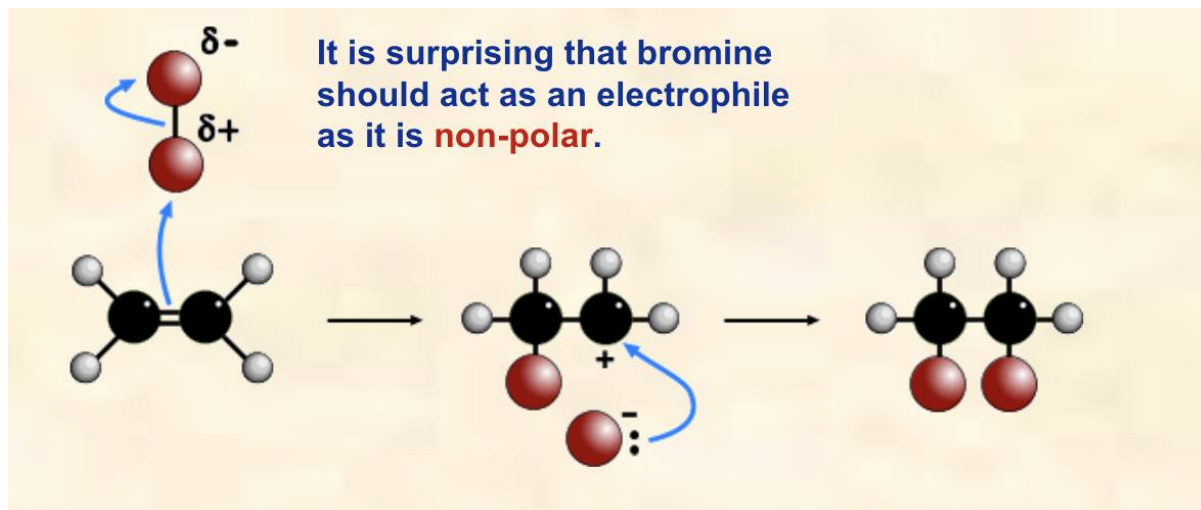


- **Inductive effect:**
 - Groups like CH_3 push e^- **density** toward nearby atoms through **+I effect**
 - Basically **share e^- density with (+) C**, making it **less e^- deprived** $\Rightarrow$ **more alkyls = more e^- density donation**
 - Therefore for **stabilities:**
 - **Tertiary > secondary > primary**



- More **electronegative** elements prefer to attach onto **tertiary carbocation**

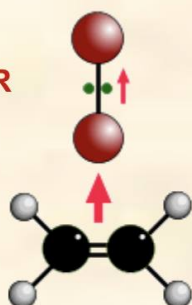
Bromination:



Explanation ...

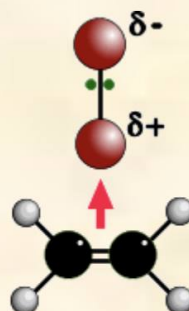
as a bromine molecule approaches an alkene, electrons in the pi bond of the alkene repel the electron pair in the bromine-bromine bond thus inducing a dipole.

NON-POLAR



AS A NON-POLAR BROMINE MOLECULE APPROACHES AN ALKENE, ELECTRONS IN THE PI ORBITAL OF THE ALKENE REPEL THE SHARED PAIR OF ELECTRONS IN THE Br-Br BOND

POLAR



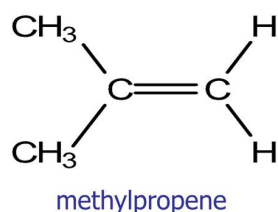
THE ELECTRON PAIR IS NOW NEARER ONE END SO THE BROMINE MOLECULE IS POLAR AND BECOMES ELECTROPHILIC.

Stereoisomers (configurational):

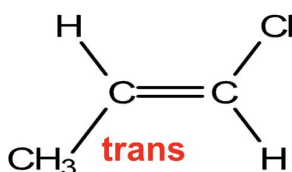
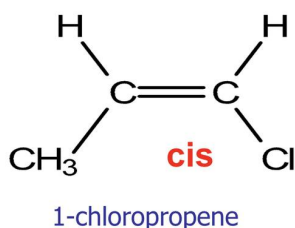
- Same **constitution** (atom identities, connectivities, bond multiplicities) with **different spatial arrangements** of atoms
- **Cis/trans & optical**

Cis/trans:

- Occurs due to **restricted rotation of C=C bond**
- Groups on each carbon are the **same** but in **different positions**
- **Both C** atoms of C=C must have **2 different groups, e.g.:**

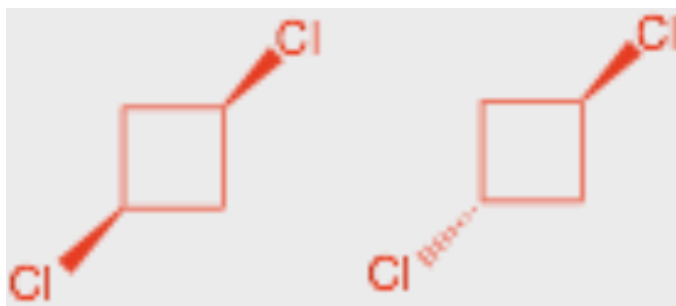


NO



YES

- In **cyclic** molecules:
 - Contain a ring of C that **restricts rotation**
 - **Cis** = groups on **same plane** (all on top/bottom); **trans** = groups on **different planes**



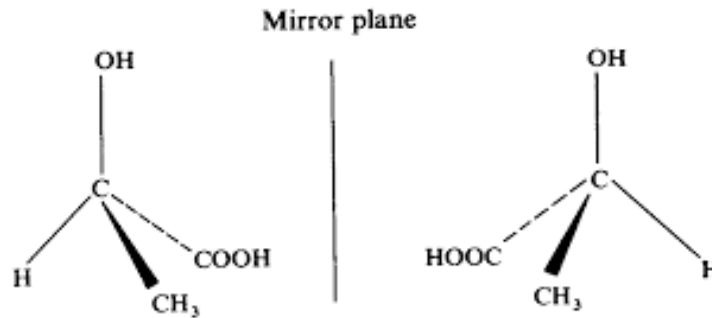
Cis

Trans

- Groups **don't need** to be on **adjacent** C atoms as in alkenes

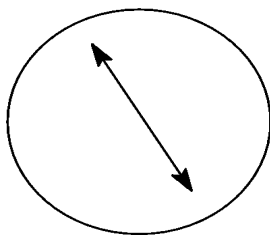
Optical/enantiomers:

- Occurs when compounds have **non-superimposable mirror images + chiral carbon**
- **Chiral carbon**: C with **4 different groups** attached

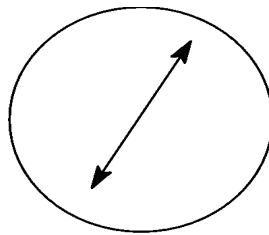


Detection:

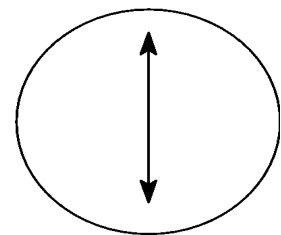
- Same chemical & physical properties but interact **differently** with **biological enzymes**
- Also **rotate plane polarised light** in different **directions** $\therefore$ **polarimeter**:
 - (+)/R/D $\rightarrow$ **clockwise**
 - (-)/S/L $\rightarrow$ **anticlockwise**
 - **Racemic mixture**: **50/50** mixture of **both** $\rightarrow$ rotations **cancel out**



(-)-enantiomer
(anticlockwise rotation)



(+)-enantiomer
(clockwise rotation)



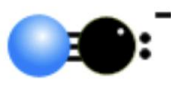
($\pm$)-racemate
(no overall effect)

Nucleophilic substitution:

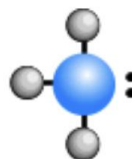
- **Nucleophile**:
 - **e⁻ donor** (Lewis **base**)
 - Possess **≥ 1 lp**; but **don't have to possess (-) charge**
 - Strength depends on (-) charge **density**
 - i.e.



OH⁻



CN⁻



NH₃

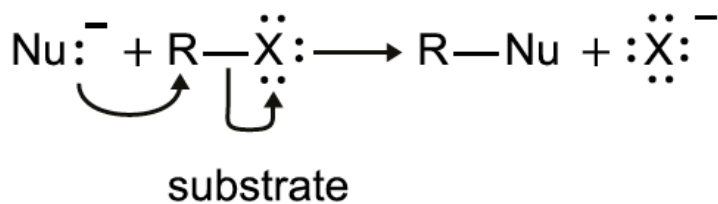


H₂O

→

Decreasing strength

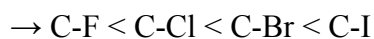
- Mechanism (in terms of e^- movement):



Nu = nucleophile
R = electrophile
X = leaving group

Rate:

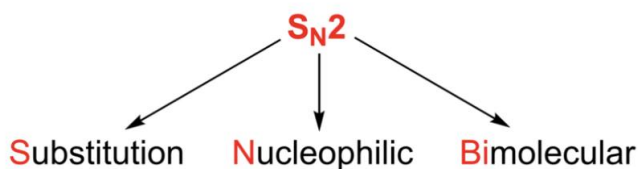
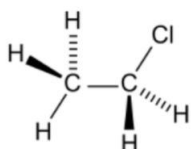
- **Bond strength between C-X** is the **most influential factor** in determining rate



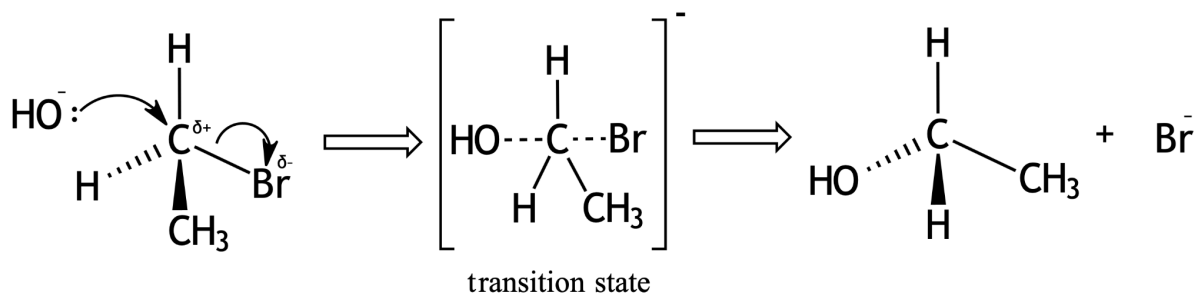
bond	C-F	C-Cl	C-Br	C-I
Bond enthalpy kJ mol ⁻¹	492	324	285	228

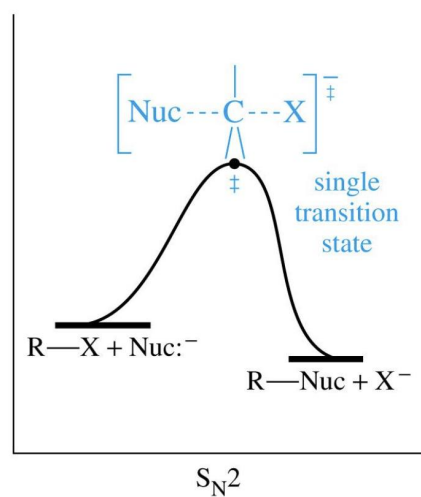
Primary - S_N2:

$$\text{Rate} = k [\text{halogenoalkane}] [\text{OH}^-]$$

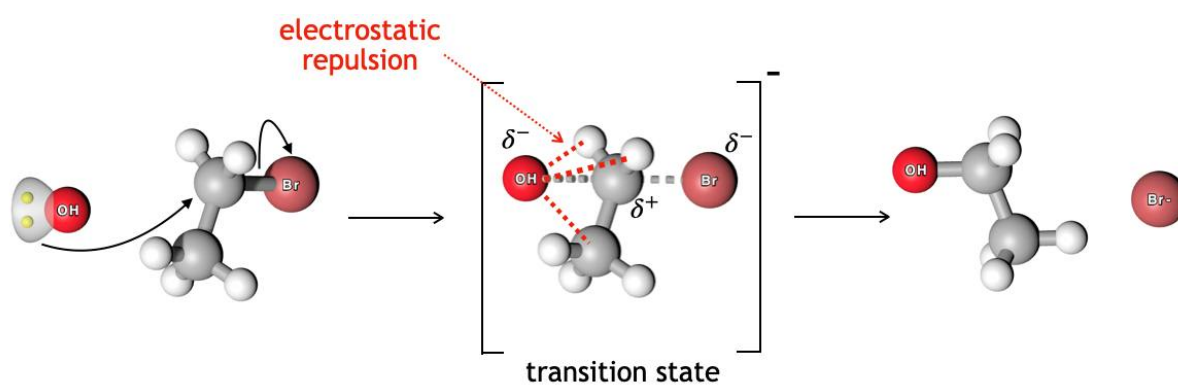


Mechanism diagram:



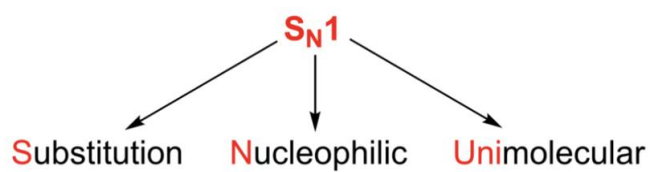
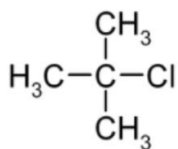


Steric inversion:



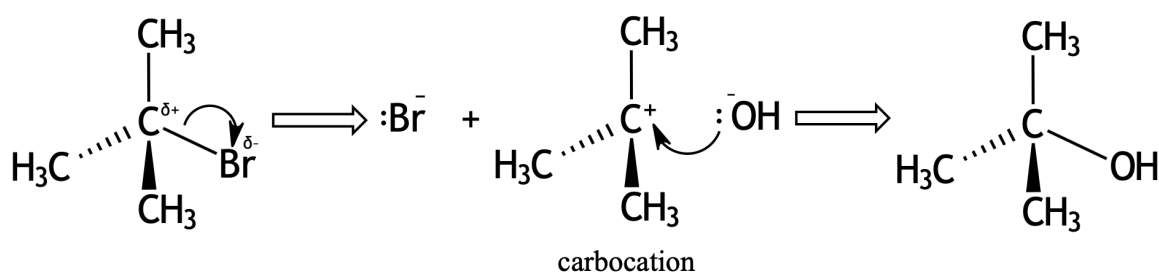
Tertiary - S_N1 :

Rate = k [halogenoalkane]

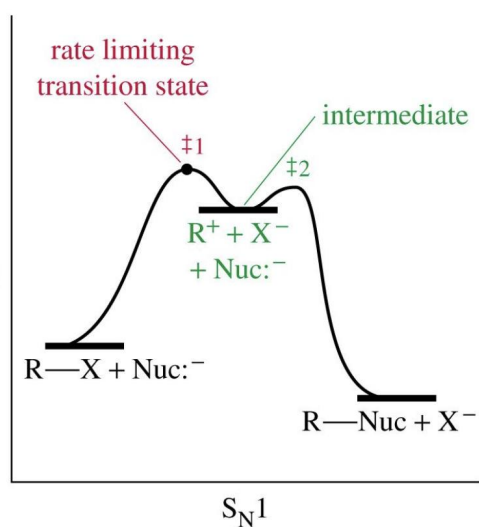


- In general, **rate of $S_N1 > S_N2$** due to:
 - **Unimolecular**
 - Carbocation is normally **more stable** than transition state intermediate of S_N2

Mechanism diagram:

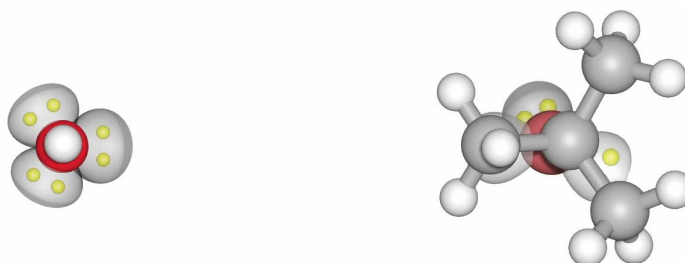


- Forms **trigonal planar** carbocation intermediate → **nucleophile** can attack from **either side of p orbital**



Steric hindrance:

- Reason why nucleophilic substitution in **tertiary** doesn't prefer S_N2
- **Congestion** caused by surrounding **alkyls** ∴ **crowding & repulsion** due to spatial arrangement:



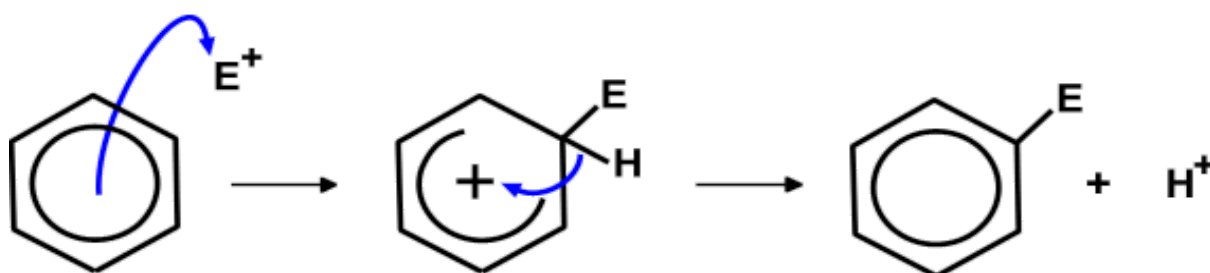
Electrophilic substitution (benzene):

- Why **substitution**?

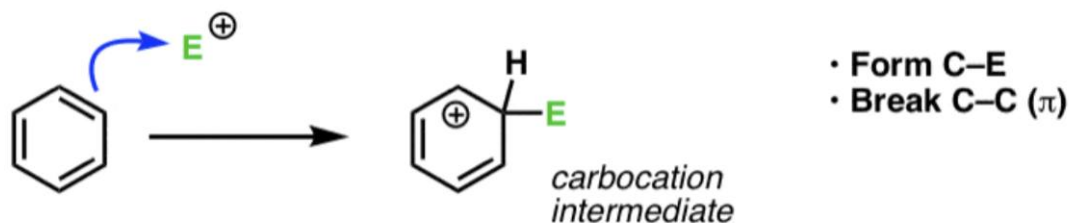
- **Addition** would **disrupt delocalised e^- system**

Mechanism:

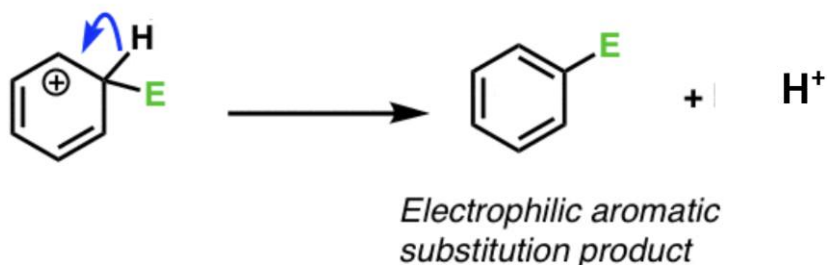
1. **High e^- density** of ring $\Rightarrow$ susceptible to **electrophilic attack**
 - Involves an initial **disruption** to ring (delocalised hence will **resist**) $\therefore$ electrophiles have to be **very powerful**
2. Pair of e^- **leaves** delocalised system $\rightarrow$ bond with **electrophile**
3. **Disrupts** stable delocalised system $\rightarrow$ forms **unstable intermediate**
4. To restore **stability** $\rightarrow$ pair of e^- in **C-H bond moves back** into ring
 - Substitution of **H atom with electrophile**



Attack of electrophile (E) by a pi bond of the aromatic ring



Breakage of C-H bond to restore aromaticity



Thermochemistry:

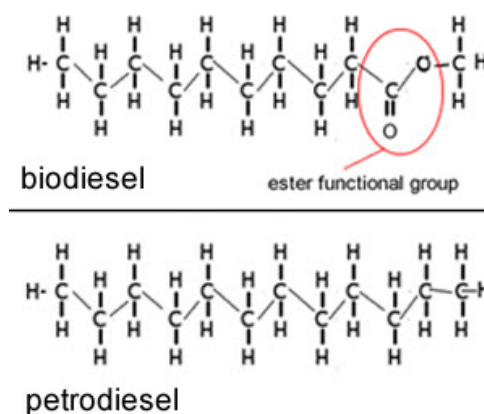
Fuels:

- Substances that react with **oxidising agent (O₂)** - combustion that **releases E** & forms new chemicals
- **Size/length** ↑ → **m.p. & viscosity** ↑ (↑ IMF)
- m.p.: **saturated** < **unsaturated** (↑ IMF)

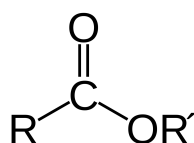
Combustion:

Products:	Complete:	Incomplete:
NO-H₂	- CO₂ - H₂O	- Same as complete - C, CO

Biofuel:



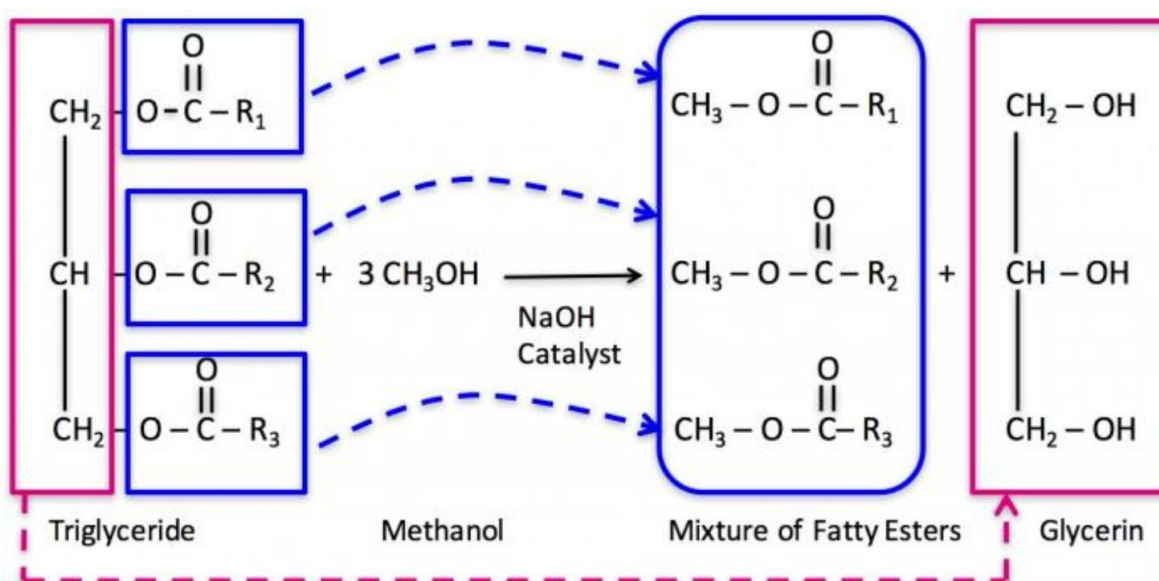
- Produced by **biological fixation of C** over short period of time through **photosynthesis**
- **Ester** functional group:



Advantages:	Disadvantages:
Release less CO ₂ than fossil fuels	Less land for agriculture
Inexpensive	Still releases CO ₂
Direct usage after synthesis (no mining/processing)	

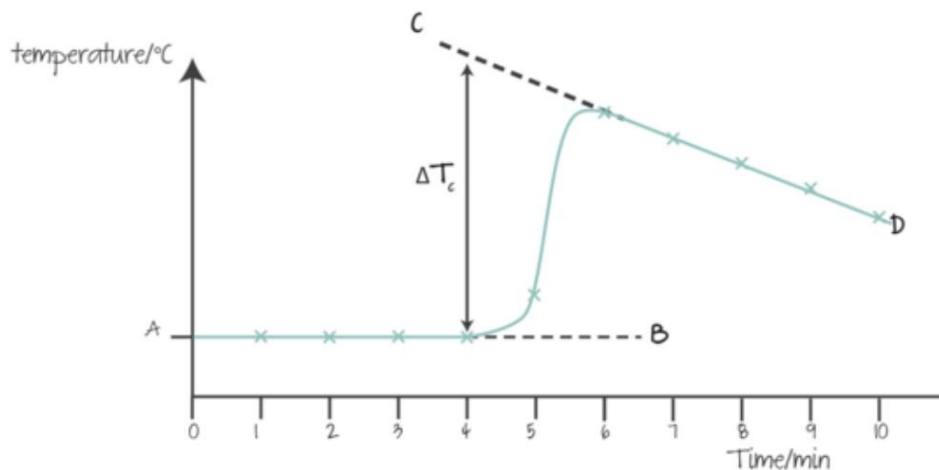
Transesterification:

- Organic reaction in which R group of alcohol ↔ R' group of ester



H:

Measuring:



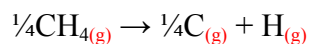
- Extrapolate
- Assumption: **no heat loss** to surrounding

Av. bond H:

- E required to break **1 mol** of a bond in **gaseous** molecule **averaged** over similar compounds
- **Less accurate:**
 - **Averaged**
 - Only in **gaseous** state
- Endo: $X-Y_{(g)} \rightarrow X_{(g)} + Y_{(g)}$
- Exo: $X_{(g)} + Y_{(g)} \rightarrow X-Y_{(g)}$

ignore HOFBrINCl

- E.g. av. bond H for CH_4 :



ΔH:

- $\Delta H = \Sigma(H_{\text{bonds broken}}) - \Sigma(H_{\text{bonds made}})$
→ $\Delta H = \Sigma(\text{bond enthalpies of reactants}) - \Sigma(\text{bond enthalpies of products})$
- $\Delta H = - \frac{Q}{n_{\text{limiting}}}$

ΔH^0 (ΔH under STP):

ΔH_c^0 :

- ΔH when **1 mol** of substance reacts **completely** with **O₂** under **STP** (100 kPa, 298.15 K) (reactants & products in **standard states** with **allotropes defined** (i.e. diamond differs from graphite))
- Reactant has to have coefficient of **1**

ΔH_f^0 :

- ΔH when **1 mol** of compound is **formed** from its **constituent elements** in **standard states**, under **STP**
- **Elements = 0**
- Product has to have coefficient of **1**

ΔH_r^0 (standard ΔH of reaction):

- $\Delta H_r^0 = \Sigma(\Delta H_f^0(\text{products})) - \Sigma(\Delta H_f^0(\text{reactants}))$
- $\Delta H_r^0 = \Sigma(\Delta H_c^0(\text{reactants})) - \Sigma(\Delta H_c^0(\text{products}))$

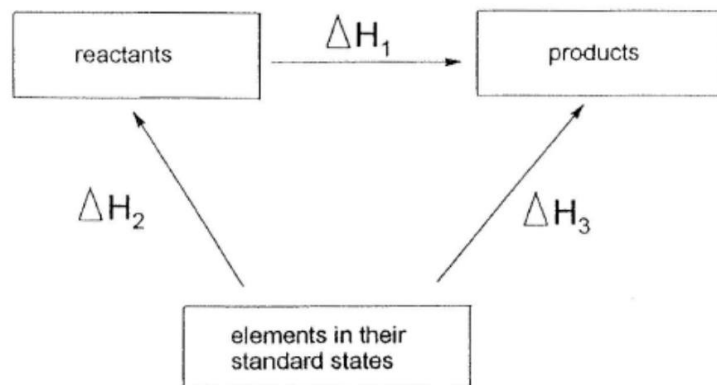
Hess stuff:

Hess's law:

- ΔH is **independent of pathway** between initial & final states

Hess's cycle:

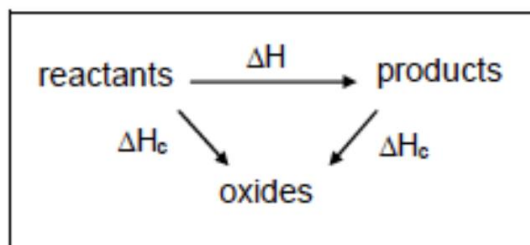
Hess cycles using enthalpies of formation (products - reactants)



$$\Delta H^\ominus = \Sigma (\Delta H_f^\ominus \text{products} - \Sigma \Delta H_f^\ominus \text{reactants})$$

- $\Delta H_1 = -\Delta H_2 + \Delta H_3$ (vectors)

Hess cycles using enthalpies of combustion (reactants - products)



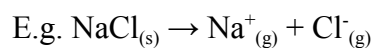
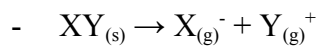
$$\Delta H^\ominus = \Sigma (\Delta H_c^\ominus \text{reactants} - \Sigma \Delta H_c^\ominus \text{products})$$

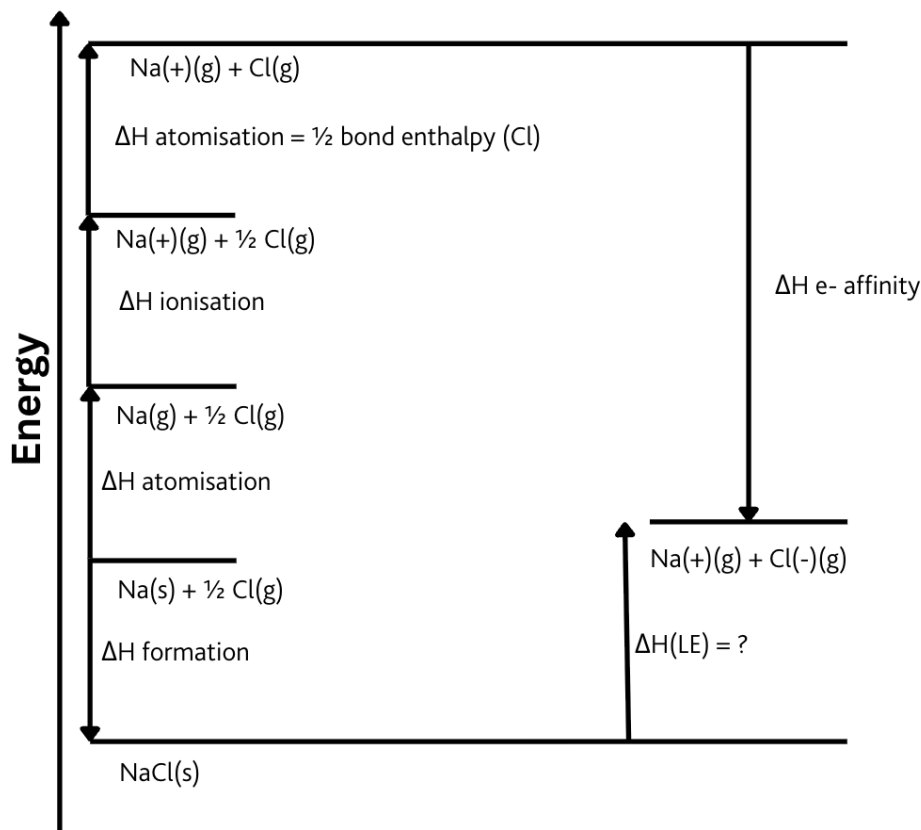
- $\Delta H = \Delta H_{cL} - \Delta H_{cR}$ (vectors)

Born Haber:

- **Standard lattice enthalpy:**

- **ΔH when 1 mol of ionic compound forms into ions under STP**





Gibb's free E:

- $\Delta G^0 = -nFE_{cell}^0$ (Gibb's free E (J) = mol. of e^- (mol)*Faraday's cst. (9.65×10^4) (mol⁻¹)*overall cell potential (V))
 - E.g. ΔG^0 for $Zn^{2+}|Zn||Cu^{2+}|Cu$ cell with $E_{cell}^0 = +1.1V$
 $\Delta G^0 = -(2)(9.65 \times 10^4)(1.1)$
 $= -212 \text{ kJmol}^{-1}$
- $\Delta G^0 = \Delta H^0 - T\Delta S^0$ (Gibb's free E (J) = enthalpy change (kJmol⁻¹) - temp. (K)*entropy (JK⁻¹mol⁻¹))
 - *Must convert S from J $\rightarrow$ kJ*
 - Spontaneity = temp. dependent
- For reaction to be spontaneous, $\Delta G \rightarrow (-)$

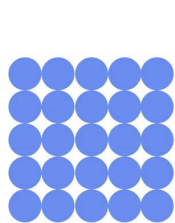
Entropy:

- Measure of **disorder**

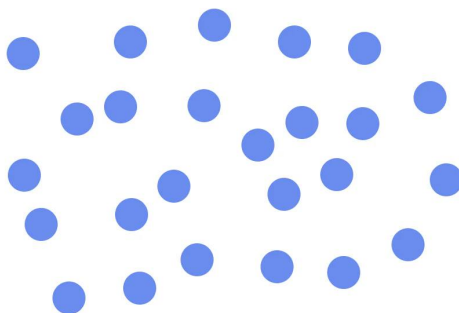
- $\Delta S = \Sigma(\Delta S_{\text{products}}) - \Sigma(\Delta S_{\text{reactants}})$
 - E.g. ΔS for $\text{CH}_{4(g)} + \text{H}_2\text{O}_{(g)} \rightarrow \text{CO}_{(g)} + 3\text{H}_{2(g)}$

$$\Delta S = [\text{S}(\text{CO}) + 3\text{S}(\text{H}_2)] - [\text{S}(\text{CH}_4) + \text{S}(\text{H}_2\text{O})]$$

$$\Delta S = 205 - 64.07 = 140.93$$



Low entropy



High entropy

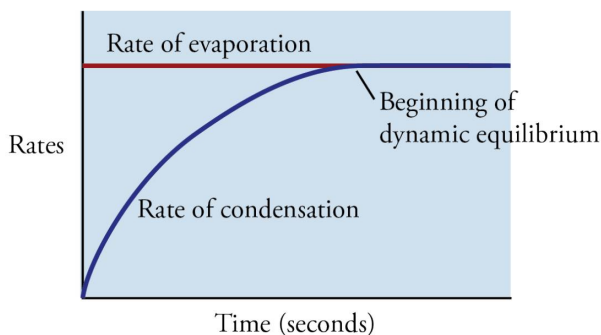
General trends:

- $S_g > S_l > S_s$
- **Greater** for **larger atoms** & molecules with **more atoms**
- Usually **increases** when pure liquid/solid **dissolves** in solvent
- Ionic compounds: **weaker ions** $\rightarrow$ **greater S**

Eq. (dynamic):

Requirements:

- **Reversible** rxn
 - **Closed system** (reactants/products cannot escape)
- $\rightarrow$ i.e. evaporation:



- **Rate = fixed** at a specific temp.
- For closed systems (vapour **can't escape**) $\Rightarrow$ **increase** in vapour **conc.**
- Eventually vapours colliding with container surface increase **rate of condensation** $\rightarrow$ **dynamic eq.**

Characteristics:

- **Rate of the forward = backward**
- **Dynamic: forward & reverse rxns are still continuing**
- **Macroscopic** properties remain **cst.** (e.g. conc. of reactants/products = cst.)

Le Châtelier's principle:

- If **conditions** of system in **eq.** are **changed** $\rightarrow$ **position of eq.** moves to **oppose** the **change** (i.e. conc. of chemical A increases $\rightarrow$ position shifts to decrease it)

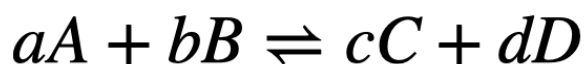
Gaseous systems:

- **Increase in pressure** $\rightarrow$ **eq.** moves to side with **fewer** moles/vice versa
- If **same no. of molecules** $\rightarrow$ **eq. can't** move to oppose

Exo vs. endo:

- **Exo + temp. increase** $\rightarrow$ eq. shifts **left (reactants)**
 $\rightarrow$ **exo: reactants $\rightarrow$ products + heat** $\therefore$ increase in temp. = increase in [right side]
- **Endo + temp. increase** $\rightarrow$ eq. shifts **right (products)**
 $\rightarrow$ **endo: reactants + heat $\rightarrow$ products** $\therefore$ increase in temp. = increase in [left side]

Eq. cst K_c :



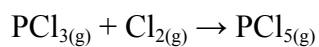
$$K_c = \frac{[C]_{eq}^c \times [D]_{eq}^d}{[A]_{eq}^a \times [B]_{eq}^b}$$

- If chemical = **solid** $\rightarrow$ **ignore** from eq.
- Purely dependent on **TEMP.**
- E.g.:

A student placed 0.20 mol of $\text{PCl}_{3(g)}$ and 0.10 mol of $\text{Cl}_{2(g)}$ into a 1.00 dm^3 flask. The reaction which produced $\text{PCl}_{5(g)}$, was allowed to come to equilibrium, at which time the flask contained 0.12 mol of PCl_3 .

What is the value of K_c ?

1. Rxn eq.:



1:1:1

2. Make & complete ICE table (base on **molar ratio**):

	Chemicals:		
	$\text{PCl}_{3(g)}$	$\text{Cl}_{2(g)}$	$\text{PCl}_{5(g)}$
Initial	0.20	0.10	0 $\leftarrow$ products = 0 for initial
Change	-0.08	-0.08	+0.08
Eq.	0.12	0.02	0.08

3. Calculate K_c :

$$K_c = \frac{0.08}{0.12 \times 0.02} = 33.3 \text{ mol}^{-1} \text{ dm}^3$$

Magnitudes:

- 10^{-10} = **not occurred**
- 10^{-3} = eq. lies to the **left**
- **1** = **neither**
- ...

Altering reaction:

Change:	Effect on K_c
Inverse rxn	$1/K_c$
Double coefficients	K_c^2
Halving coefficients	$K_c^{1/2}$
Combine forward & backward	$K_{c1} \cdot K_{c2}$

Rxn. quotient Q:

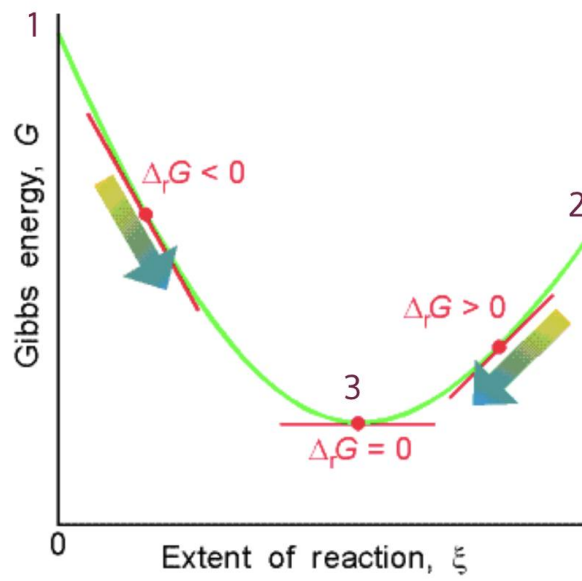
- Relation to K_c :

Relation:	<	=	>
Net rxn. direction	$\rightarrow$	N/A	$\leftarrow$
[reactants]/[products]	More reactants	Same	More products

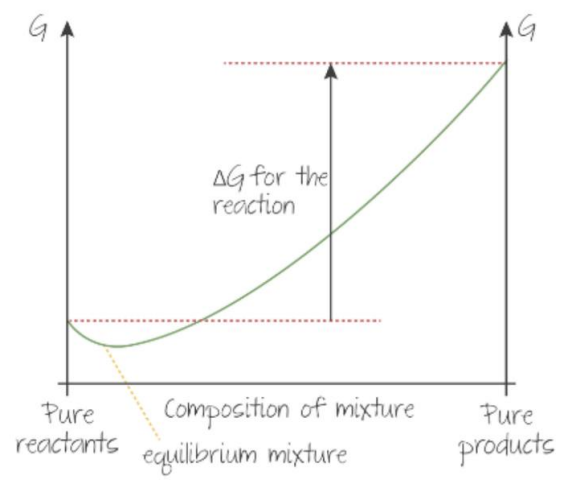
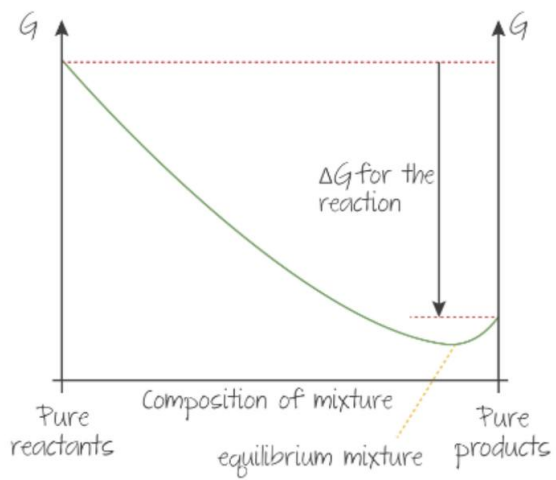
Gibb's free E:

- $\Delta G = \Delta G^0 + RT \ln Q$
- $\Delta G^0 = - RT \ln K$
- $K_c = e^{-\frac{G^0}{RT}}$

Graphical:



- 1: start of rxn ($\Delta G = (-)$)
- 2: end of rxn ($\Delta G = (+)$)
- 3: eq.



- Rxn. with **large negative ΔG** will be mainly **products** at eq.; vice versa

Redox reactions:

	Oxidation	Reduction
Oxygen	Gain	Loss
Hydrogen	Loss	Gain
e ⁻	Loss	Gain
Oxidation state	Increase	Decrease

Oxidation states:

Elements:	Oxidation state:	Exceptions:
Most grp 1s	+1	N/A
H	+1	-1 → less electronegative (e.g. metal hydrides)
O	-2	-1 → peroxides (H ₂ O ₂)
		+2 → when combined with more electronegative element (eg. OF ₂)
F	-1	N/A
Cl	-1	when combined with O/F

Trend:

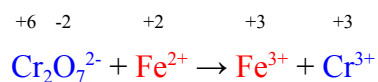
Metals:	Non-metals:
- Larger = better reducing agent	- Smaller = better oxidising agent

(lower IE)	(more exo e ⁻ affinity)
------------	---

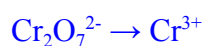
Balancing in acidic:

Cr₂O₇²⁻ react with Fe²⁺ to form Fe³⁺ & Cr³⁺ under acidic conditions

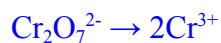
1. Assign oxidation numbers



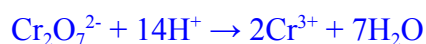
2. Write half equations (unbalanced):



3. Use coefficients to balance atoms (ignore H & O):



4. Use H₂O & H⁺ to balance H & O atoms:



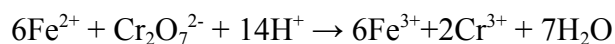
5. Balance charges:



6. Balance e⁻ to cancel out:

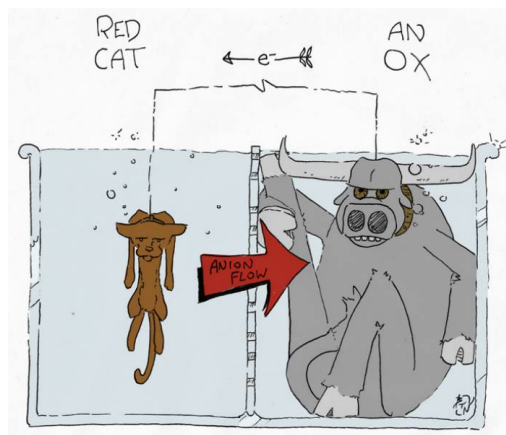
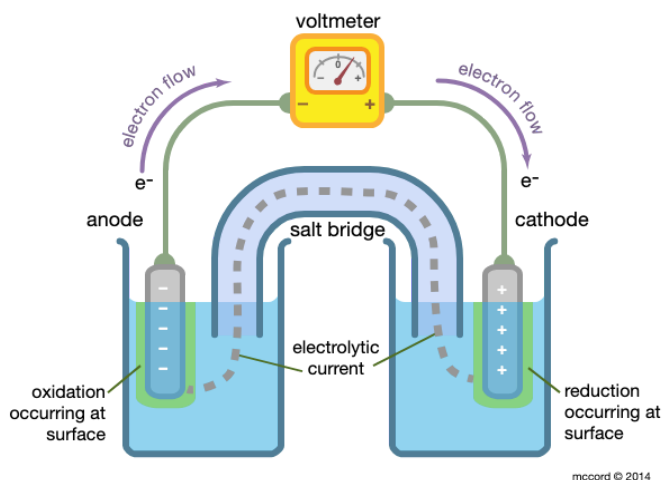


7. Final:



Voltaic cells:

- Converts **chemical** E → **electrical** E (spontaneous redox rxns.)



- **Salt bridge:**
 - Complete circuit by allowing flow of ions
 - Neutralise build of charges $\rightarrow (-) \rightarrow$ anode; $(+) \rightarrow$ cathode

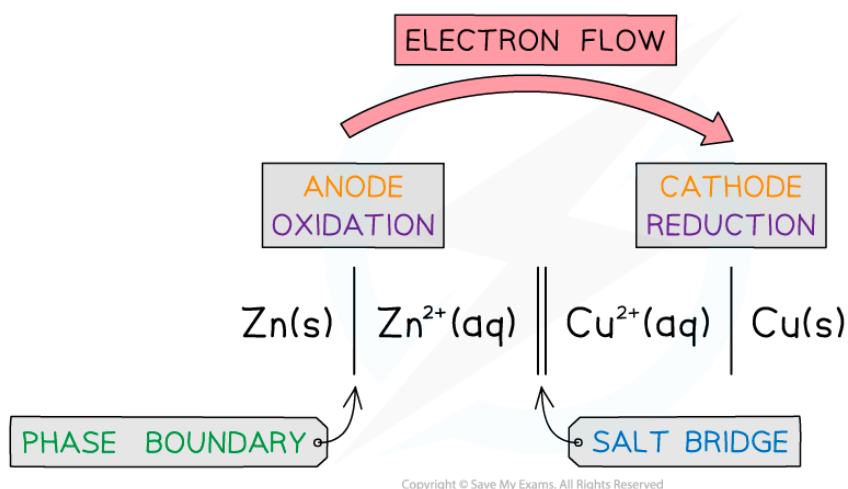
Anode:	Cathode:
(-)	(+)
Oxidation	Reduction
Decrease in mass	Increase in mass

- **Overall cell potential:**
 - $E_{cell}^{\ominus} = E_{cathode}^{\ominus} - E_{anode}^{\ominus}$ (overall cell potential (V) = cell potential_{cathode} (V) - cell potential_{anode} (V))
 - E.g. find $E_{cell}^{\ominus}$ for:

$$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightleftharpoons \text{Cu}(\text{s}) \quad +0.34 \text{ V}$$

$$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^{-} \rightleftharpoons \text{Zn}(\text{s}) \quad -0.76 \text{ V}$$
 - $E_{cell}^{\ominus} = E_{cathode}^{\ominus} - E_{anode}^{\ominus}$
 - Cathode = Cu; anode = Zn
 - $E_{cell}^{\ominus} = 0.34 - (-0.76) = 1.10 \text{ V}$
 - For reaction to occur, $E^{\ominus} = (+)$ (not 0 or (-))

Cell convention:



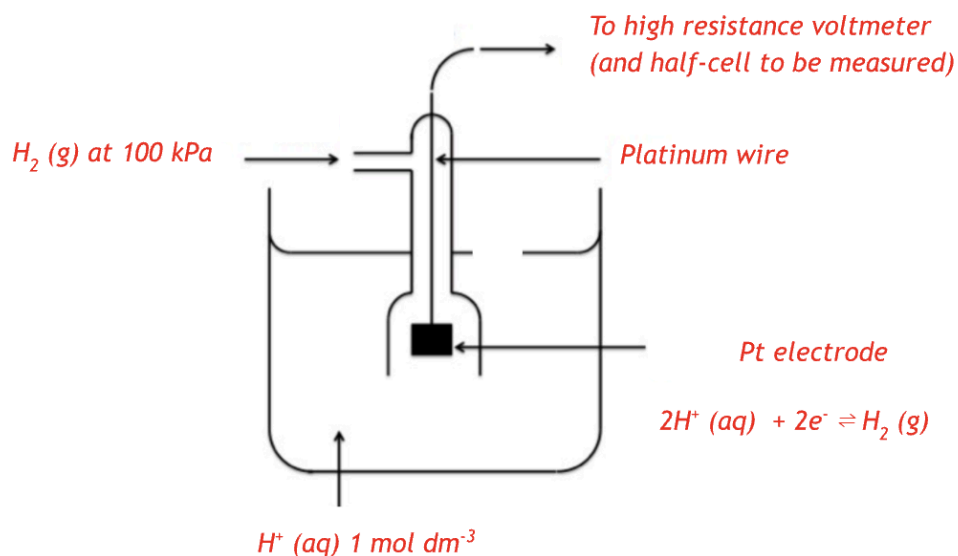
add other ions in solution with “,”

Voltaic vs. electrolytic:

Voltaic:	Electrolytic:
Chemical → electrical	Electrical → chemical
Based on redox → spontaneous	Non-spontaneous → only occurs when energy is supplied
Anode = (-); cathode = (+)	Anode = (+); cathode = (-)
Anode = oxidation; cathode = reduction	

Standard electrode potential:

- V obtained when **half-cell** connected to **SHE (standard hydrogen electrode)** under **STP (298K, 1mol dm⁻³)**
 - **SHE: standard electrode** used to **compare** all other half cells against (standard electrode potential = 0V at **298.15K, 1 mol dm⁻³ H⁺_(aq) & H_{2(g)}** at **100 kPa**); inert **Pt** electrode = site for e⁻ transfer → **2H⁺_(aq) + 2e⁻ → H_{2(g)}**



19. Standard reduction potentials at 298.15 K

Oxidized species	Reduced species	E° / V
$\text{Li}^+(\text{aq}) + e^-$	$\rightleftharpoons \text{Li}(\text{s})$	-3.04
$\text{K}^+(\text{aq}) + e^-$	$\rightleftharpoons \text{K}(\text{s})$	-2.93
$\text{Ca}^{2+}(\text{aq}) + 2e^-$	$\rightleftharpoons \text{Ca}(\text{s})$	-2.87
$\text{Na}^+(\text{aq}) + e^-$	$\rightleftharpoons \text{Na}(\text{s})$	-2.71
$\text{Mg}^{2+}(\text{aq}) + 2e^-$	$\rightleftharpoons \text{Mg}(\text{s})$	-2.37
$\text{Al}^{3+}(\text{aq}) + 3e^-$	$\rightleftharpoons \text{Al}(\text{s})$	-1.66
$\text{Mn}^{2+}(\text{aq}) + 2e^-$	$\rightleftharpoons \text{Mn}(\text{s})$	-1.18
$\text{H}_2\text{O}(\text{l}) + e^-$	$\rightleftharpoons \frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-(\text{aq})$	-0.83
$\text{Zn}^{2+}(\text{aq}) + 2e^-$	$\rightleftharpoons \text{Zn}(\text{s})$	-0.76
$\text{Fe}^{2+}(\text{aq}) + 2e^-$	$\rightleftharpoons \text{Fe}(\text{s})$	-0.45
$\text{Ni}^{2+}(\text{aq}) + 2e^-$	$\rightleftharpoons \text{Ni}(\text{s})$	-0.26
$\text{Sn}^{2+}(\text{aq}) + 2e^-$	$\rightleftharpoons \text{Sn}(\text{s})$	-0.14
$\text{Pb}^{2+}(\text{aq}) + 2e^-$	$\rightleftharpoons \text{Pb}(\text{s})$	-0.13
$\text{H}^+(\text{aq}) + e^-$	$\rightleftharpoons \frac{1}{2}\text{H}_2(\text{g})$	0.00
$\text{Cu}^{2+}(\text{aq}) + e^-$	$\rightleftharpoons \text{Cu}^+(\text{aq})$	+0.15
$\text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2e^-$	$\rightleftharpoons \text{H}_2\text{SO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+0.17
$\text{Cu}^{2+}(\text{aq}) + 2e^-$	$\rightleftharpoons \text{Cu}(\text{s})$	+0.34
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) + 2e^-$	$\rightleftharpoons 2\text{OH}^-(\text{aq})$	+0.40
$\text{Cu}^+(\text{aq}) + e^-$	$\rightleftharpoons \text{Cu}(\text{s})$	+0.52
$\frac{1}{2}\text{I}_2(\text{s}) + e^-$	$\rightleftharpoons \text{I}^-(\text{aq})$	+0.54
$\text{Fe}^{3+}(\text{aq}) + e^-$	$\rightleftharpoons \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{Ag}^+(\text{aq}) + e^-$	$\rightleftharpoons \text{Ag}(\text{s})$	+0.80
$\frac{1}{2}\text{Br}_2(\text{l}) + e^-$	$\rightleftharpoons \text{Br}^-(\text{aq})$	+1.09
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2e^-$	$\rightleftharpoons \text{H}_2\text{O}(\text{l})$	+1.23
$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6e^-$	$\rightleftharpoons 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$	+1.36
$\frac{1}{2}\text{Cl}_2(\text{g}) + e^-$	$\rightleftharpoons \text{Cl}^-(\text{aq})$	+1.36
$\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5e^-$	$\rightleftharpoons \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$	+1.51
$\frac{1}{2}\text{F}_2(\text{g}) + e^-$	$\rightleftharpoons \text{F}^-(\text{aq})$	+2.87

Strongest reducing agent

Strongest oxidising agent

✓ What to Use (Quick Decision Guide)

Situation	Anode (Oxidation)	Cathode (Reduction)
Acidic	$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$	$2H^+ + 2e^- \rightarrow H_2$
Neutral	$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$	$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$
Alkaline	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$	$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$
Oxygen present (e.g. fuel cell)	—	$\frac{1}{2}O_2 + H_2O + 2e^- \rightarrow 2OH^-$

Transition metals:

- **Incomplete d subshell**
- **Exceptions: Sc, Zn:**
 - **Zn = full d subshell**
 - **Sc = only 1 oxidation state (+3) → Ar electronic configuration (energetically unfavourable to lose more)**

Properties:

Variable oxidation states:

- Occupy both **s & d orbitals** → **successive I.E. close in values**

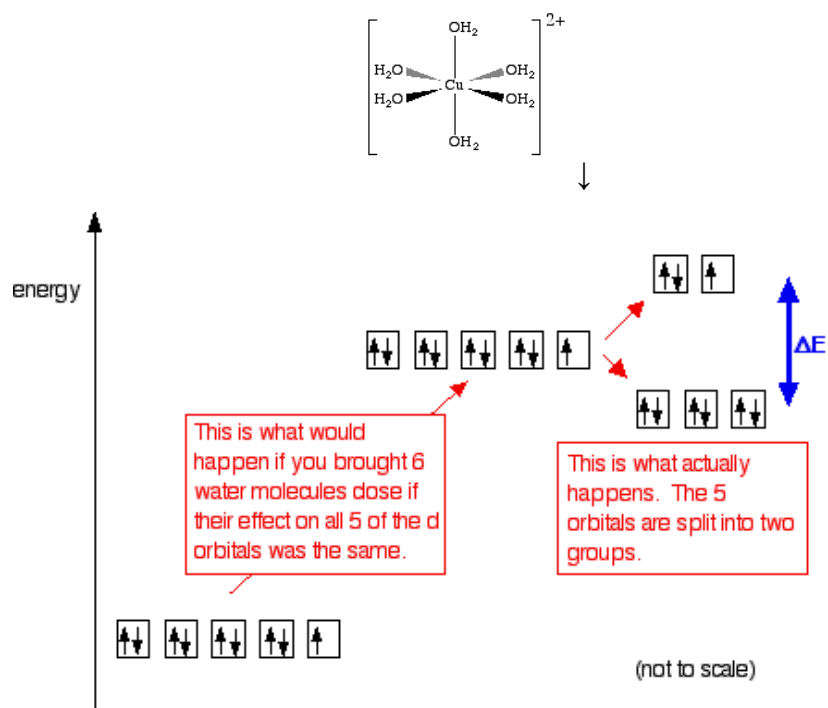
Oxidation States Of First Transition Series Common ● ; Less Common ●

+7					●					
+6				●	●	●				
+5			●	●	●	●				
+4		●	●		●	●	●			
+3	●	●	●	●	●	●	●	●		
+2		●	●	●	●	●	●	●	●	
+1									●	●
	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn

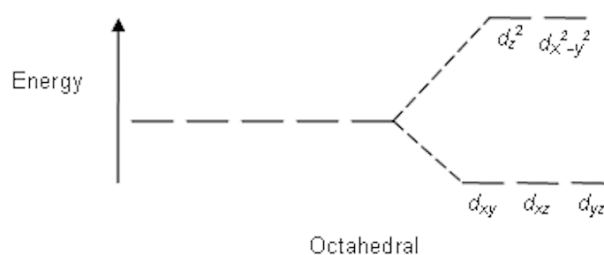
Forms complex ions:

- **Coordinate covalent bonds** formed by **ligands**

- **Ligands** give **lone pair** to transition cation
- Create electric field around ion & interferes with d-orbital
- **Ligands cause d-orbital splitting:**
 - Free metal ion $\rightarrow$ all 5 d-orbitals **equal in E**
 - Lone pair e^- from **ligand** **repel d-orbital** $e^- \rightarrow$ some **orbitals** get **pushed higher (e_g) & lower (t_{2g}) in E** (depends on **shape** of complex)
 - E.g.:



Crystal field splitting E:



- **Difference in E** between **higher & lower d-orbitals** after they **split** due to **ligands**
- Influenced by:
 - **Geometry** of complex
 - Identity of **metal**:

- Increases **down** a group with metals of **same** oxidation state

Group 9 complex	$\Delta_o / \text{cm}^{-1}$
$[\text{Co}(\text{NH}_3)_6]^{3+}$	22900
$[\text{Rh}(\text{NH}_3)_6]^{3+}$	34100
$[\text{Ir}(\text{NH}_3)_6]^{3+}$	41100

- **Oxidation state** of ion:
 - **Increase with oxidation state:**
 - **Charge increases** → more **attraction** to ligand
 - Metal-ligand **distance decreases** → **more overlap** of orbitals hence **repulsion**
 - Nature of **ligand**:

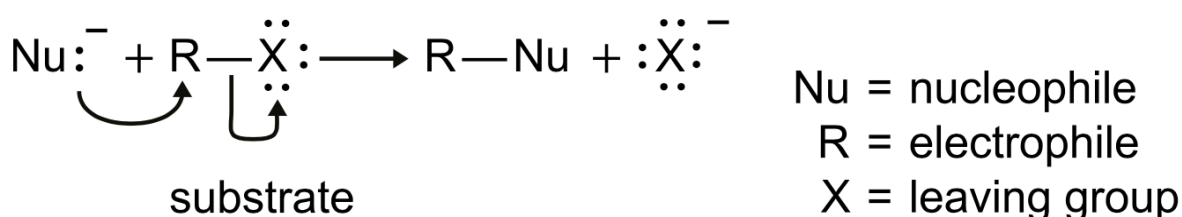
← strong weak →	
CN ⁻ , CO	NO ₂ ⁻ en NH ₃ H ₂ O ox OH ⁻ F ⁻ SCN ⁻ , Cl ⁻ Br ⁻ I ⁻
Relative ligand field strengths	

Low spin:	High spin:
Strong field ligand → higher repulsion	Weak field ligand → smaller repulsion
Higher Δ_o	Lower Δ_o
Greater charge density	Lower charge density
Energetically favourable to pair up	Energetically favourable to spread out
Diamagnetic	Paramagnetic

Magnetic properties:

- **Ferromagnetism:** long range alignment of unpaired e^- → strong magnetic interactions (Fe, Co, Ni only)
- **Paramagnetism:** occurs in substances with unpaired e^- that temporarily aligns with magnetic fields
 - Strongest = **highest** no. of unpaired e^-
- **Diamagnetism:** no unpaired e^- that repels magnetism

Lewis base:	Lewis acid:
e^- donor	e^- acceptor
Nucleophile	Electrophile
Ligand	Metal ion



Coloured compounds:

- Absorbed wavelengths (of **complementary** color) are used to **promote e^- from lower → higher energy d-orbitals**
 - E.g. if **blue** is seen, **orange** is absorbed
- Strong field ligands have **greater charge density** ∴ **higher splitting energy**

← strong weak →											
CN ⁻ , CO	NO ₂ ⁻	en	NH ₃	H ₂ O	ox	OH ⁻	F ⁻	SCN ⁻ , Cl ⁻	Br ⁻	I ⁻	
Relative ligand field strengths											

Act as catalysts:

- Provide surface for **ions to lose/gain e^- → variable oxidation states**

- Ions of **transition** elements can react with **1 reactant (A)** → **increase oxidation state** & then react with **other reactant (B)** → return to its **original oxidation state**
- **2 reactants gain/lose e^-** → **easier to react**

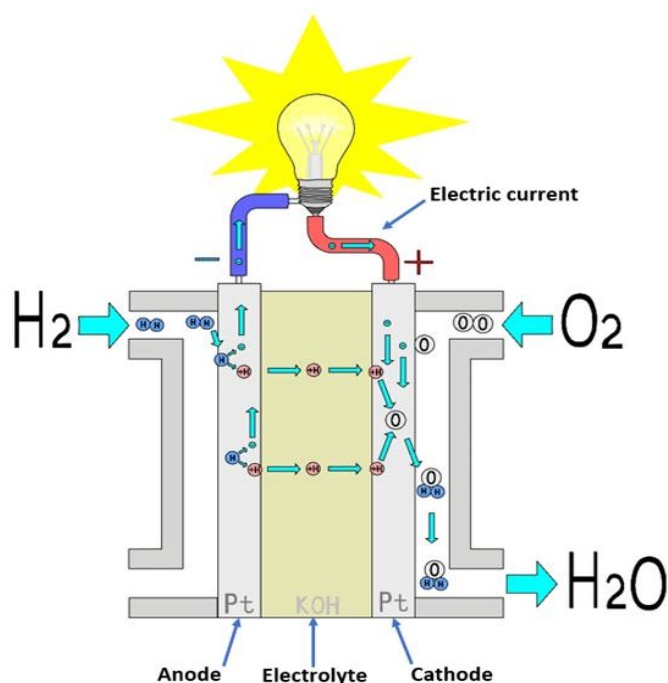
High mp:

- Form ions of **varying oxidation states** → **large** number of **delocalised e^-** surrounding cations in metallic lattice $\therefore$ increased strength of metallic bond

Fuel cells:

- Type of **primary** cell → generate **electricity** by **consuming reactants** (typically **replenished** rather than being stored **internally**)
- Don't have a mechanism to **store electrical E generated**
- **Different** from other primary cells → can operate **continually** provided there's a source of fuel

Hydrogen:

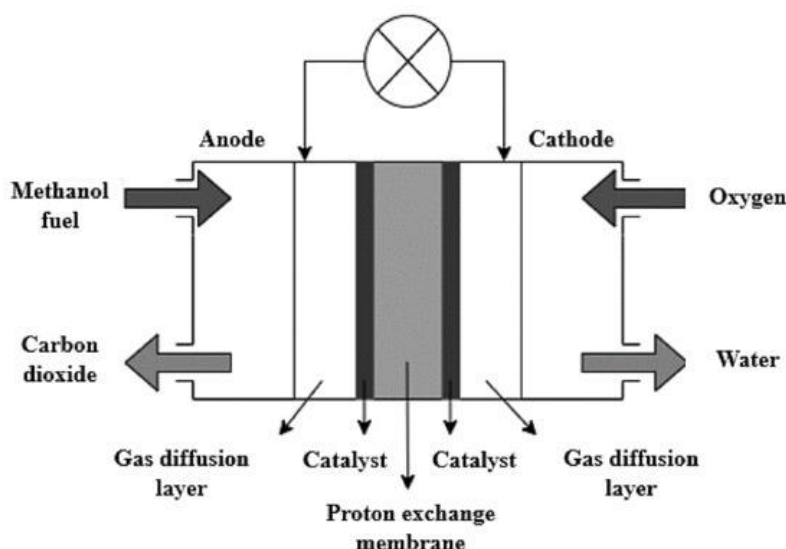


Anode:

Cathode:

$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$
$\text{O}_2 + 2\text{H}_2 \rightarrow 2\text{H}_2\text{O}$	

Methanol:



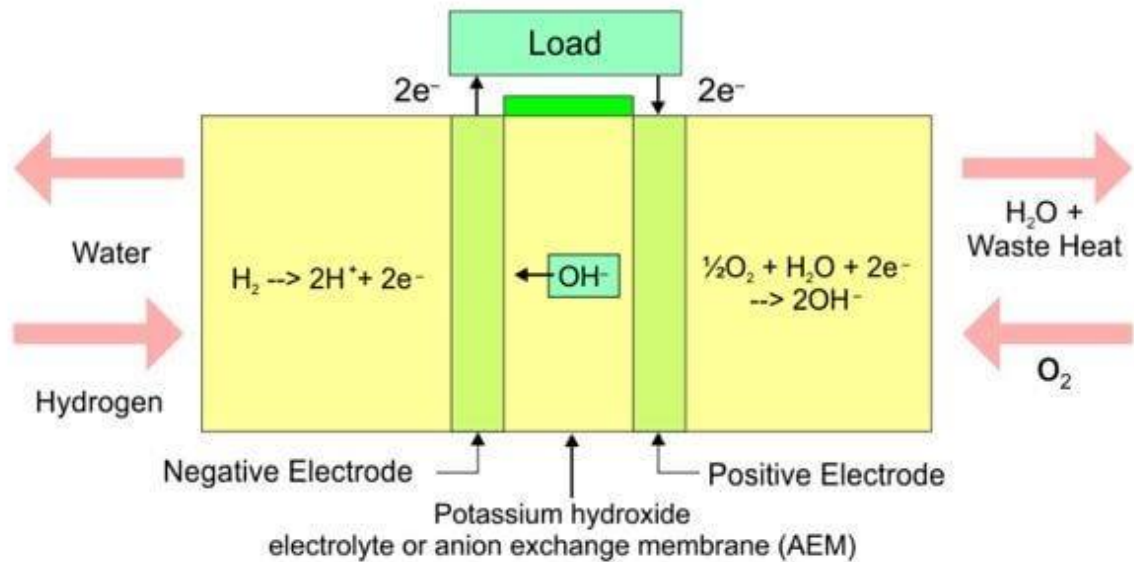
Anode:	Cathode:
$\text{CH}_3\text{OH} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + 6\text{H}^+ + 6\text{e}^-$	$\frac{3}{2}\text{O}_2 + 6\text{H}^+ + 6\text{e}^- \rightarrow 3\text{H}_2\text{O}$
$\text{CH}_3\text{OH} + \frac{3}{2}\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$	

Comparison with hydrogen:

<u>Advantages:</u>	<u>Disadvantages:</u>
Easier to store & transport than hydrogen	Toxic & highly flammable
Doesn't require high pressure/temp.	Most commonly made from non-renewable fossil fuels
Can be produced from renewable resources through fermentation → cleaner than	Fuel cell produces greenhouse gas

hydrogen as less greenhouse gases	
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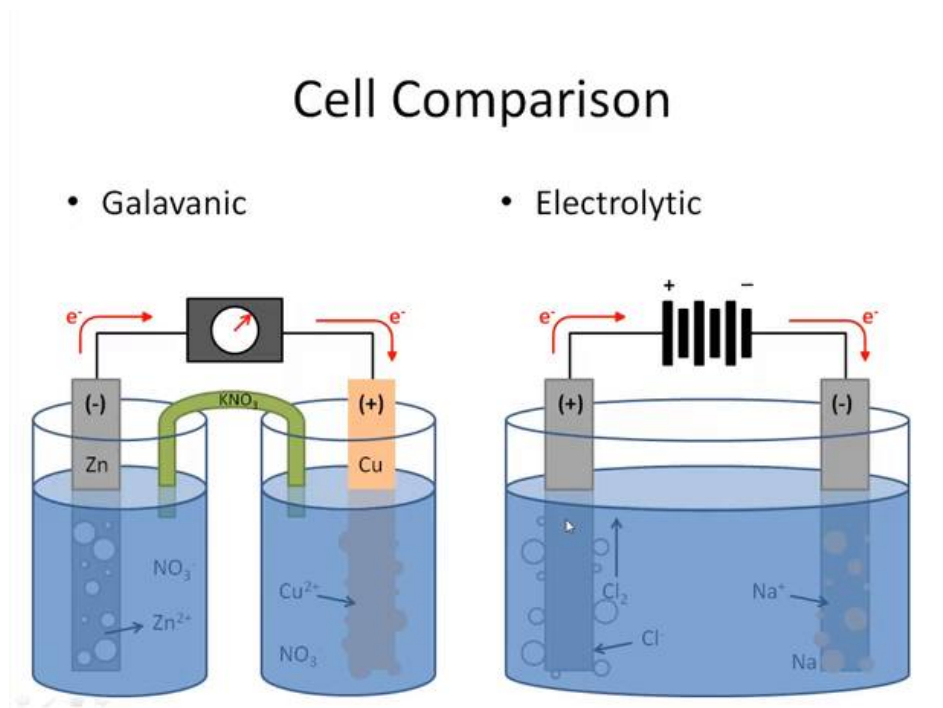
Alkaline based:



- Porous carbon electrodes with Pt/Pd catalyst

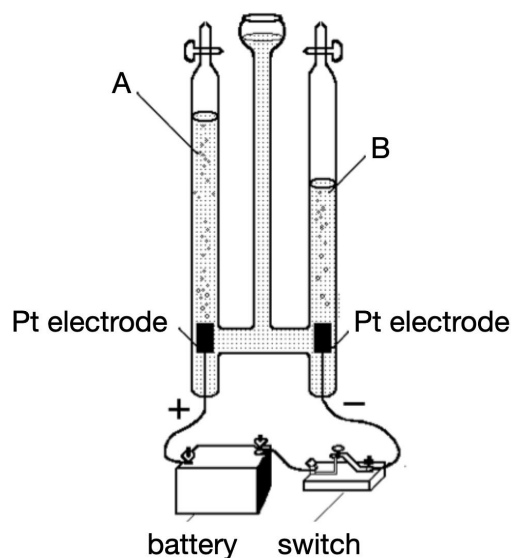
Anode:	Cathode:
$2\text{H}_2 + 4\text{OH}^- \rightarrow 4\text{H}_2\text{O} + 4\text{e}^-$	$2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^- \rightarrow 4\text{OH}^-$
$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$	

Electrolytic cells:



- Converts **electrical** → **chemical** by bringing about **non-spontaneous** reactions/changes
- **Active** electrodes: **take part in reaction - consumed/deposited** (i.e. electroplating, refining)
- **Inert** electrodes: **do not take part in reactions** (i.e. electrolysis of water)
- Factors affecting mass of products:
 - **Time**
 - **Charge of ion**
 - **Current**
 - **SA** of electrode
 - **Temp.** of solution

Hoffman voltameter:

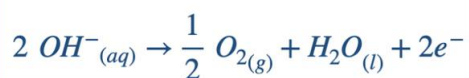
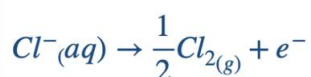
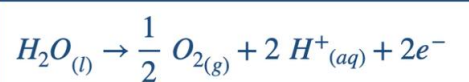


- Electrolysis of **water**
- Instead of pure water, solution of **sulphuric acid** is used:
 - Water only **self-ionises** to a very **small** extent $\therefore$ **low ion conc.**
 - **H₂SO₄ fully dissociates in water** $\therefore$ provides **ions** for **conductivity**

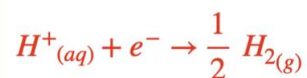
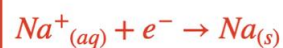
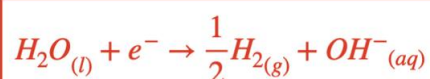
Anode:	Cathode:
$2\text{H}_2\text{O} \rightarrow \text{O}_{2(g)} + 4\text{H}^+ + 4\text{e}^-$	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_{2(g)}$
$2\text{H}_2\text{O(l)} \rightarrow 2\text{H}_{2(g)} + \text{O}_{2(g)}$	

Aqueous NaOH:

Anode (oxidation)



Cathode (reduction)



Anode:	Cathode:
$\frac{1}{2} \text{Cl}_2(\text{g}) + \text{e}^- \rightleftharpoons \text{Cl}^-(\text{aq}) \quad E^\ominus = +1.36 \text{ V}$ $4\text{H}^+(\text{aq}) + \text{O}_2(\text{g}) + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}(\text{l}) \quad E^\ominus = +1.23 \text{ V}$ $(\text{Cl}^-(\text{aq}) \rightarrow \frac{1}{2} \text{Cl}_2(\text{g}) + \text{e}^-)$ $(2\text{H}_2\text{O}(\text{l}) \rightarrow 4\text{H}^+(\text{aq}) + \text{O}_2(\text{g}) + 4\text{e}^-)$	$\text{Na}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{Na}(\text{s}) \quad E^\ominus = -2.71 \text{ V}$ $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \quad E^\ominus = -0.83 \text{ V}$
H₂O has lower reduction potential → less (-) $E^\ominus \therefore \text{O}_{2(\text{g})}$ produced	H₂O has higher reduction potential → less (-) $E^\ominus \therefore \text{H}_{2(\text{g})}$ produced

	Dilute:	Concentrated:
Anode:	O ₂	Cl ₂
Cathode:	H ₂	H ₂

- **Concentrated** → Cl⁻ in **high enough conc.** to be **major** product
- **Dilute** → basically electrolysis **water**

Electroplating:

- **Coating** a metal object with a thin layer of **another metal** using electrolysis
 - Improve **appearance/corrosion resistance**

Component:	Material/function:
Cathode	To be electroplated
Anode	Pure metal used for coating
Electrolyte	Contains metal ions of plating metal → maintains est. ion conc. in electrolyte
Power supply	Supply e⁻ to cathode

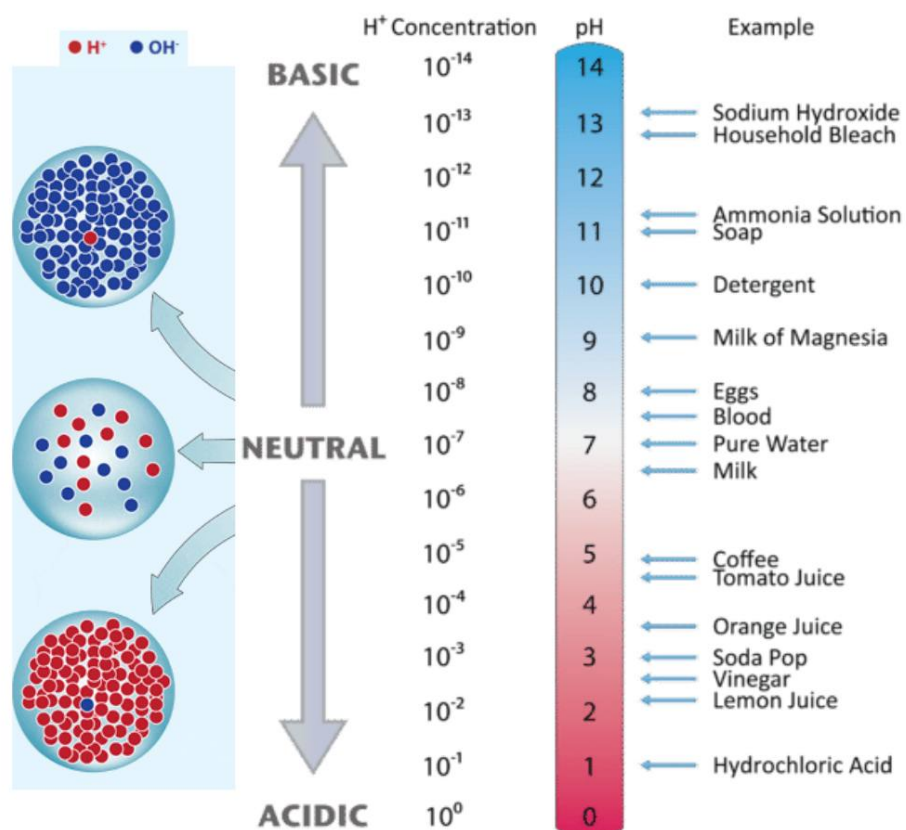
Refining:

Component:	Material/function:
Cathode	Small pure metal (desired)
Anode	Impure/raw metal
Electrolyte	Aq. solution of metal's salt

- Impurities fall off **anode** as **sludge**

Acids & bases:

pH scale:



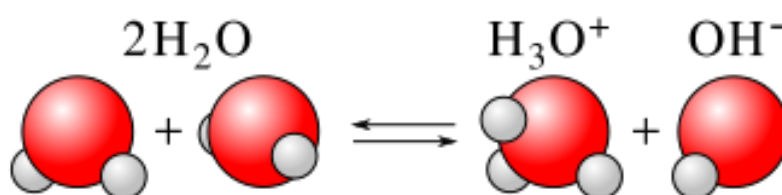
- Neutral: $[H^+] = [OH^-]$

Advantages over exp. in conc.:

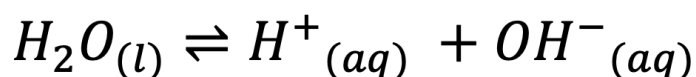
- More **intuitive** range

K_w :

- **Self-ionisation of water:**
- Eq. system:



- Simplified:

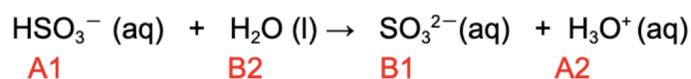


- IRL: lone H^+ will be attracted to electronegative oxygen in $H_2O \rightarrow H_3O^+$
- $K_w = [H^+][OH^-] \leftarrow [H_2O]$ not included as heterogenous & large conc. compared to ions
- $K_w = 1 \cdot 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ at 298.15K
- Eq. cst. $\therefore$ temp. dependent
 - Endo + temp. increase $\rightarrow K_w$ increase $\rightarrow [H^+]$ increase $\therefore$ pH decrease because of (-) sign
- Calculating pH:
 - $\rightarrow [H^+] = [OH^-]$, as water is **neutral**
 - $\rightarrow \therefore K_w = [H^+]^2$
 - $\rightarrow \text{pH} = -\log_{10}[H^+]$

Different definitions:

Acids:

- **Arrhenius: dissolve** to form H^+
 - Limitations:
 - Substances need to be an **aqueous solution** to form **ions**
 - **Bases** need to be **soluble**
 - **B-L: p^+ donor**
 - **Conjugate base = same specie** but with **1 less H^+**
- E.g.:



- **Lewis: e^- acceptor (electrophile)**

Acidic reactions:

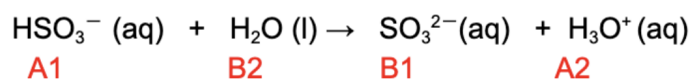
- **Metal + acid $\rightarrow$ salt + H_2**
- **Metal oxide/hydroxide + acid $\rightarrow$ salt + H_2O**
- **Metal carbonate + acid $\rightarrow$ salt + H_2O + CO_2**

Oxides:

- **Metal oxide + water $\rightarrow$ metal hydroxide (basic)**
- **Non-metal oxide + water $\rightarrow$ acid**
- **Exceptions: NO , CO , $\text{N}_2\text{O} \Rightarrow$ neutral**

Bases:

- **Arrhenius: dissolve** to form OH^-
 - **B-L: p^+ acceptor**
 - **Conjugate acid = same specie** but with **1 more H^+**
- E.g.:



- **Lewis: e^- donor (nucleophile)**

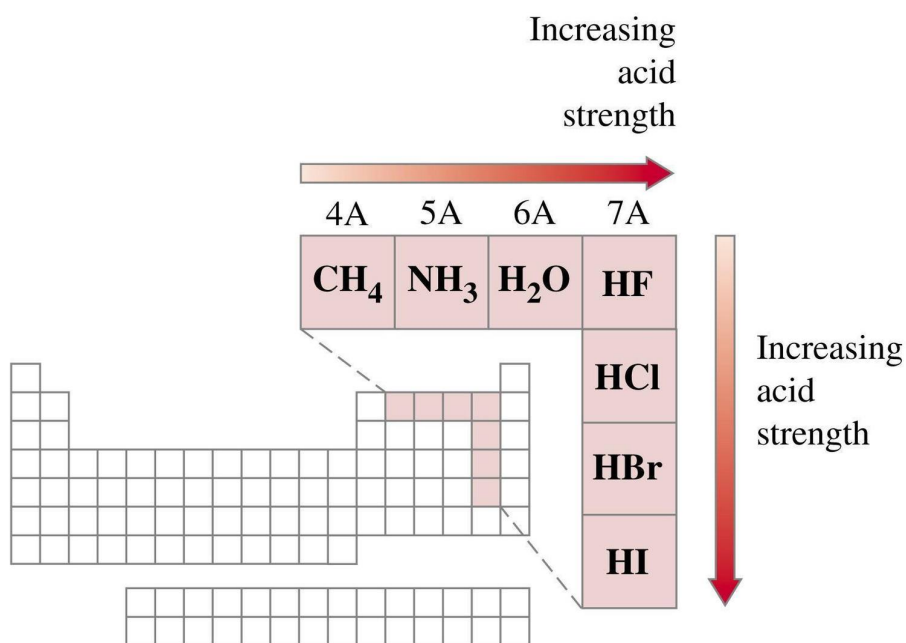
Strength:

- **Concentration** (how many **mol.**) $\neq$ **strength** (how much **dissociation in water**)
- **Strong** = **complete** dissociation in water; $\text{HA} \rightarrow \text{H}^+ + \text{A}^-$
- **Weak** = **partial** dissociation in water; $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$

Strong & weak...

Acids:

- Acid strength $\propto \frac{1}{H-X \text{ bond dissociation energy}} \propto \text{anion radius (shielding)}$
- $K_a = \frac{[\text{H}_{(aq)}^+][\text{A}_{(aq)}^-]}{[\text{HA}_{(aq)}]}$
- $pK_a = -\log_{10} K_a$



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Strong:	Weak:
HCl	HF
HNO ₃	H ₂ S
H ₂ SO ₄	CH ₃ COOH
HBr	...

HI	
HClO ₄	

Bases:

$$- K_b = \frac{[BH^+_{(aq)}][OH^-_{(aq)}]}{[B_{(aq)}]}$$

$$- pK_b = -\log_{10} K_b$$

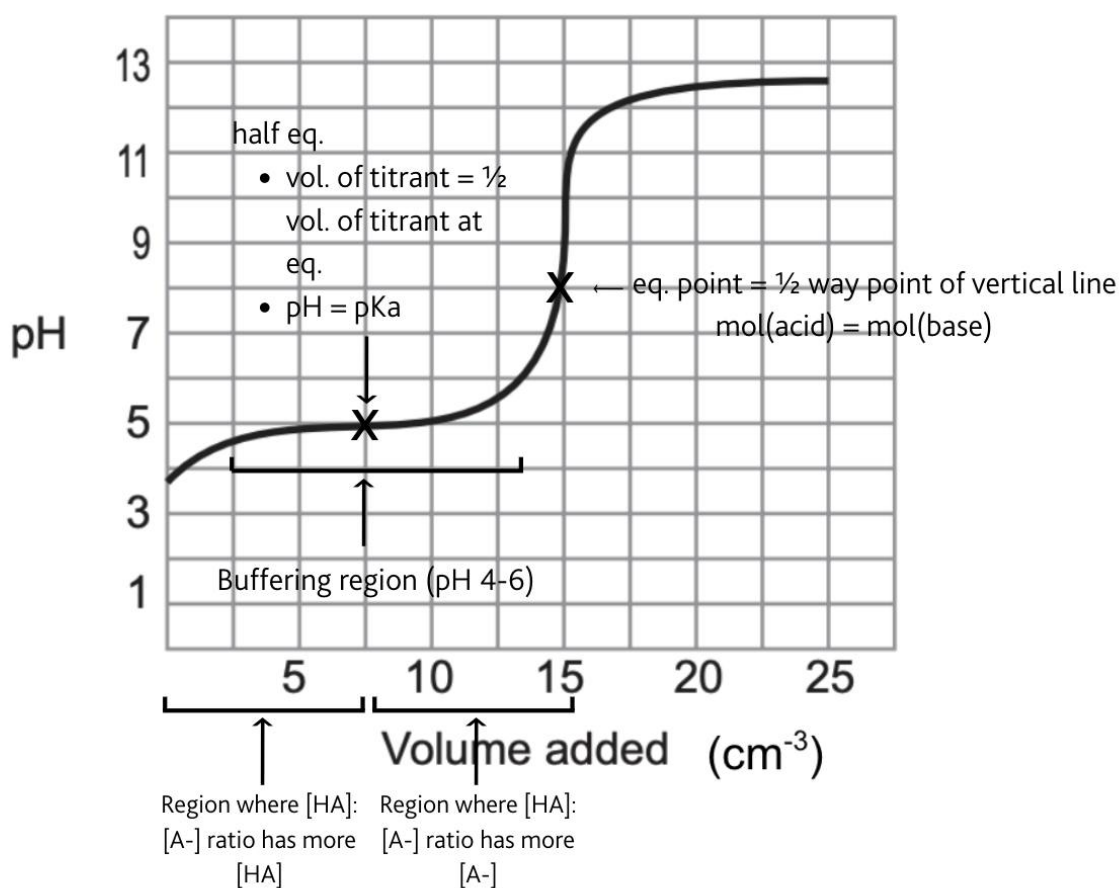
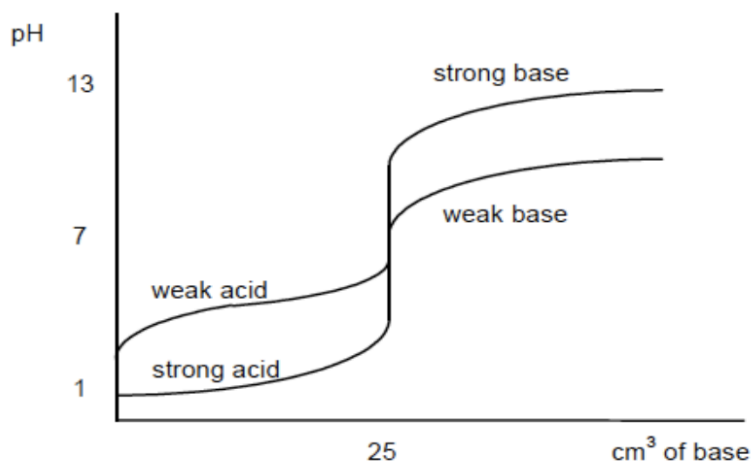
Strong:	Weak:
Group 1 hydroxides (Li, Na, K, Rb, Cs)	Mg(OH) ₂
Ca(OH) ₂	NH ₃
Sr(OH) ₂	NH ₄ OH
Ba(OH) ₂	...

- General trend: **strong acid** → **weak conjugate base**; **weak acid** → **strong conjugate base**

Properties:

Strong:	Weak:
Higher conductivity	Lower conductivity
More exo neutralisation	More endo neutralisation
Faster RoR	Slower RoR
Larger K _a /K _b (eq. lies more right)	Smaller K _a /K _b (eq. lies more left)

pH curves:



- **Equivalence point:** stoichiometrically equal amounts of acid & base & solution contains only salt & water ($\text{mol}_{\text{acid}} = \text{mol}_{\text{base}}$)
- **Half-equivalence point:** pH = pKa; vol. titrant = 1/2 vol. of titrant at eq. point

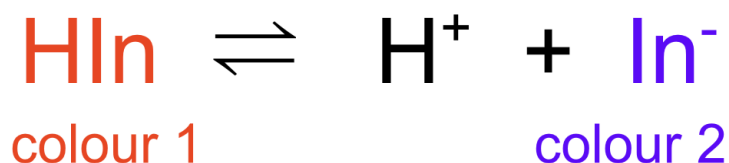
Salt hydrolysis:

- Soluble salts will **dissociate** into respective ions when dissolved in **water**
- Resulting ions are **conjugate** acids/bases of **known species in neutralisation**
- Some ions may **interact** with **water molecules** → **hydrolysis** ∴ affect **acidity**
- **SA + SB** → **neutral**
- **WA + SB** → **basic**
- **SA + WB** → **acidic**
- **WA + WB** → **unpredictable** (both conjugates **strong** enough to **hydrolyze water**)

Basic salts:	Acidic salts:
Formed between weak acid & strong base neutralisation	Formed between strong acid & weak base neutralisation
Basicity due to conjugate of weak acid (strong conjugate base)	Acidity due to conjugate of weak base (strong conjugate acid)
F^- , NO_2^- , CO_3^{2-} , PO_4^{3-} , SO_4^{2-} , HCO_3^- , $C_2H_3O_2^-$	Al^{3+} , Fe^{3+} , transition metal ions with high charges, NH_4^+ , HSO_4^+ , $H_2PO_4^+$

Indicators:

- WAs with **different colour** to **conjugate base**



- **Low pH** → eq. shifts **left** (more H^+) → see **colour 1**; vice versa
- **Endpoint**: point at which **indicator changes colour** during acid-base titration
- **Suitable indicator** → **endpoint pH** ≈ **equivalence point pH**
 - **Endpoint pH** = $pK_{a_{\text{indicator}}} \pm 1$

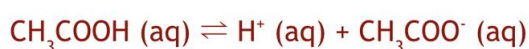
Calculating pH of WA/WB:

Assumptions:

1. $[H^+] = [A^-]$
2. $[HA]_{eq.} = [HA]_{initial} - [H^+]$, but since $[H^+] \ll [HA]_{initial} \Rightarrow [HA]_{eq.} \approx [HA]_{initial}$
3. Value of K_a given in question is for same temp. of the reaction

Worked example: Calculate the pH of 0.1 mol dm^{-3} ethanoic acid at 298.15 K

Step 1 : Write the balanced equation for the dissociation of ethanoic acid



Step 2: Write the acid dissociation expression, K_a , for dilute ethanoic acid

$$K_a = \frac{[H_3^+ \text{ (aq)}] [CH_3COO^- \text{ (aq)}]}{[CH_3COOH \text{ (aq)}]}$$

Step 3: Based on the stoichiometry of the dissociation reaction, $[H^+] = [CH_3COO^-]$.
Rewrite the K_a expression replacing $[CH_3COO^-]$ with $[H^+]$

$$K_a = \frac{[H^+ \text{ (aq)}]^2}{[CH_3COOH \text{ (aq)}]}$$

Step 4: As K_a is small assume that $[CH_3COOH]_{initial} = [CH_3COOH]_{equilibrium}$

$$[CH_3COOH \text{ (aq)}]_{equilibrium} = 0.1 \text{ mol dm}^{-3}$$

Step 5: Input the values to determine $[H^+]$. K_a value will be given in the question as it is no longer in the data book. In this case, $K_a = 1.8 \times 10^{-5}$

$$1.8 \times 10^{-5} = [H^+]^2 / 0.1$$

$$\text{So } [H^+]^2 = 1.8 \times 10^{-5} \times 0.1 = 1.8 \times 10^{-6}$$

$$[H^+] = \sqrt{1.8 \times 10^{-6}} = 1.3 \times 10^{-3} \text{ mol dm}^{-3}$$

Step 6: determine $\text{pH} = -\log_{10} [H^+]$

$$\text{pH} = -\log_{10} (1.3 \times 10^{-3}) = 2.9$$

Buffers:

- Substance that **resists change in pH** on addition of **small amounts** of acid/alkali
- Made from **WA/WB + conjugate A/B** (often added as **salt**)

- Acidic buffers: WA + SB; basic buffers: WS + SA

Method A: The Direct Method (Mixing Two Separate Bottles)

You simply take a bottle of a weak acid and mix it with a bottle of its salt.

- *Example:* Mix Ethanoic acid (CH_3COOH) and Sodium ethanoate (CH_3COONa).

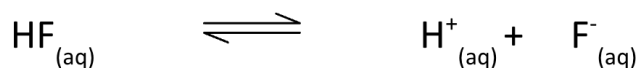
Method B: The Partial Neutralization Method (The Exam Question Method)

You start with *only* a weak acid, and you deliberately destroy **some** of it by adding a strong base (NaOH).

- To make this work, the **weak acid must be in excess**.
- The strong base is completely used up, and the chemical reaction transforms it into the conjugate base salt.

Common ion effect:

- When conjugate salt of WA/B is added to that A/B $\rightarrow$ **reverses dissociation (Le Chatelier)**
- e.g.:



the addition of $\text{NaF}_{(\text{aq})}$

shifts equilibrium this way $\leftarrow$

Lowering the % dissociation of HF

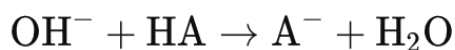
Process behind resistance:

H^+ :



- Added H^+ **removed** through reaction with $\text{A}^- \rightarrow \text{HA}$

OH^- :



- OH^- **neutralized** through H^+ **supplied by WA** $\rightarrow$ **water**

Buffer pH:

- Depends on **pKa/pKb** of WA/B
 - When choosing: choose acid with **pKa close to desired pH**

Buffer capacity:

- Depends on **conc. of A/B & conjugate**
- Determined by **ratio** $[\text{A}^-]/[\text{HA}]$
- Largest buffer capacity have ratio of 1 $\rightarrow \text{pH} = \text{pKa}$

Influencing factors:

- **Temp.** $\rightarrow$ **Ka**, affects both **pH + capacity**
- **Dilution** $\rightarrow$ **conc.** (only affects **capacity**)

Calculating pH of buffers:

Henderson-Hasselbach:

$$\bullet \text{pH} = \text{pKa} + \log([\text{A}^-]/[\text{HA}])$$

$$\bullet \text{pH} = \text{pKa} + \log(\text{base/acid})$$

$$\bullet \text{pOH} = \text{pKb} + \log([\text{M}^+]/[\text{MOH}])$$

$$\bullet \text{pOH} = \text{pKb} + \log(\text{acid/base})$$

- E.g. Calculate pH of a solution that is 0.50mol dm^{-3} ethanoic acid ($K_a = 1.8 \times 10^{-5}$) & 0.25mol dm^{-3} sodium ethanoate

$$\text{pH} = -\log(1.8 \times 10^{-5}) + \log(0.25/0.5) = 4.44$$

Calculating pH of mixtures:

Strong acids and strong alkalis (either in excess)

1. Calculate the initial number of moles of H^+ and OH^- ions in the solutions
2. As H^+ and OH^- ions react in a 1:1 ratio; calculate unreacted moles species in excess
3. Calculate the volume of solution by adding the two original volumes
4. Convert volume to dm^3 (divide cm^3 by 1000)
5. Divide moles by volume to find concentration of excess the ion in mol dm^{-3}
6. Convert concentration to pH

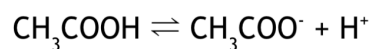
If the excess is H^+ $\text{pH} = -\log[\text{H}^+]$

If the excess is OH^- $\text{pOH} = -\log[\text{OH}^-]$ then
 $\text{pH} + \text{pOH} = 14$

or use $K_w = [\text{H}^+][\text{OH}^-] = 1 \times 10^{-14}$ at 25°C therefore
 $[\text{H}^+] = K_w / [\text{OH}^-]$ then
 $\text{pH} = -\log[\text{H}^+]$

Alternative:

- Made through **partial neutralisation** of WA/B by a SA/B



If we add more acid to this solution...

- $[\text{H}^+]$ would obviously increase.
- So the equilibrium moves to the left.
- This would remove the added H^+ by combining it with CH_3COO^-
- $[\text{H}^+]$ decreases therefore minimising change to the pH

If we add more alkali to this solution...

- $[\text{OH}^-]$ would obviously increase.
- OH^- reacts with H^+
- $[\text{H}^+]$ decreases
- So the equilibrium moves to the right.
- $[\text{H}^+]$ increases therefore minimising change to the pH.